COMPUTERIZED ENROLLMENT SYSTEM FOR PRECIOUS LITTLE LIGHTS ACADEMY

A Software Engineering Project by

Hofileña, Paul April A.
Mejia, Neil Sean D.
Raoet, Christian Louie C.

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Dedication

We, the proponents, Hofileña, Paul April A., Mejia, Neil Sean D., and Raoet, Christian Louie C. humbly dedicate this study to Precious Little Lights Academy to modernize their enrollment process.

The students aim to provide a reliable reference for the future researchers who potentially want to continue implementing a Computerized Enrollment System.

We, the proponents, also dedicate this to our family and friends, who have been supporting us in our education journey.

Acknowledgement

With the proponents' continuous perseverance and patience, this study is completed. The proponents are grateful to those who helped with the completion of this study.

First and foremost, the proponents would like to thank their adviser, Dr. Karren V. De Lara, for the continuous guidance throughout the study. This paper would not be complete without her expertise and feedback, which has greatly contributed to the completion of this paper.

Secondly, the proponents would also like to thank the beneficiary, Mrs. Annalie T. Amadore, for allowing the proponents to use Precious Little Lights Academy as the subject of this study. Her trust and cooperation to the proponents were vital in pushing the researchers to gather data and insights that contributed to the success of this study. The proponents deeply appreciate her willingness to open the doors of Precious Little Lights Academy.

Lastly, the proponents would like to express their gratitude to God, who guided the proponents to always believe in themselves. God is Good, God is Great, all the time.

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CHAPTER 1

Introduction

Traditional systems across various industries and fields have long relied on manual processing and paper documentation of data and information. While some systems have been digitalized using spreadsheet applications like Excel and Google Sheets, though these traditional systems are reliable and effective, they encounter some limitations. Utilizing the technologies available today can provide improvements that are beneficial in the long run. However, the need to change an existing working system is often seen as unnecessary, as it would cost time and resources.

The manual processing of student enrollment, through the use of pen and papers, still exists in many schools. The disadvantages of these traditional methods are often related to the processing and efficiency of the whole enrollment process. For example, human-related errors can be encountered, where a wrong input in crucial information of the student could restart the whole enrollment process of the student, leading to additional processing time. Another disadvantage is the security of the information, when documents are stored in file cabinets that are accessible to any staff members, the sensitive information and documents are vulnerable to being viewed or mishandled.

It is important to recognize that these manual systems are susceptible to automation by transitioning to a computerized system. Such systems can be integrated into computer software, which acts as an information processor that generates, manages, and displays information to the user (Canevska, 2024). This is where software development is required in order to transition from a paper-based system to a computerized system. Software development is the process of planning and developing a software program (Wolny, 2024).

A computerized enrollment system consists of features that will automate most of the enrollment process without human intervention, with improvements in the accuracy of the information that has been provided in the system. Those features could be the generation of filled out forms that are print-ready for less workload for the registrar.

Transitioning from a manual paper-based system to an automated enrollment system offers significant advantages. The advantages include efficiency, security, and reliability. Unlike the pen-and-paper approach, which is prone to human-related errors that can restart the entire process, a computerized system uses automated data validation to ensure the accuracy and consistency of information. By automating the enrollment process, the staff can be relieved of work, which can be utilized in other aspects of operating the school, and can improve the efficiency of the enrollment process. Furthermore, while paper documents that are stored in file cabinets are vulnerable to being viewed or mishandled, a computerized system enhances security by protecting sensitive documents with management and access controls and password protection. Ultimately, a computerized system provides a far more reliable, secure, and efficient system that improves the enrollment process for both students and staff.

The methodology that the proponents chose to follow in the development of this system is the Waterfall software development model. The tasks that will be handled in the development of the software will be done sequentially based on the phases of the model, which are Requirement Definition, System and Software Design, Implementation and Unit Testing, Integration and System Testing, and Operation and Maintenance. By handing the development process one at a time, the developers will be able to assess the overall quality and completeness of the task that is developed for a phase before moving on to the succeeding stages. Though some criticized the Waterfall model due to the less involvement of the users in the development process (Tsui et al., 2023), the proponents will still be able to moderate this limitation by creating a detailed documentation of the requirements that were gathered from the interview conducted at the beneficiary school.

Background of the Study

The proponents conducted this study for an institution that still uses a traditional enrollment system, which is Precious Little Lights Academy, a Christian School that was founded in 1988, providing quality education from nursery up until to the grade school level. The school is administered by Rev. Leoncio L. Amadore Jr. as the Administrator, and Mrs. Annalie T. Amadore, M.A.Ed, as the Principal. As of 2025, the school is solely based in Burgos, Rodriguez, Rizal, and currently accommodates 119 students.

The current enrollment procedure in Precious Little Lights Academy starts with the appointment of the guardian for a meeting, which is scheduled through email or by messaging the school's Facebook page. After the consultation, a diagnostic test of the student is conducted to look for signs of psychological disorders like ADHD and similar disorders, as the institution is not certified to provide education for students with such conditions. Upon confirmation of enrollment and payment, the guardian is required to submit the required documents, such as SF10, and they are tasked to personally fill out a form that includes the student and guardian information section and a statement of account. Once the enrollment period concludes, the Principal personally manages and organizes each student's and teacher’s schedules and room assignments from Nursery to Grade level 6 for the academic year.

Precious Little Lights Academy is facing several problems with its current enrollment system and procedures. One problem that they acknowledge is how the current system, they are utilizing both Excel-based and Paper-based procedures, is inefficient. The school has to print out an empty form, which is to be filled out by the client or guardian, and this process is unsystematic and slow. Another problem is that the storage for these sensitive documents is a single filing cabinet, which is prone to security risks. Lastly, the scheduling of each section per grade level is personally done by a single individual, which is labor-intensive and can be a huge burden.

Statement of the Problem

Precious Little Lights Academy was founded in 1988 by Rev. Leoncio L. Amadore Jr., ever since then, the school has depended on a manual enrollment process, which required in-person appointments and presentation of vital documents, such as the SF-10 and photocopies of the PSA. The paper documents are also kept in a box and filing cabinets by the academy, easily accessible to unauthorized individuals. Furthermore, the use of Excel in manually entering the data needed for long-term storage of student information is ineffective in storing, editing, and retrieving student information. As stated, these issues reduce the school's overall operational effectiveness and ability to manage its enrolled students across multiple grade levels.

This system is designed to solve the following:

1. Poor student information management, as the academy totally depends on manual methods of enrollment, where the parents are required to submit

documents physically, and Microsoft Excel is employed for data input for long-term storage of data. This is very inefficient for storing, editing, and retrieving student information, hindering the establishment of a computerized and user-friendly system to handle records.

2. Too much time and energy in enrollment processes because the present manual enrollment process requires a great deal of time and effort on the part of school employees and parents, who have to make face-to-face visits for diagnostic tests and interviews. This results in major delays, especially during high-enrollment periods, directly preventing the reduction of the overall enrollment process time.
3. High risk of inaccuracy and insecurity of data, the lack of a specialized digital system causes an increased likelihood of data inaccuracies and errors in student records. At the same time, physical storage of confidential student files in unsecured boxes and cabinets exposes them to unauthorized access, without enhancing data accuracy or creating proper security measures.
4. Absence of strong security for sensitive data, including certain confidential information of students, both hard copy and digitally stored in plain Excel files, is without proper security mechanisms and open to misuse by unauthorized users. This reinforces the imperative necessity for creating strong security mechanisms to safeguard sensitive student data.

Objectives of the Study

The proponents will design a system that will address the challenges faced by the Precious Little Light Academy by replacing the existing manual operation of the school with a LAN-based Computerized Enrollment System. The proposed system will focus on speeding up the operation of the school, improving data integrity, and strengthening the handling of information security. This will enable the school to maintain accurate student records and will undoubtedly improve the school's efficiency, especially during enrollment periods.

The specific objectives are:

1. To create a user-friendly LAN-based computerized enrollment system capable of efficiently recording, updating, and retrieving student information.
2. To automate and optimize the student enrollment process in order to significantly minimize the time and manual effort required.
3. To implement validation techniques and data integrity checks to improve the accuracy and reliability of student records.
4. To establish robust security protocols, including access controls and data protection mechanisms, to safeguard sensitive student information from unauthorized access or breaches.

Significance of the Study

This particular study aims to propose a Computerized Enrollment System as an efficient and modern means to address the cumbersome challenges posed to smaller institutions in manually enrolling students, which include tons of paperwork, unnecessary error possibilities, and numerous queues, causing frustration to parents, staff, and students. By automating some of the most important steps, it makes operations more effective, minimizes the chances of human error, and provides increased security to student records. This system is accessible and beneficial to key stakeholders, including school administrators, non-teaching staff, teachers, parents, and students.

Data entry and payments form the core of the enrollment process, and the system greatly assists administrative tasks, allowing staff to focus more on the school's primary objectives. Staff members receive relief from manual workload since the system streamlines the enrollment domain under their control. Parents and students appreciate the seamless, straightforward process that reduces unnecessary stress caused by long queues. Furthermore, the system's configurable design supports other small institutions, such as community training programs and community-run centers, making these community learners additional beneficiaries.

This work illustrates a first-of-its-kind approach in IT and software engineering by developing an inexpensive, easily scalable solution tailored for under-resourced, smaller schools. It applies contemporary software engineering practices such as modularity, data privacy, and interface usability to solve real-world educational problems. By prioritizing affordability and simplicity, this study promotes the democratization of technology, demonstrating how resource-constrained environments can benefit from IT without heavy infrastructure deployment.

The proponents of the project, however, gain experience in the practical perspective of software engineering. The system, which emphasizes good database design, the essence of strong security, and a user-oriented approach toward those with little computer skills, really has an impact on areas of IT, computer science, and software engineering.

In essence, this study empowers smaller schools and similar institutions to prosper in the digital era by replacing traditional enrollment practices with a reliable, efficient system that strengthens the registration processes and streamlines operations. It shows how even the smallest institutions can create a brighter, more efficient future for their communities, setting the stage for ongoing growth in educational technology.

Scope and Delimitations

Scope

This study intends to document the design, development, and deployment of a LAN-based Computerized Enrollment System exclusively for the only branch of Precious Little Lights Academy, in Burgos, Rodriguez, Rizal. This system primarily aims to digitalize the school's enrollment system, and is to operate within the school's local network without requiring an internet connection.

The system is to be developed as a desktop application using Java SE (Standard Edition) version 25. Data will be stored and managed by PostgreSQL, accessed via JDBC, and hosted on a local machine acting as the central server within the LAN environment.

The system will include the following core modules:

The Security Module handles the protection of the data and operations and authentication of the users logging into the system in order to prevent unauthorized access, modification, and deletion. The implementation of this module will include a Login page where the user can log into the system with their registered account credentials. Change password functionality will be processed by generating a One Time Code that will be sent to the registered email address of the user for verification. The Level of Access security will be implemented by enforcing access and permission based on three predefined roles, which are Admin, Registrar, and Teacher. The Admin role can access all the modules in the system. The Registrar role can process only enrollments and payments. The Teacher role can only view section schedules and enter if a student is qualified for moving up or recognition to the next grade level. The protection and security of the user's credentials is attained by implementing password hashing using bcrypt in the PostgreSQL database. Upon successful login, the application loads a role-specific GUI, disabling unauthorized functions based on the assigned role.

The Account Registration Module handles the creation of new accounts for new users which only the Admin is authorized to use. The implementation of this module is done by providing an Account Registration page where the Admin creates an account with specified level of access. The Account Registration page will have fields that are required to provide necessary information for the account. The system will verify the credentials in the PostgreSQL database in case of duplication, only if there are no issues will prompt the system to store all the user's credentials.

The Search Module is a core function in the system which provides a fast and efficient way to find relevant information and data within the system's database. The implementation of this module begins with a search bar, where a user can input their query. The request for this information is then processed by the system, which constructs a database query that will be sent to the PostgreSQL database, where the returned results will be retrieved by the system and displayed in an organized, formatted list depending on the type of information requested by the end-user. The display of this list will vary depending on the specific information requested by the end-user.

The Enrollment Module guides the Registrar through capturing and storing mandatory data for new or returning students during an in-person appointment. This includes: full student name, birthdate, address, educational background details, and guardian contact details (name, phone, email, relationship). It also schedules the mandatory diagnostic test (date/time) and tracks its status (Scheduled, Completed, Failed). Upon completion, the student will then be enrolled containing all entered data. Data is written to normalized PostgreSQL tables: students, guardians, and enrollments. Client-side validation in the Java GUI ensures fields like email and phone number follow correct formats before submission. Server-side NOT NULL and UNIQUE constraints in PostgreSQL enforce data integrity. The PDF is generated using the iText 7 library, populating a pre-designed template with data fetched via a JDBC SELECT query.

The Class Schedule Management Module allows the Admin or Principal (acting as Admin) to define the class schedule per section. This involves assigning specific Subject (e.g., Math, English), Teacher, Classroom, and Time Slot (e.g., Mon/Wed/Fri 8:00-9:00 AM) combinations for each Grade Level and Section. Data is stored in interrelated tables: subjects, teachers, classrooms, grade_levels, and a central class_schedule table. The Java application implements a Greedy Algorithm that, when invoked, attempts to auto-generate the schedule of a section for a given grade level by iterating through available time slots and selecting combinations of subjects while avoiding conflicts. The UI presents a weekly timetable schedule for viewing and can be manually adjusted if prompted.

The Payment and Billing module calculates the total tuition and miscellaneous fees for a student based strictly on their Grade Level and the selected Payment Scheme (e.g., “Full Payment,” “Quarterly - 4 Installments”). It records each payment transaction (date, amount, method: Cash or GCash/Bank + reference number) and maintains a running balance. It can generate a detailed “Statement of Account” PDF showing all charges, payments, and the current outstanding balance. Fee structures are defined in a fee_schedules table (columns: grade_level_id, payment_scheme, total_amount). Each payment is recorded as a new row in a payments table, linked to the student’s enrollment_id. The current balance is calculated dynamically by the Java application via SQL: Total Fees - SUM(Payments). The Statement of Account PDF is generated using iText 7, pulling data from fee_schedules, payments, and students tables via a JOIN query.

The Reports Module generates two standardized official documents on demand, such as the Registration Form and Statement of Account. All documents are exported as print-ready PDF files saved to the local workstation. Each report type triggers a specific, predefined SQL query. For example, the Registration Form retreives data from the Students, Enrollment, and Class tables. The Java application formats the query results into the exact layout mandated by DepEd for the SF9 or the school’s template for the Certificate. The final output is rendered as a PDF using iText 7 and saved to a configurable local directory (e.g., C:\PLL_Reports\).

The Maintenance Module equips the Admin with essential system administration tools to ensure data resilience and system integrity. It allows the Admin to manually initiate a database backup via a guided interface that executes a local pg_dump command, browse and restore from previously saved backup files, and perform cleanup tasks such as archiving inactive user accounts or purging test records. While full automated disaster recovery remains out of scope, this module provides a controlled, user-friendly way to manage critical maintenance operations within the school's LAN environment.

The Help Module displays a user manual and a frequently asked questions section, which differ by user role, making it fully accessible offline within the LAN environment.

The About Module displays the details of the institution, which is Precious Little Lights Academy, the details include the school's main Bible verse that signifies their core value, school's contact information and address, credits to the beneficiary or the founders, and other details of the school.

Delimitiations

To ensure feasibility and the project's achievement of intended objectives, the following delimitations are identified:

The system is developed exclusively as a desktop application for Windows operating systems as its platform and OS restriction. It requires Windows 10 or higher and Java Runtime Environment (JRE) version 25 or later to run. Compatibility with macOS, Linux, mobile devices, or web browsers is explicitly excluded.

The system operates strictly within a Local Area Network (LAN) environment. All data communication occurs between client machines and a central PostgreSQL server hosted on-premise at the school as its network environment. No internet connectivity, cloud services, or remote access features are included. The system cannot be accessed from outside the school's physical network.

The system is designed, configured, and deployed solely for the use of Precious Little Lights Academy's single branch located in Burgos, Rodriguez, Rizal as its institutional scope. It is not scalable or configurable for multi-branch institutions or other schools without significant re-engineering.

The system does not integrate with any live digital payment gateways (e.g., GCash API, bank APIs) as its payment processing limitation. All financial transactions must be recorded manually by the Registrar. The system supports logging payments as either Cash or GCash/Bank Transfer — the latter requiring manual entry of a transaction reference number and optional screenshot upload for staff verification. Automatic payment validation or reconciliation is not implemented.

While the system includes features to schedule diagnostic test appointments and track their status (e.g., Scheduled, Completed, Failed), it does not administer, score, or interpret any diagnostic, psychological, or academic assessments. The actual test administration and evaluation remain entirely manual and offline, conducted by school staff as per current practice.

The system supports four predefined user roles only: Administrator, Registrar, Teacher, and Parent. Custom roles, role inheritance, or dynamic permission assignment are not supported. The Parent access role is limited to viewing their own child's information, as no multi-child or cross-student access is permitted.

The system assumes stable electrical power and functioning hardware. It includes no automated backup, failover, or disaster recovery mechanisms. Data loss due to power outages, hardware failure, natural disasters, or accidental deletion is not mitigated by the software. School staff are responsible for performing regular manual backups and maintaining hardware integrity.

The system generates only three predefined official documents: SF9 Report Cards, Certificates of Enrollment, and Statements of Account as all exported as PDFs. Custom report creation, ad-hoc querying, or export to formats like Excel or Word is not supported. Templates are fixed and aligned with DepEd or school-specified layouts.

The Class Schedule Management module uses a Greedy Algorithm to suggest conflict-free schedules based on teacher availability, room capacity, and time slots as its scheduling constraints. However, it does not guarantee globally optimal solutions or support complex constraints such as teacher preferences and subject sequencing. Final schedule approval and manual adjustments remain the responsibility of the Principal.

The system does not include tools for bulk import or migration of historical student records from existing Excel files or paper archives. Initial data population must be performed manually by school staff during deployment. Only records created within the system are managed and tracked.

CHAPTER 2

System Design

This chapter presents the system design for the proposed Computerized Enrollment System for Precious Little Lights Academy. Software design is a process in which the requirements gathered from stakeholders are converted into a blueprint, serving as the foundation for building the system (Canevska, 2024).

This documentation includes essential diagrams such as the Procedural Flowcharts, Context Diagram, Data Flow Diagrams, Use Case Diagram, Hierarchical Input-Process-Output Diagrams, Input-Process-Output Diagrams, and the Entity Relationship Diagram, which illustrates the system procedures, flow of data, actor interactions, functions of the modules, and the database structure with the relationship between the entities and the data (Canevska, 2024; Tsui et al., 2023).

Additionally, screen designs and user interface layouts are unveiled to ensure accessibility for all user roles in the system, which include Admin, Registrar, and Teacher.

Narrative Description of the Existing System

The current Enrollment System that is used by Precious Little Lights relies on both paper-based systems and Excel-based systems to manage the enrollment and information processing from the establishment of the school to the present time. The admin or the principal primarily uses this system in order to complete the enrollment process and continue the education of the student.

The enrollment process begins when a parent or guardian contacts the school to schedule an appointment during the enrollment period. During this appointment, the staff conducts a diagnostic test on the student to detect signs of psychological disorders. The reason for conducting this test is that the institution is not certified to educate these students. Once the student has passed this diagnostic test, the parents or guardians are required to submit necessary documents, such as the SF10 and PSA Copy. The parents or guardians are also asked to fill out student and guardian information forms, which are created using Excel and printed out. The parents are also asked for the payment for the tuition, depending on the payment scheme they chose. Once the requirements have been submitted, the admin or the registrar stores these documents in a filing cabinet for future reference and safekeeping. Lastly, before the school year begins, the admin or the principal personally organizes the sections, schedules, and room assignments in all the grade levels to ensure smooth operation and education in the school during the school year.

Although this system has been used by the institution for years and has proven functional and effective, it has provided them with several drawbacks and consequences. The manual encoding of data is slow and prone to input errors that may repeat the enrollment process if not validated properly. Although retrieving, editing, and updating records on paper is possible through the use of correction tapes or the re-printing of the forms, the resources used in this process are substantial if we are dealing with a huge number of enrollments and errors. The sensitive information stored on the filing cabinet without any form of security can induce risks such as malicious actions, unauthorized modification, and damage from external factors. Overall, the existing system is observed as inefficient, error-prone, and lacking the security required to effectively manage the enrollment process and the school operations.

Narrative Description of the Proposed System

The proposed Computerized Enrollment System for Precious Little Lights Academy is a LAN-based desktop application developed using Java SE version 25 with a PostgreSQL database accessed through JDBC. It is designed exclusively for the school's single branch in Burgos, Rodriguez, Rizal, and operates entirely within the school's local network without requiring internet connectivity.

The system replaces the current paper and Excel-based enrollment process with a secure, role-based digital solution that supports four predefined user roles: Administrator, Registrar, and Teacher. Each role has specific access rights to ensure data integrity and confidentiality.

The Security Module enforces authentication through bcrypt-hashed passwords and loads a role-specific interface upon login. Password recovery is handled via one time codes sent to pre-registered email addresses.

The Registration and Verification Module allows the Administrator to create new accounts in case of new Registrars and Teachers, in order for them to access the system.

The Search Module provides a fast and efficient way to retrieve student information and records through a structured query interface.

The Enrollment Module guides the authorized user in capturing and storing complete student information during in-person appointments, scheduling diagnostic tests, tracking the student's status, and the selection of the chosen payment plan of the student.

The Class Schedule Management Module enables the Administrator to define the academic timetable by assigning subjects, teachers, and time slots for each grade level and section, supported by a Greedy Algorithm to minimize scheduling conflicts.

The Payment and Billing Module calculates tuition and fees based on grade level and selected payment plan, records manual cash or GCash/bank transfer transactions by using a reference number, and the updating of the student's Statement of Account per transaction.

The Reports Module produces two official documents on demand, which is the Statement of Account and the Registration Form of a student, all exported as print-ready PDFs.

The Maintenance Module allows the Administrator to manually back up or restore the database of the system and archive student records.

The Help Module offers a role-filtered FAQ section and a User Manual to assist users who have inquiries on the usage of the system .

Lastly, the About Module displays the details of the institution, which is Precious Little Lights Academy, the details include the school's main Bible verse that signifies their core value, school's contact information and address, credits to the beneficiary or the founders, and other details of the school.

The system does not support internet access, cloud integration, live payment processing, multi-branch operations, bulk data import, or custom reporting, and is confined strictly to the operational context of Precious Little Lights Academy.

Procedural Flowchart of the Proposed System

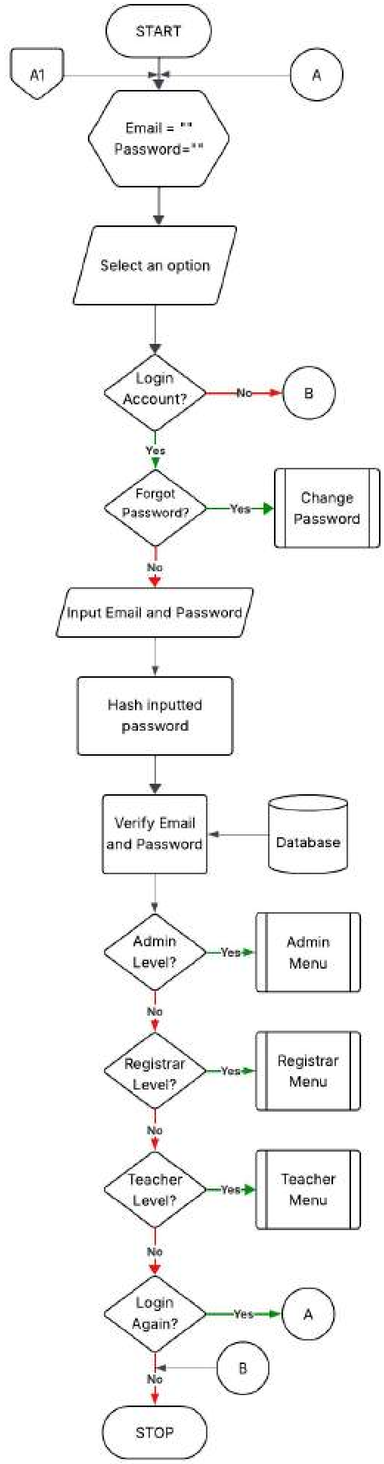


Figure 1 Login Flowchart

This figure shows the login process in the Computerized Enrollment System

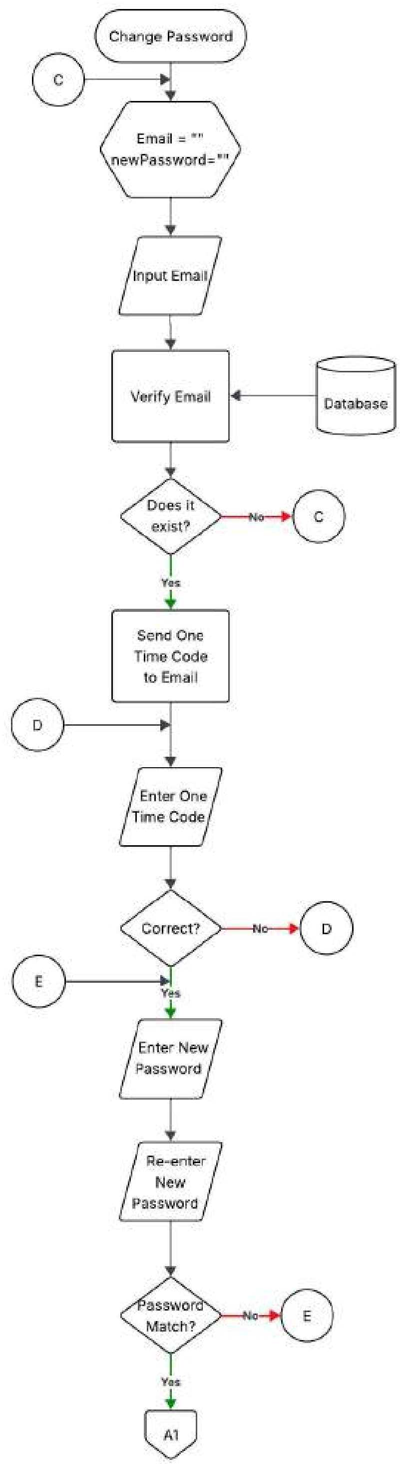


Figure 2 Change Password Flowchart

This figure shows the change account password process in the Computerized Enrollment System

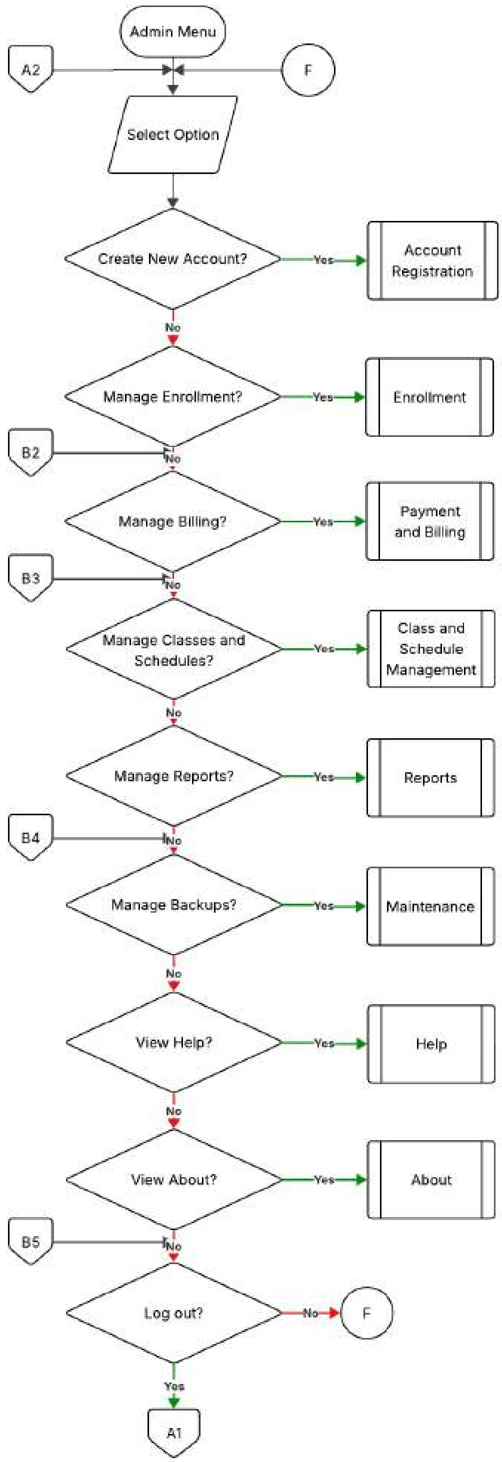


Figure 3 Admin Menu Flowchart

This figure represents the main navigation menu for the Administrator in the Computerized Enrollment System.

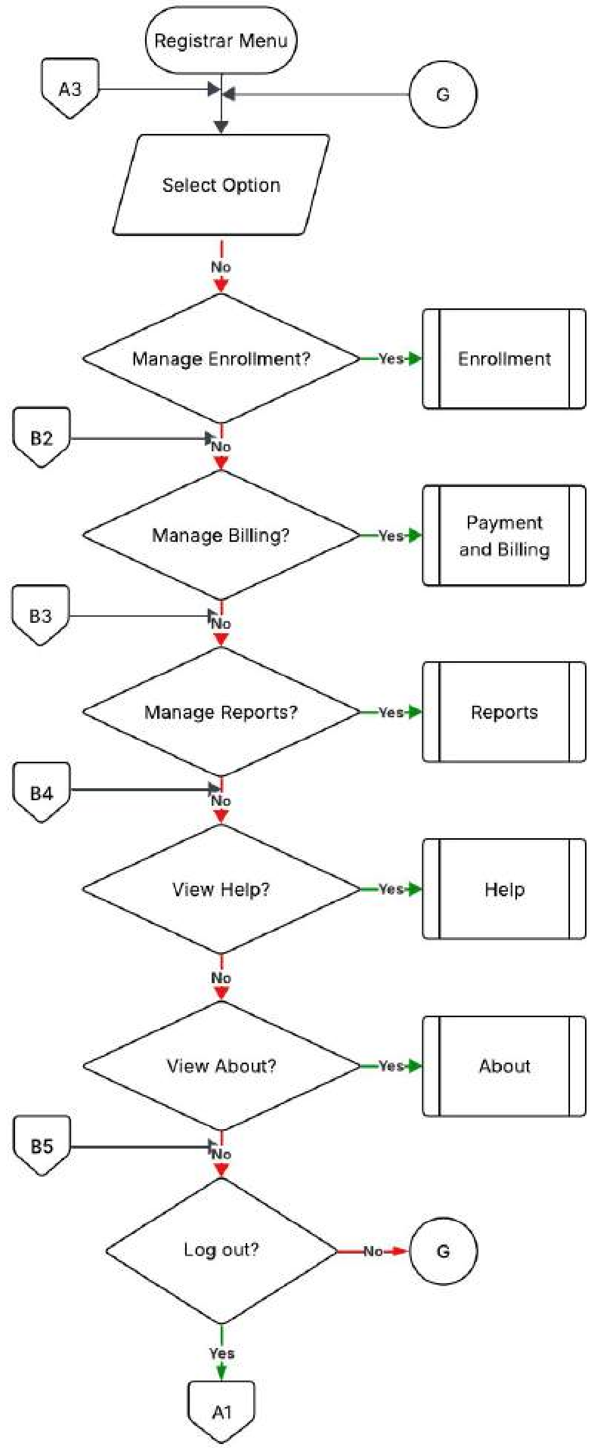


Figure 4 Registrar Menu Flowchart

This figure represents the main navigation menu for the Registrar in the Computerized Enrollment System.

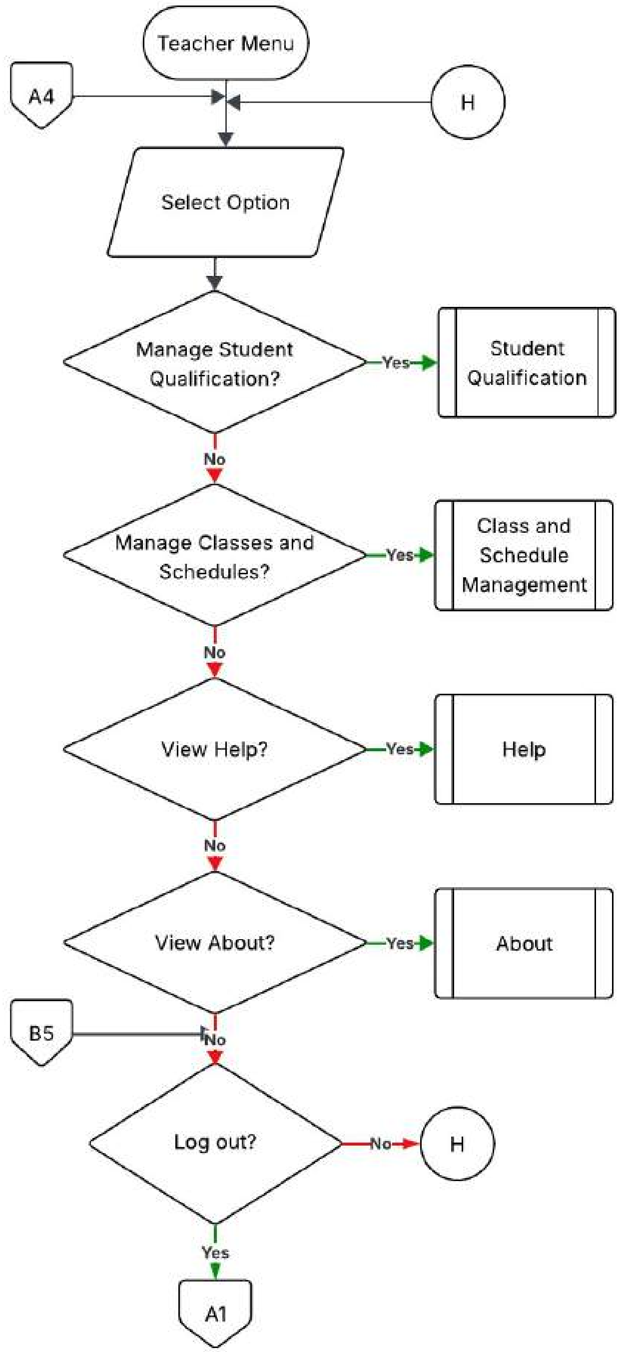


Figure 5 Teacher Menu Flowchart

This figure represents the main navigation menu for the Teacher in the Computerized Enrollment System.

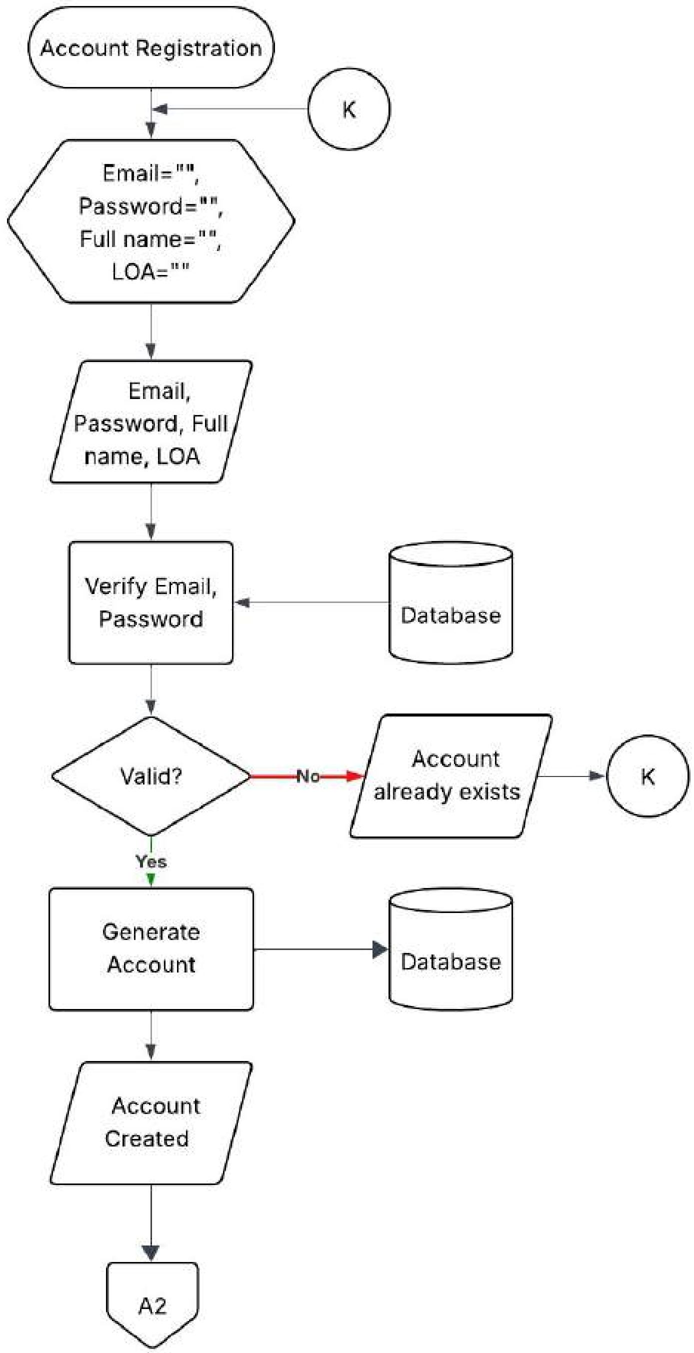


Figure 6 Account Registration Module Flowchart

This figure shows the account registration module of the proposed system where the user can register a new account for a new user.

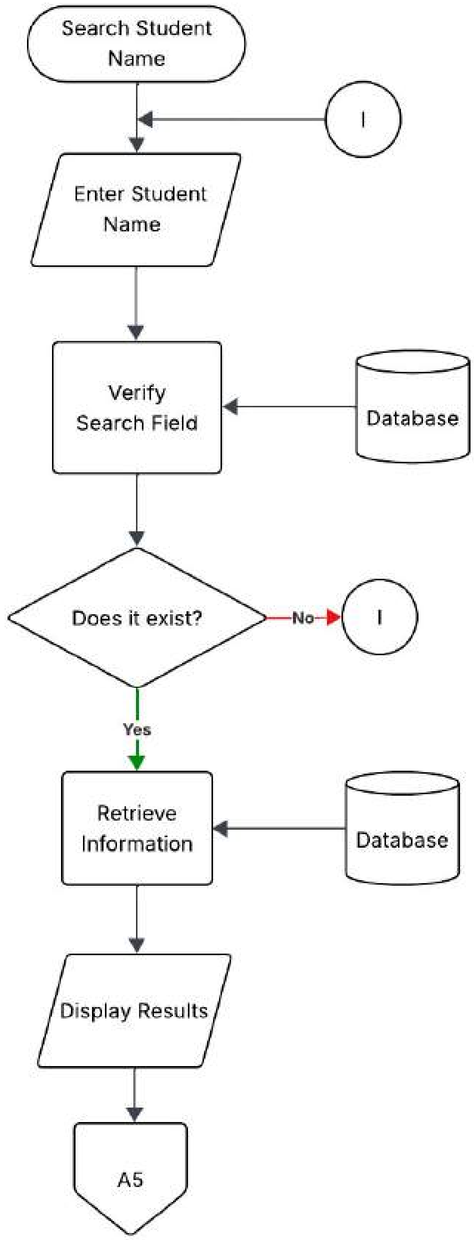


Figure 7 Search Student Name Module Flowchart

This figure shows the search process which allows a user to search a student's name in order to retrieve related information.

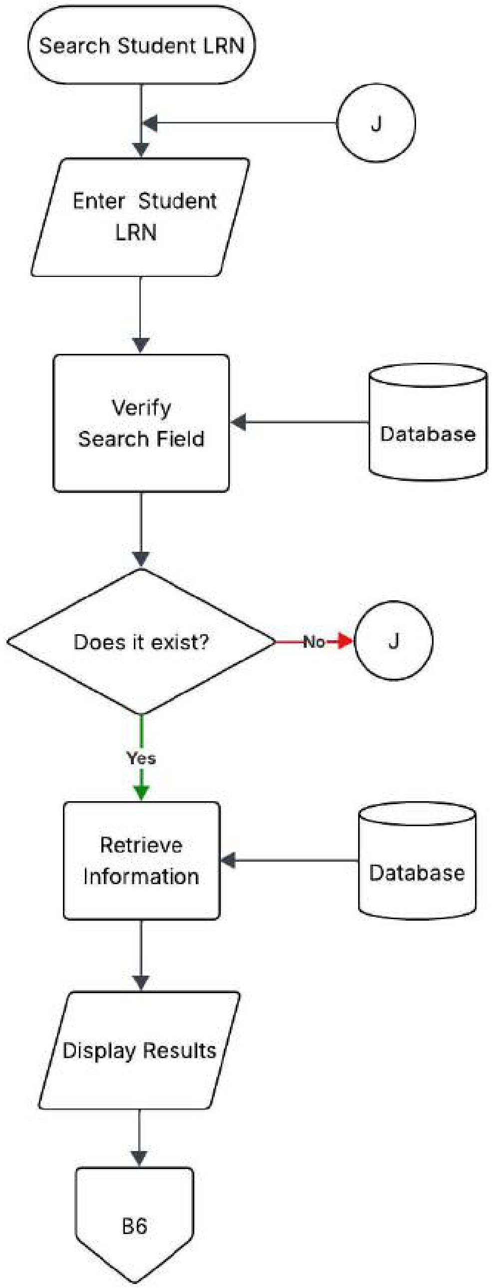


Figure 8 Search Student LRN for Reports Module Flowchart

This figure shows the search process, which allows a user to search a student's Learner Reference Number (LRN) in order to retrieve related information.

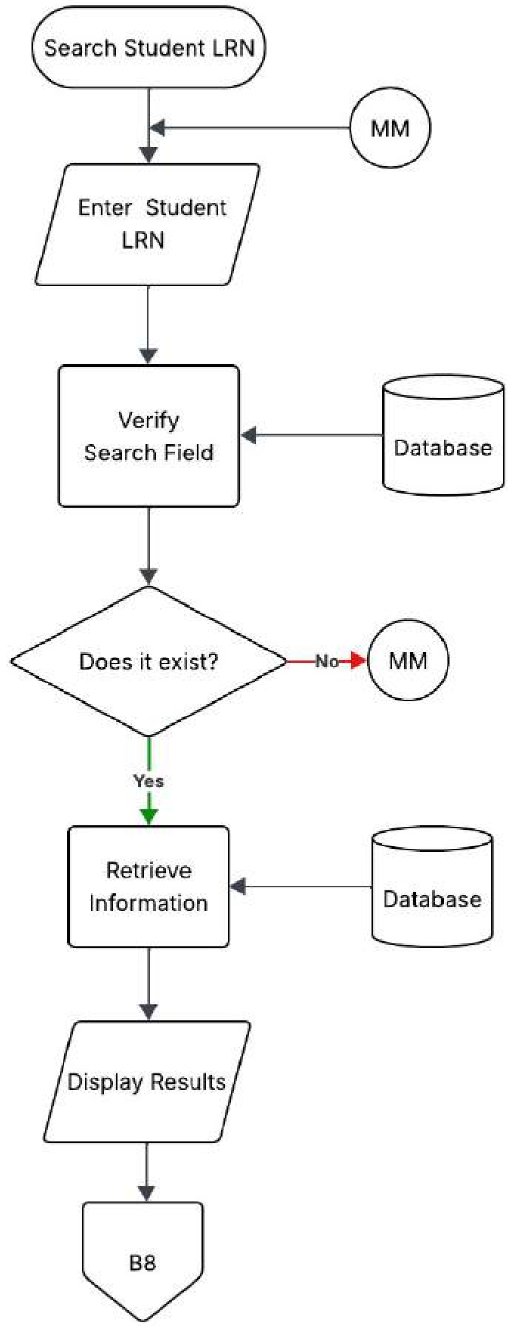


Figure 9 Search Student LRN for Payment and Billing Module Flowchart

This figure shows the search process, which allows a user to search a student's Learner Reference Number (LRN) in order to retrieve related information.

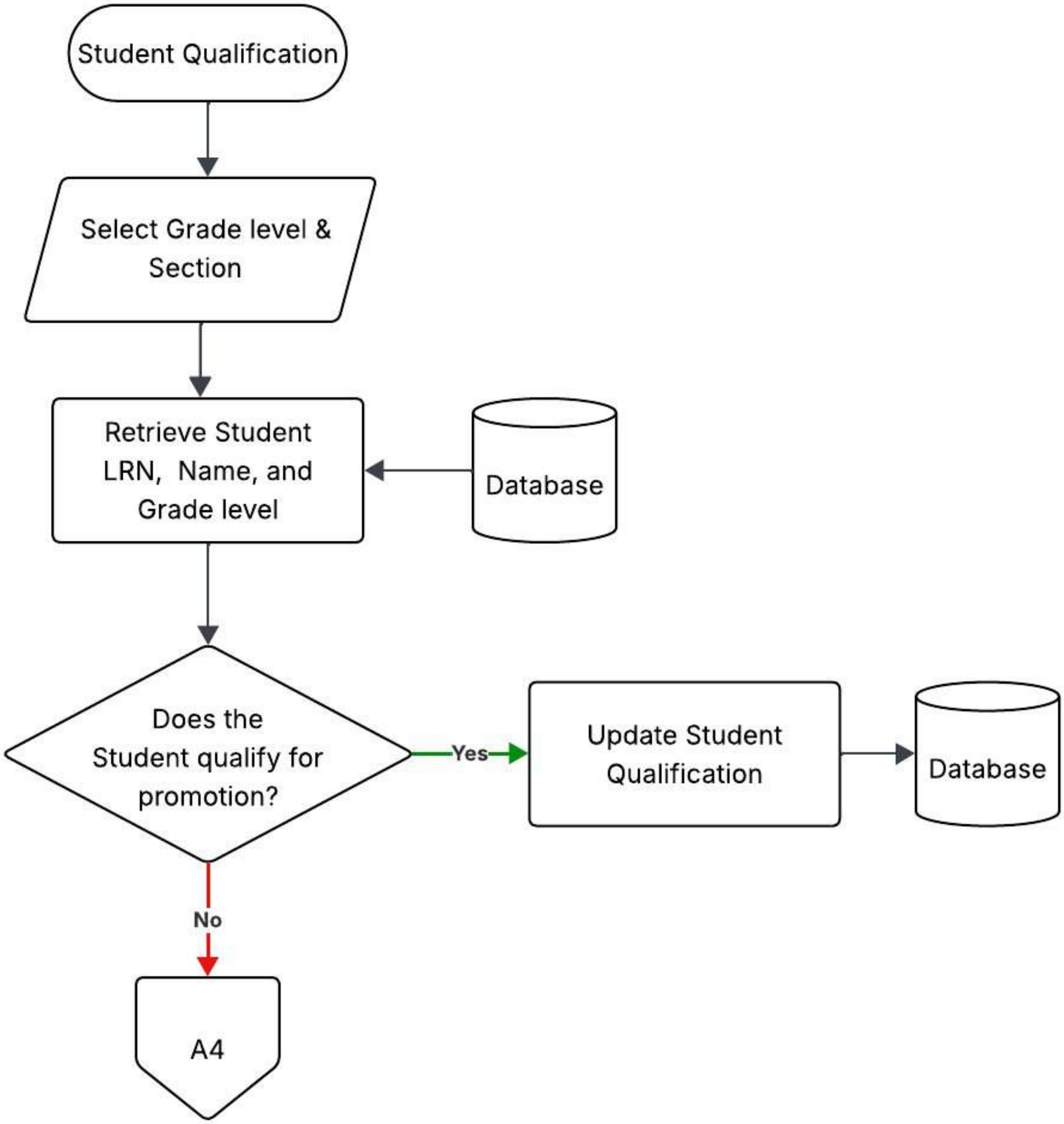


Figure 10 Student Qualification Module Flowchart

This figure shows the student qualification module of the proposed system, where the user can set if a student is qualified for recognition/moving up.

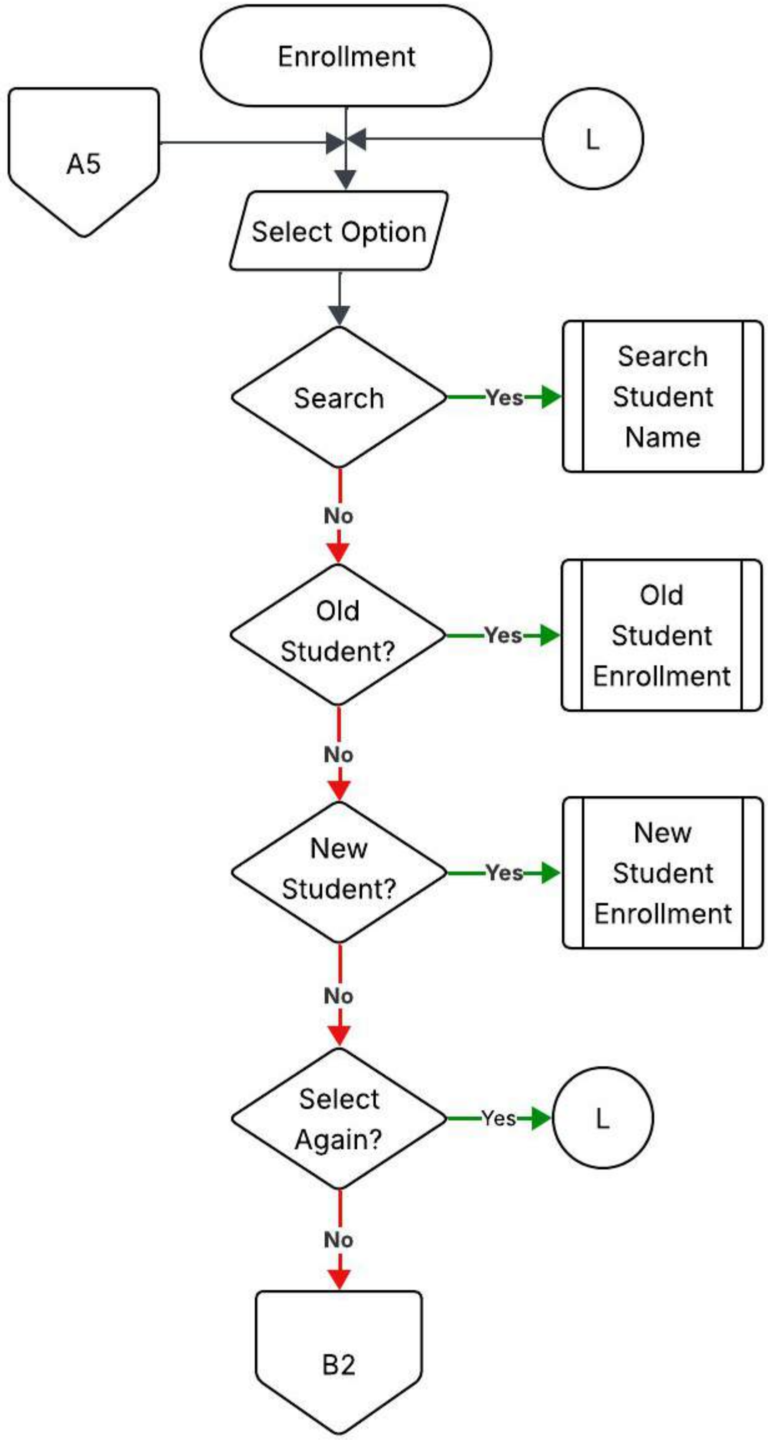


Figure 11 Enrollment Module Flowchart

This figure shows the enrollment process, which includes searching for students and managing new and old student enrollments.

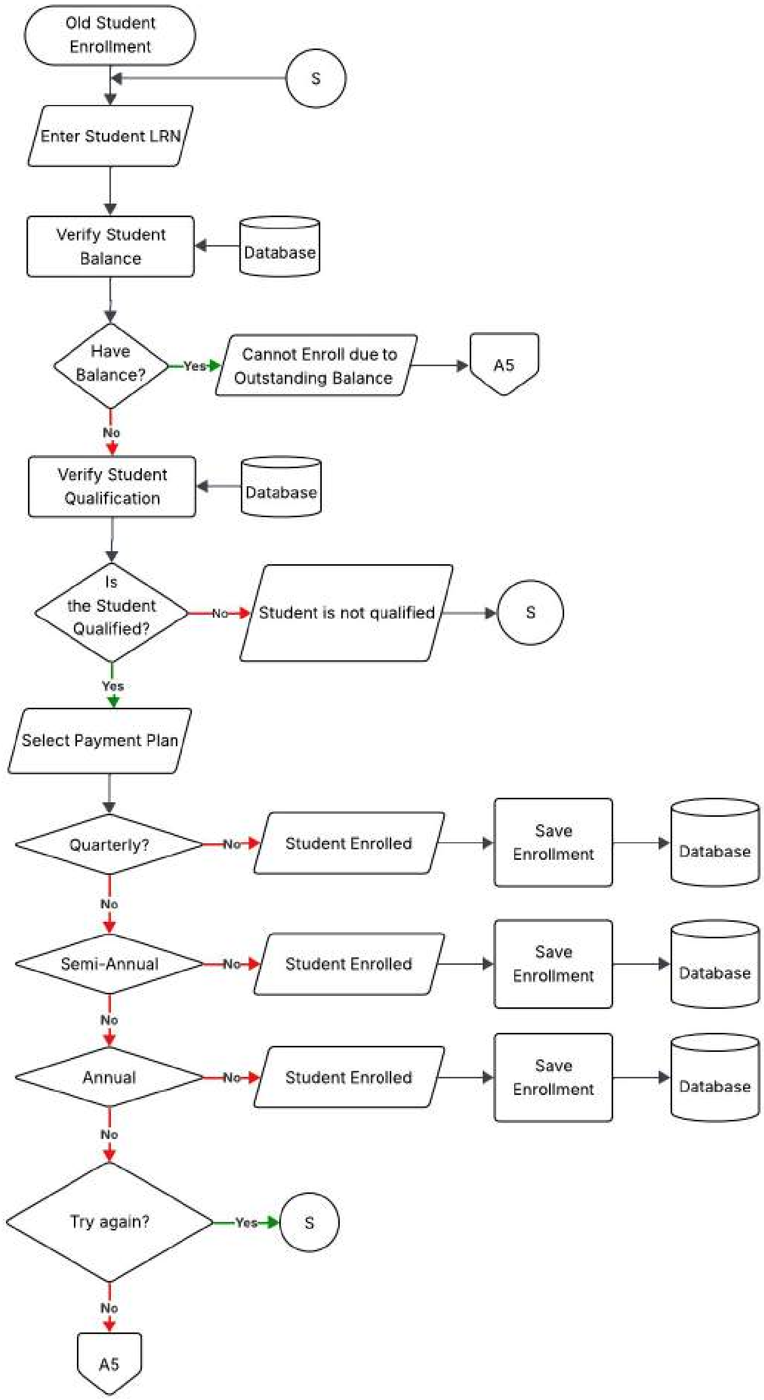


Figure 12 Old Student Enrollment Sub-Module Flowchart

This figure shows the enrollment process for old students' enrollment.

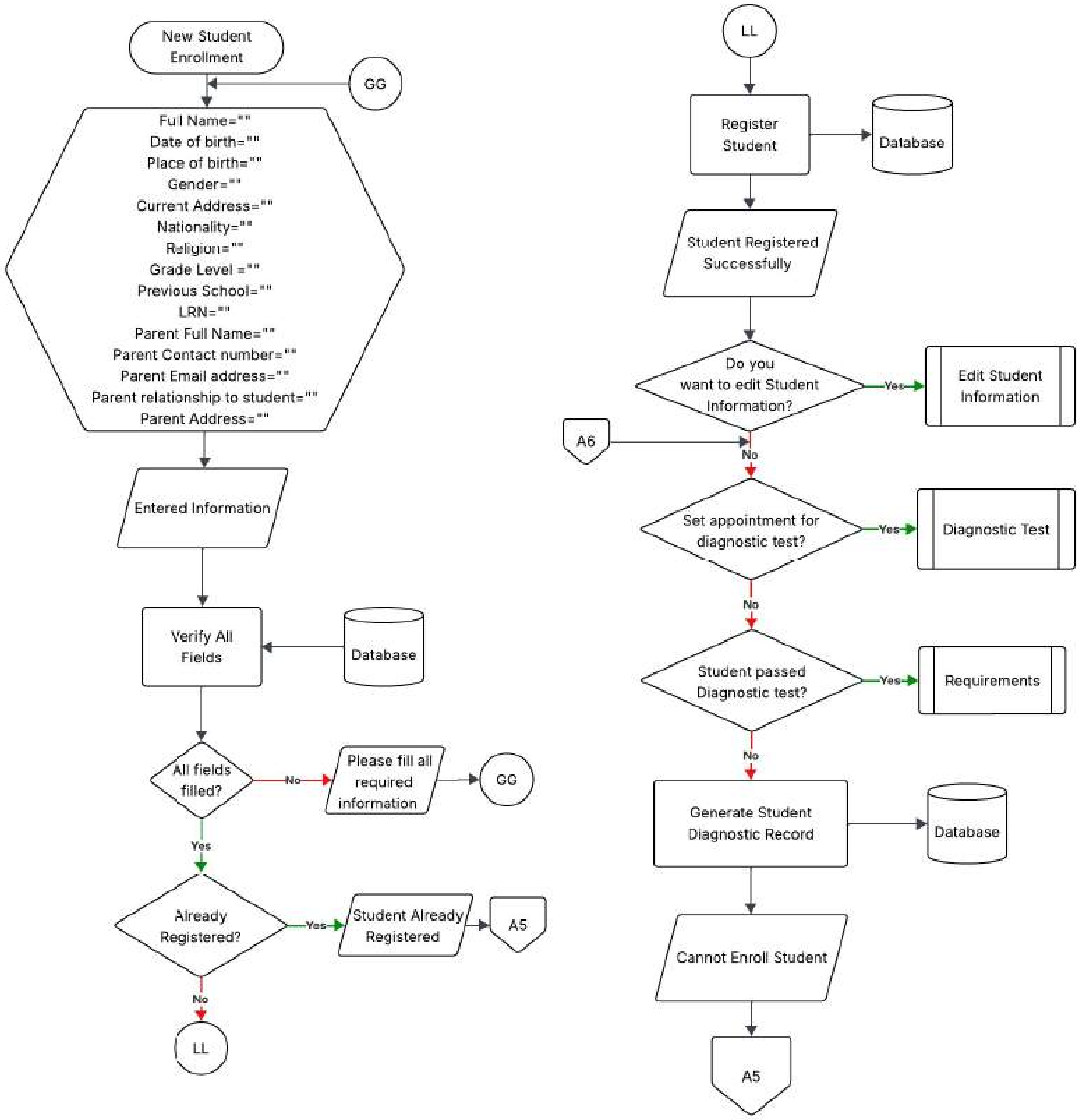


Figure 13 New Student Enrollment Sub-Module Flowchart

This figure shows the enrollment process for new students’ enrollment.

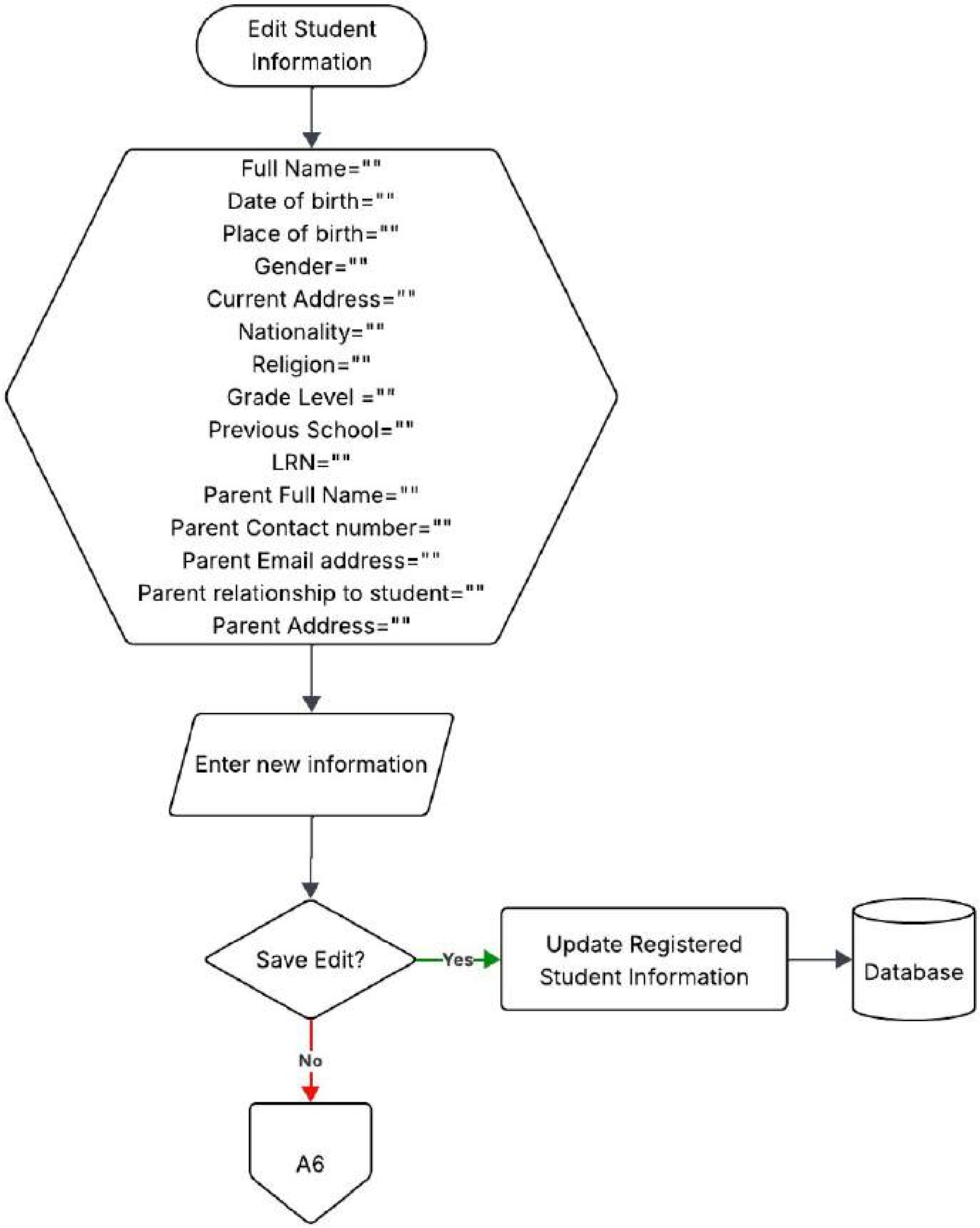


Figure 14 Edit Student Information Module Flowchart

This figure shows the process of editing a student's information.

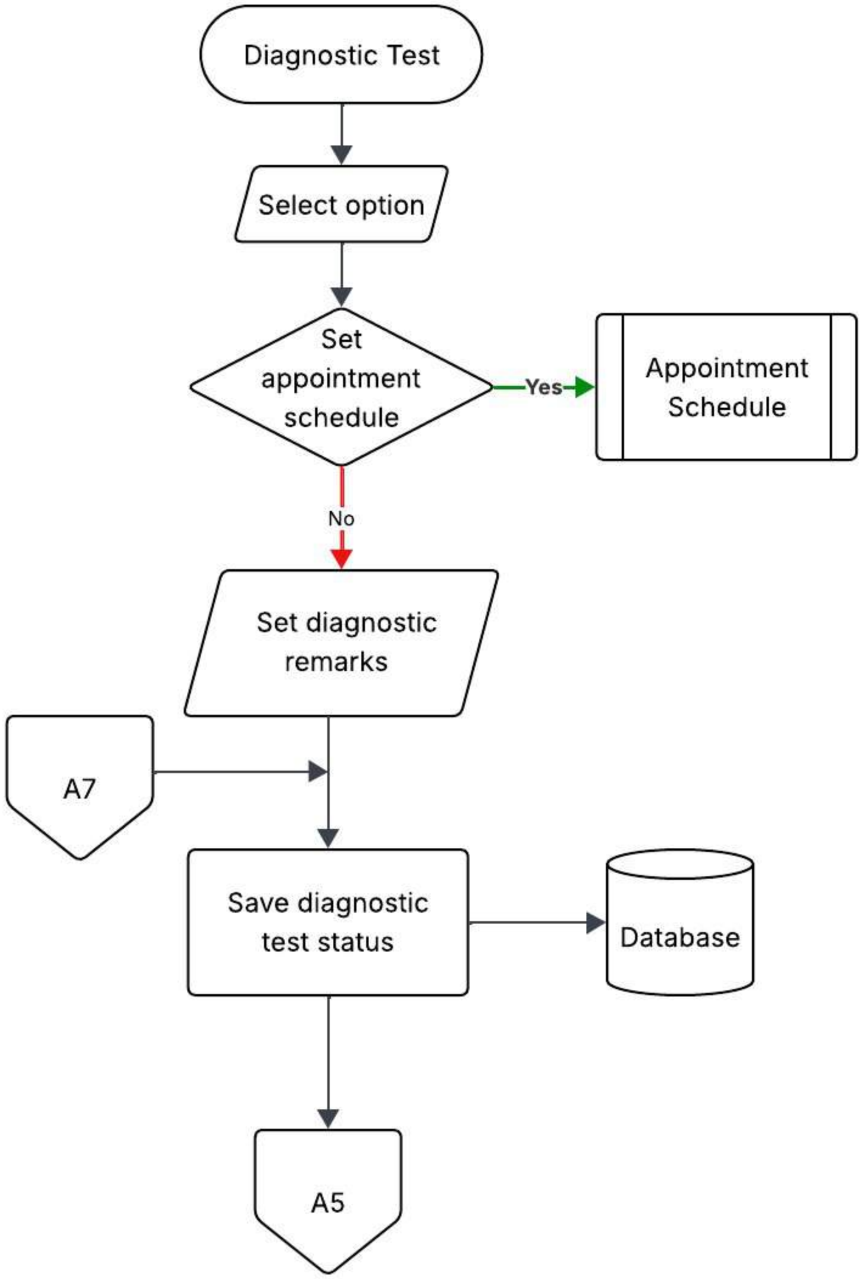


Figure 15 Diagnostic Testing Module Flowchart

This figure shows the process of managing diagnostic tests, where the user can set an appointment schedule or provide diagnostic remarks for a student.

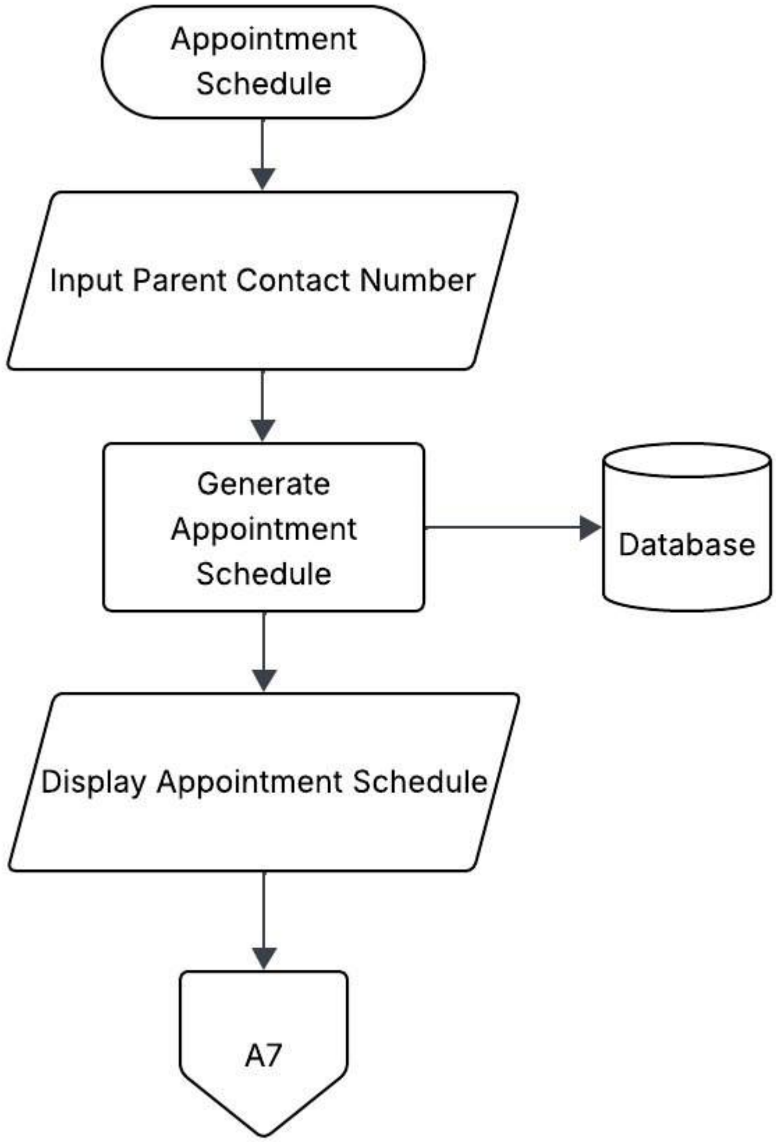


Figure 16 Appointment Schedule Module Flowchart

This figure shows the process of creating a new diagnostic test appointment.

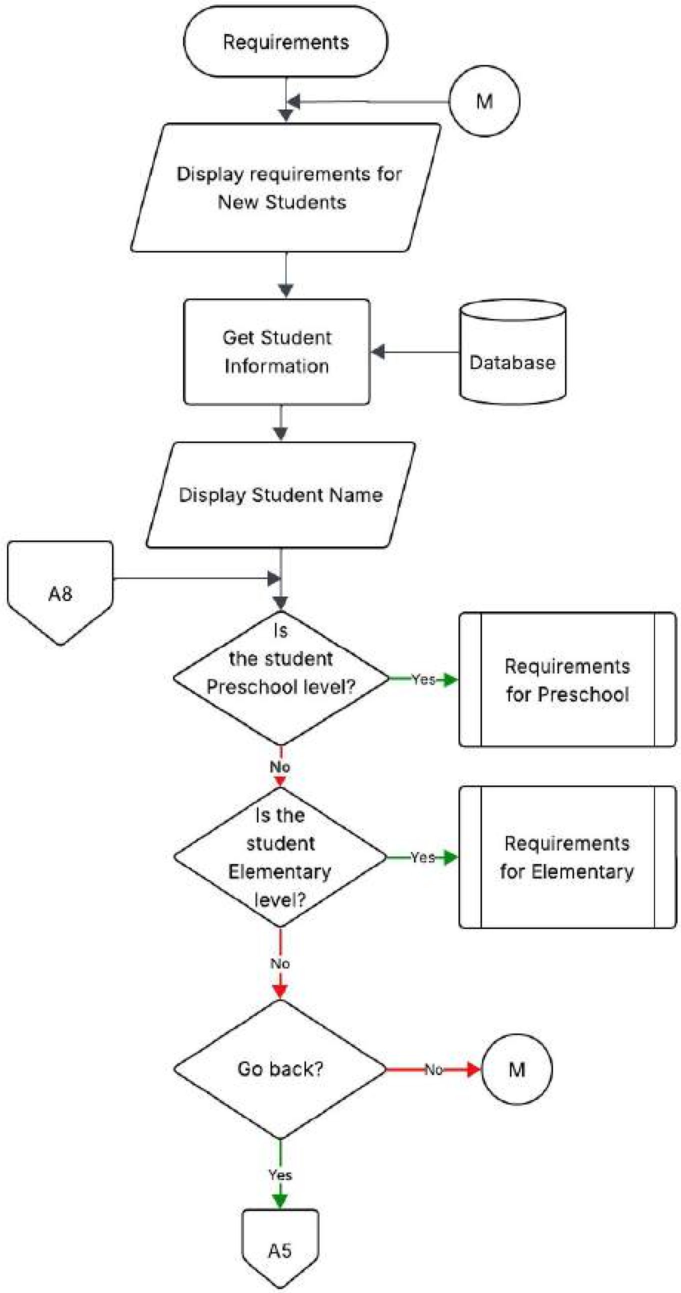


Figure 17 Requirements Module Flowchart

This figure shows the procedure of remarking requirements for enrollment where the user can select if the enrollee is Preschool or Elementary level, as they have different requirements.

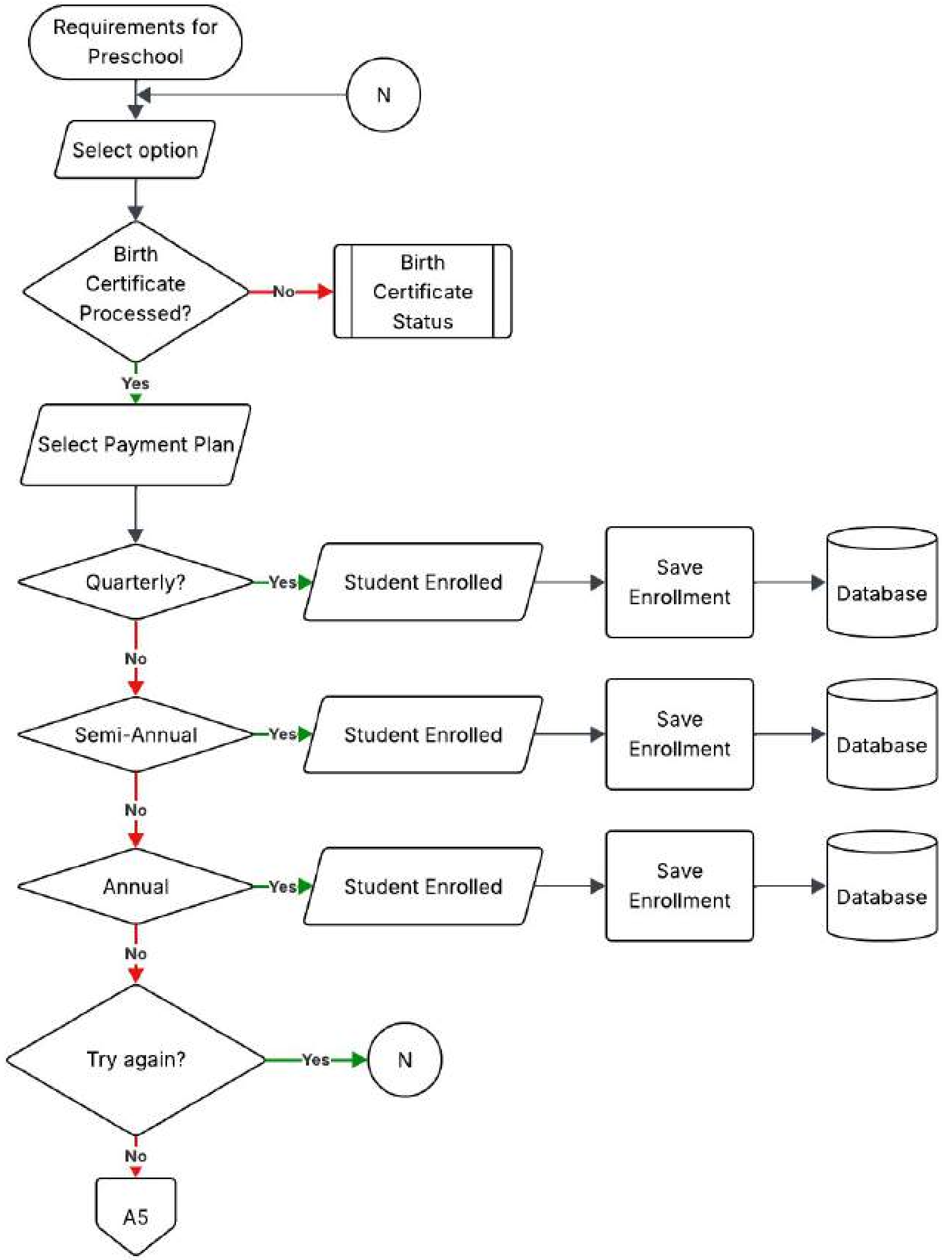


Figure 18 Requirements for Preschool Sub-Module Flowchart

This figure shows the processing of requirements and payment plan selection for Preschool enrollees.

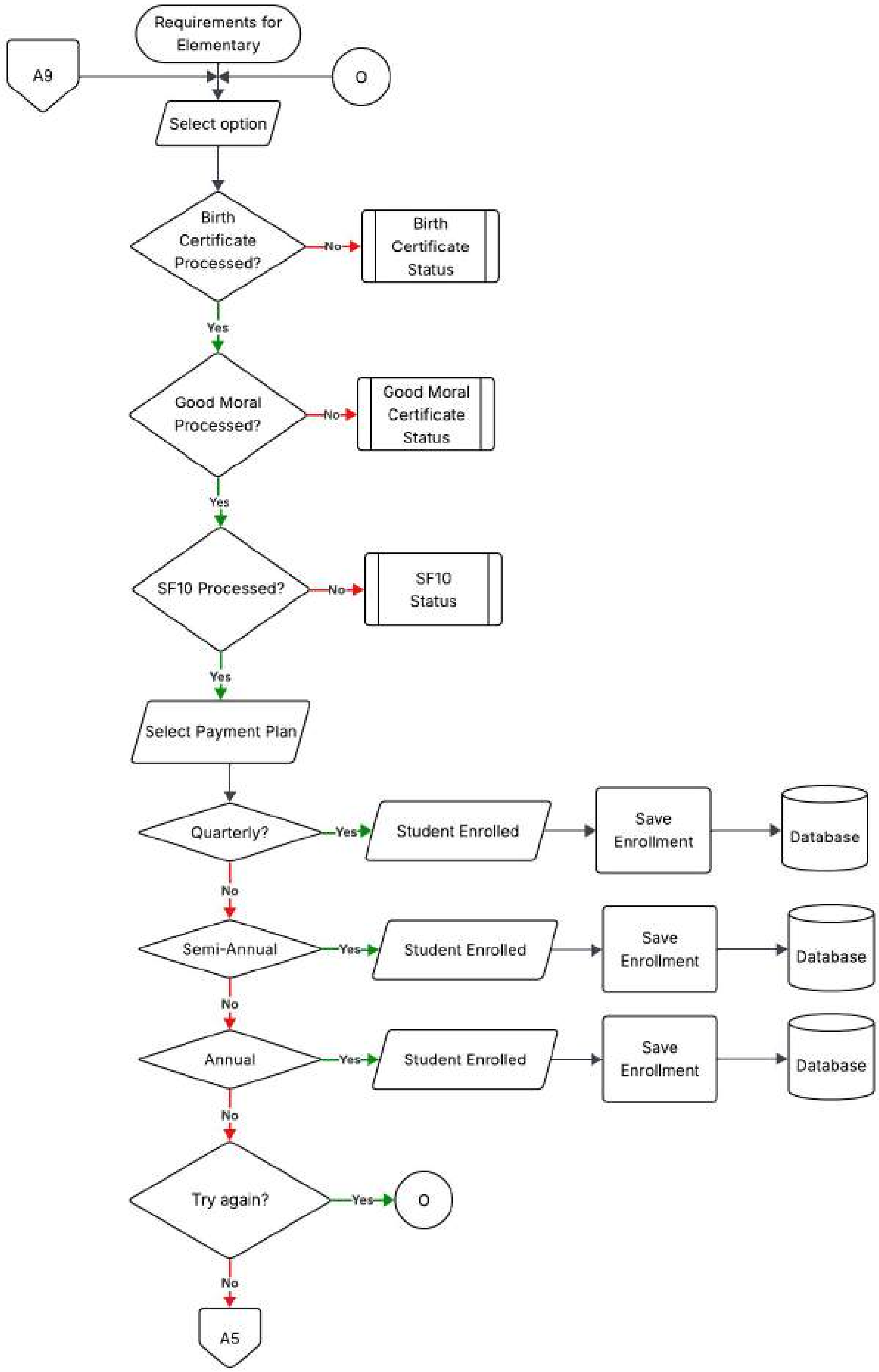


Figure 19 Requirements for Elementary Sub-Module Flowchart

This figure shows the processing of requirements and payment plan selection for Elementary enrollees.

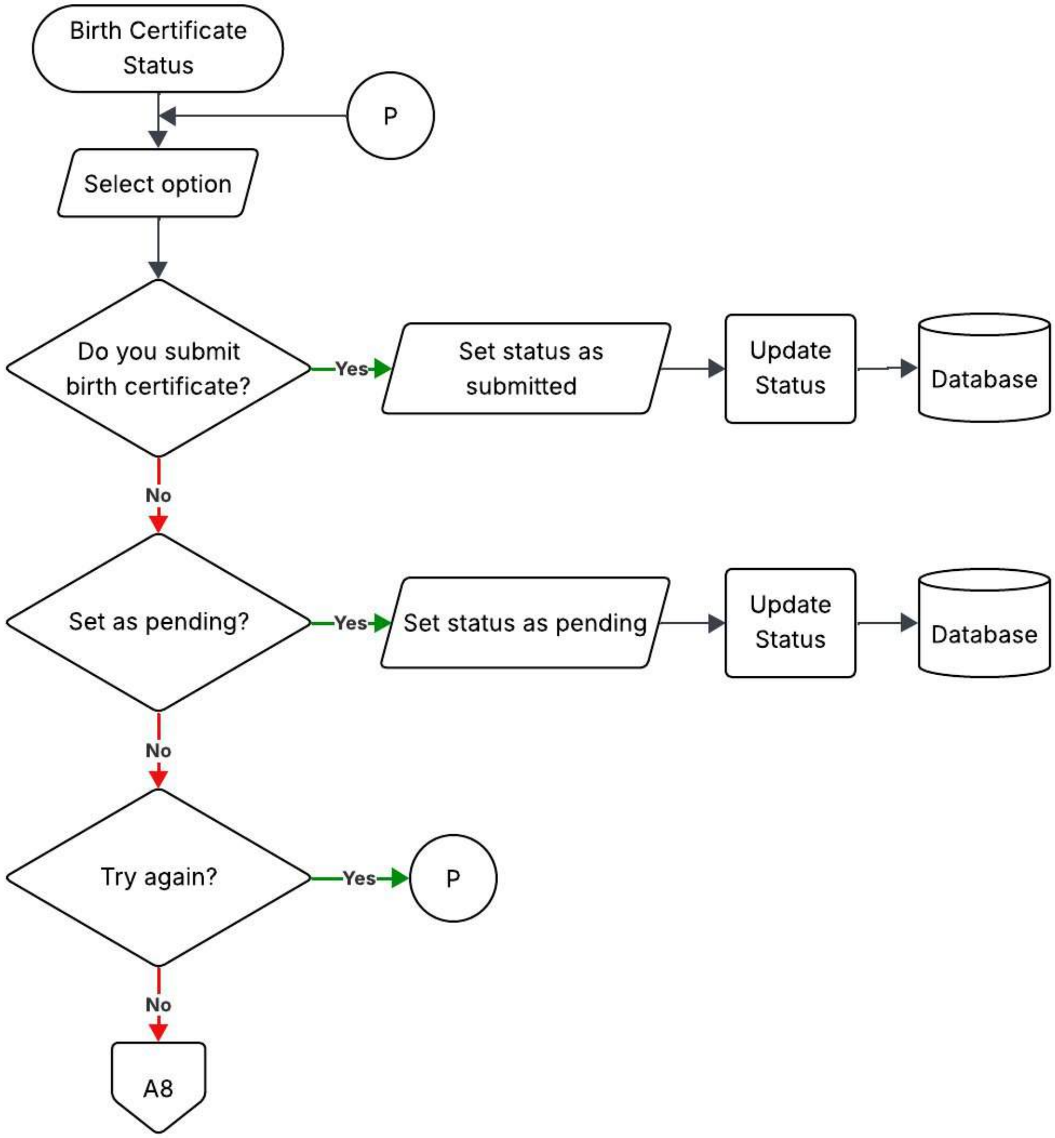


Figure 20 Birth Certificate Status Flowchart

This figure shows the remarking of the Birth Certificate status process.

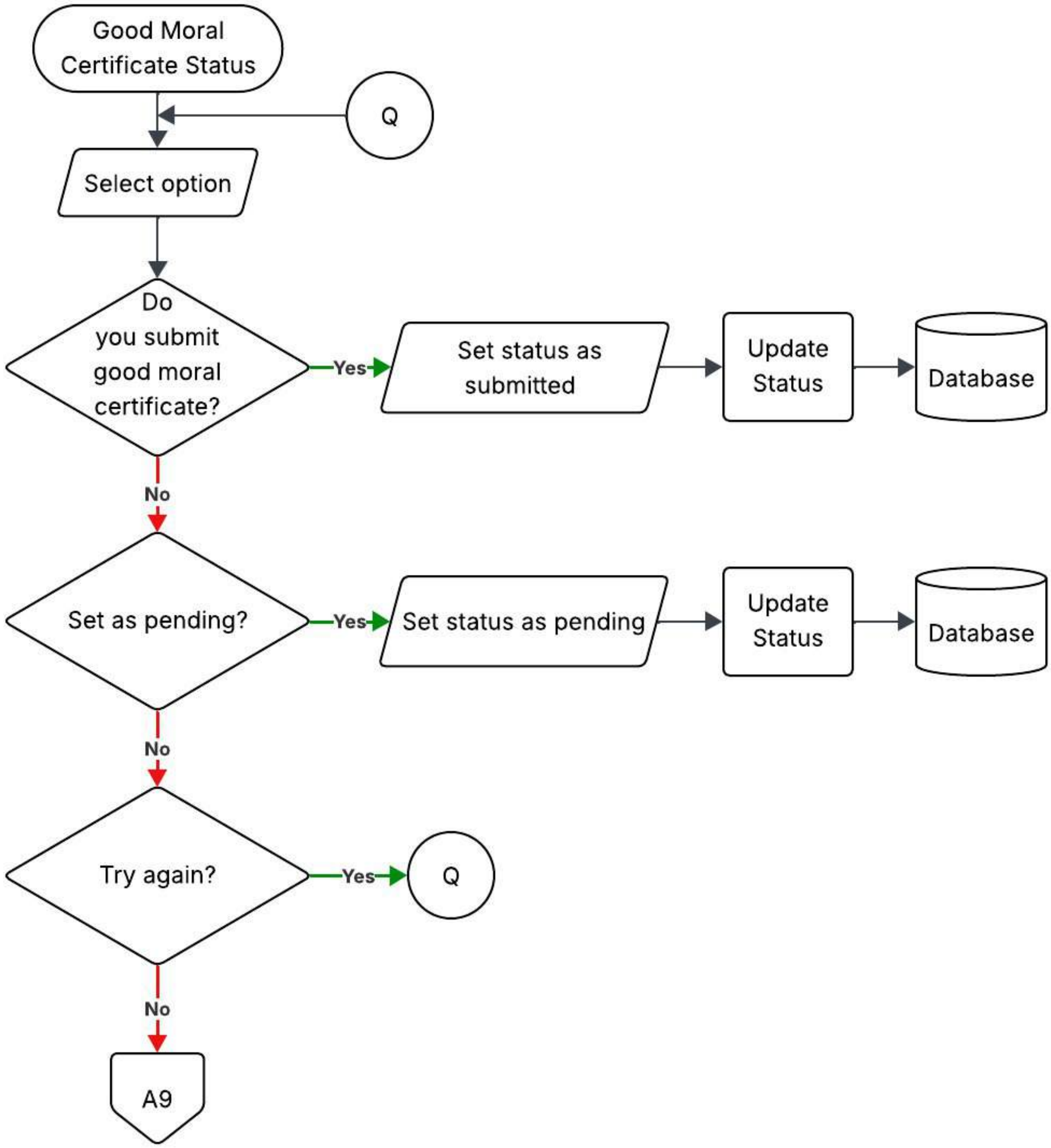


Figure 21 Good Moral Certificate Status Flowchart

This figure shows the remarking of the Good Moral Certificate status process.

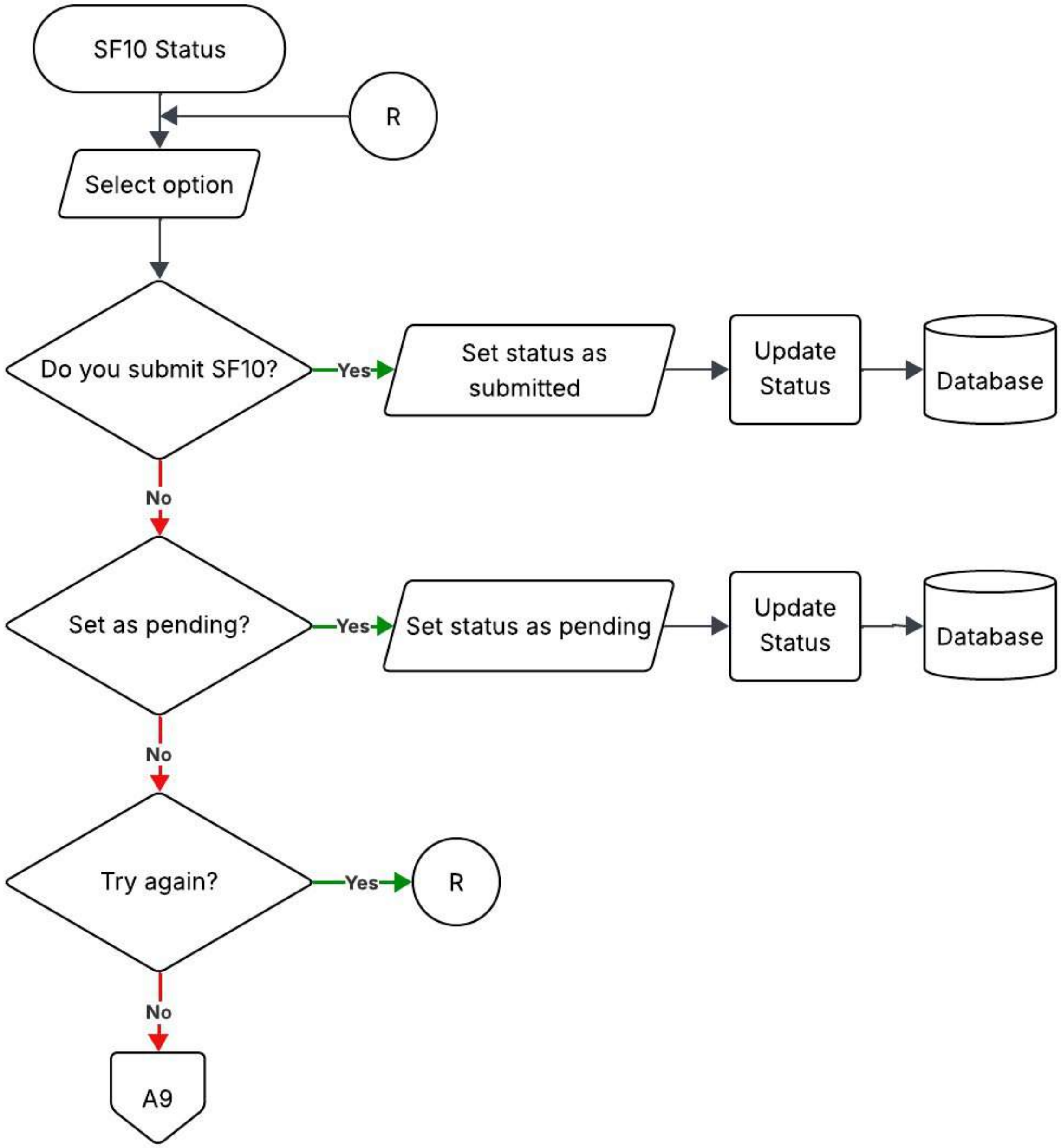


Figure 22 SF10 Status Flowchart

This figure shows the remarking of the SF10 status process.

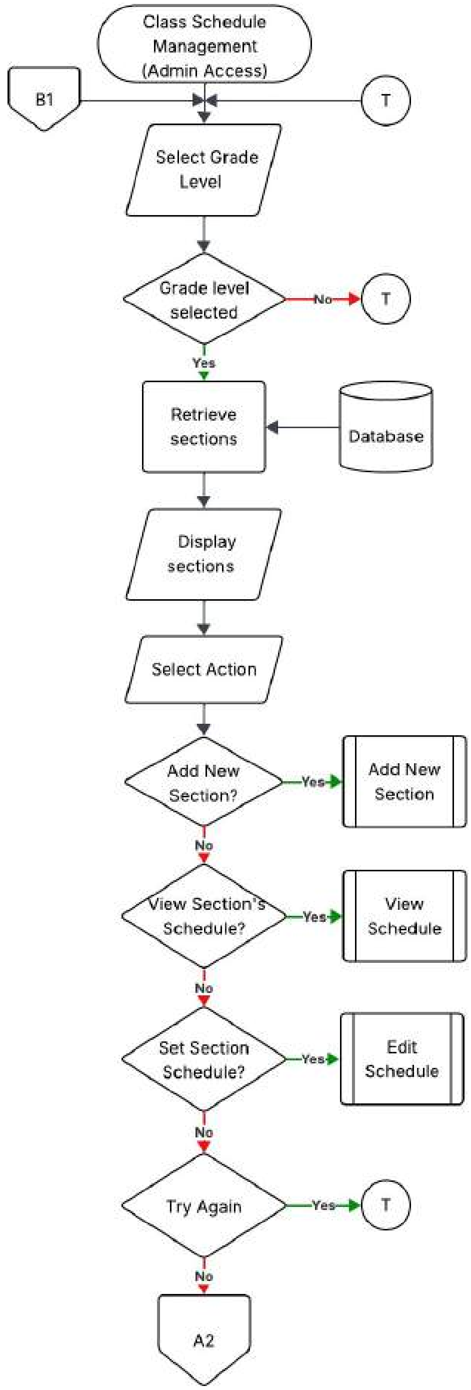


Figure 23 Class Schedule Management (Admin Access) Module Flowchart

This figure describes the Admin's menu for the class schedule management module of the system

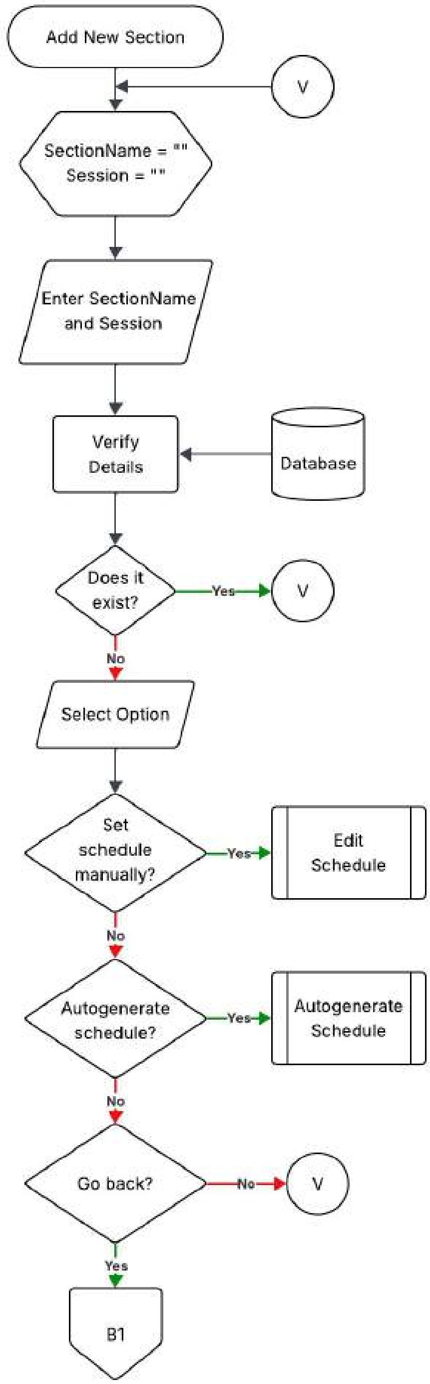


Figure 24 Add New Section Sub-Module Flowchart

This figure shows the process of adding new sections where the user can specify if they want the schedule to be autogenerated or manually set.

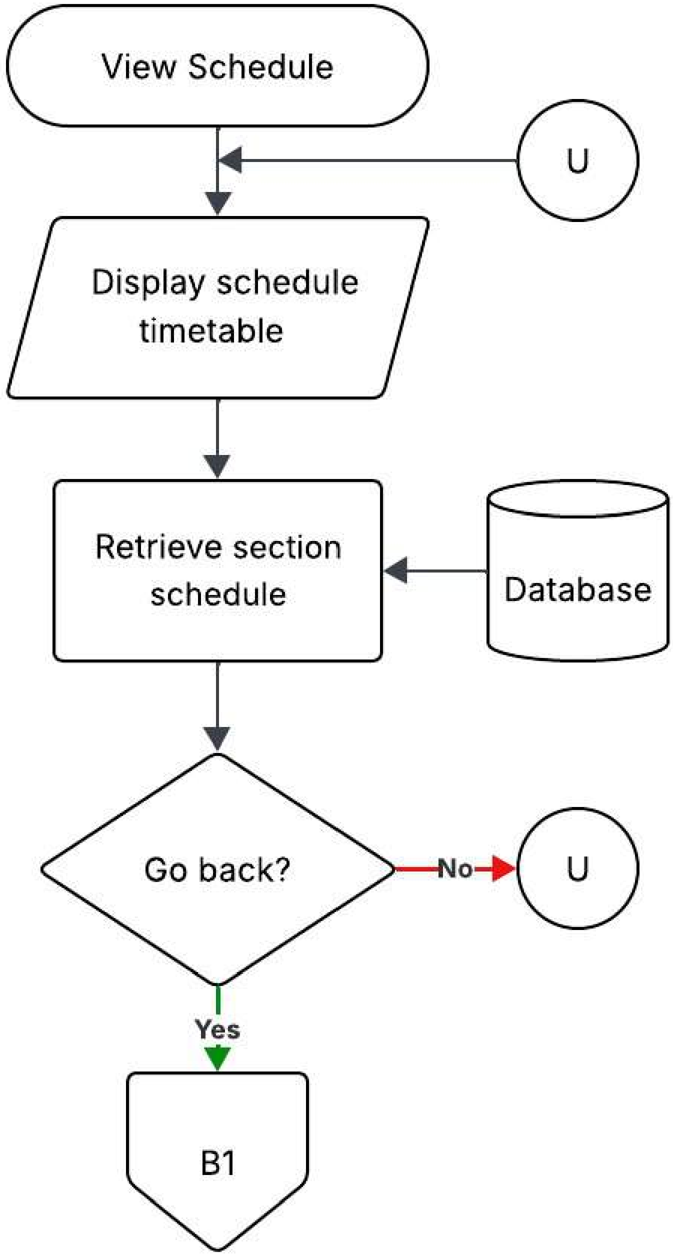


Figure 25 View Schedule Sub-Module Flowchart

This figure shows the viewing of a section's schedule process.

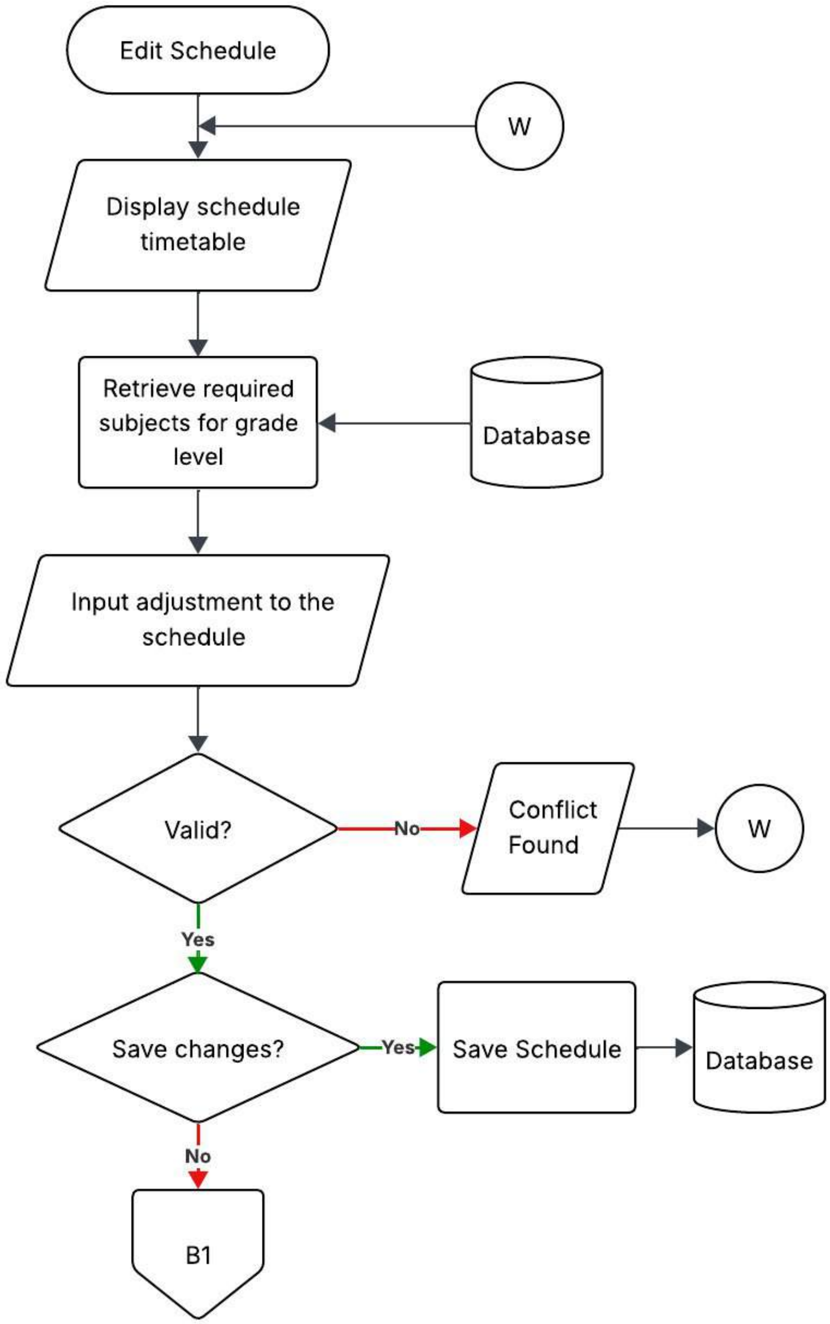


Figure 26 Edit Schedule Flowchart

This figure shows the manual creation of a section's schedule process.

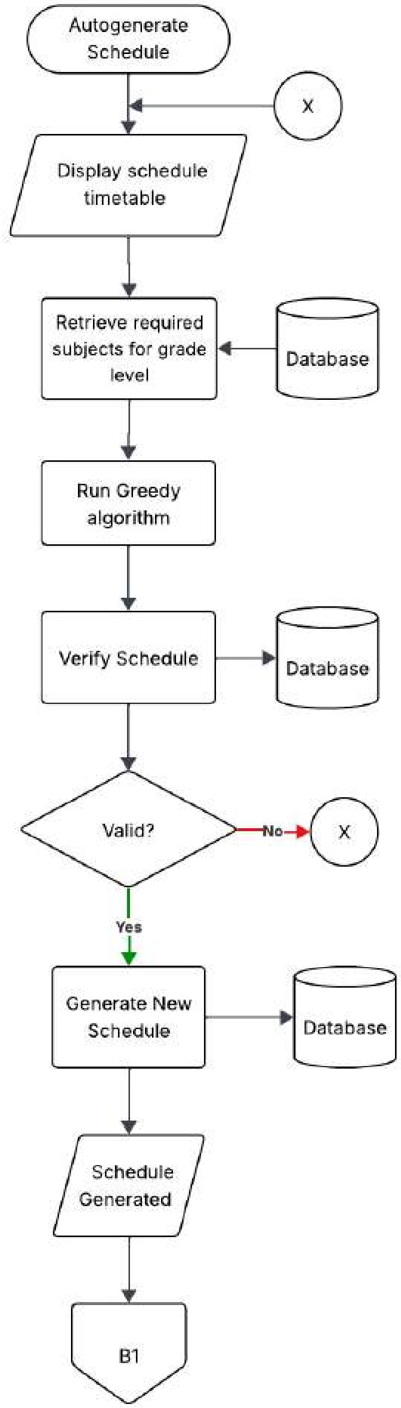


Figure 27 Autogenerate Schedule Flowchart

This figure shows the autogeneration of a sections' schedule process.

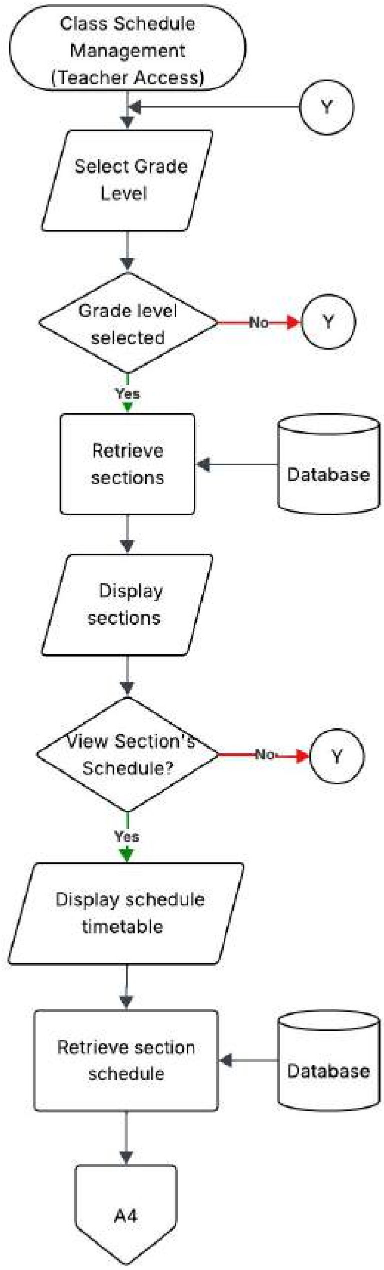


Figure 28 Class Schedule Management (Teacher Access) Module Flowchart

This figure shows the Teacher's menu for the class schedule management module of the system.

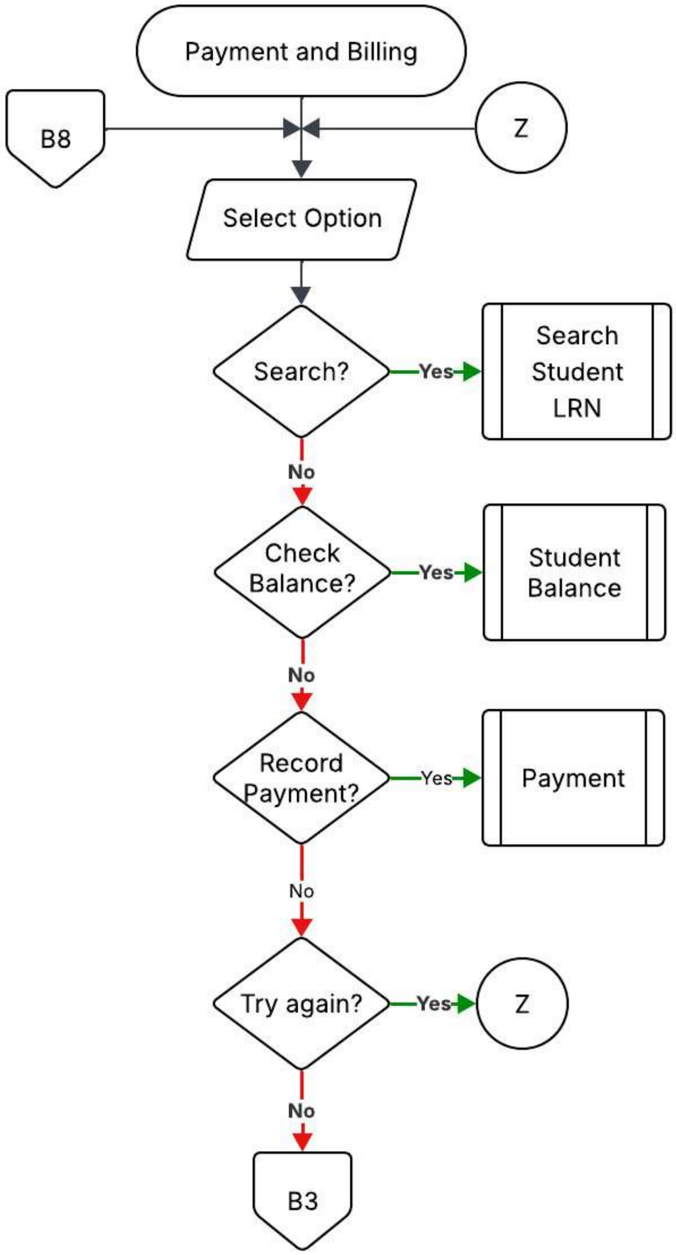


Figure 29 Payment and Billing Module Flowchart

This figure shows the payment and billing module of the system where the user can check a student's remaining balance or record a payment.

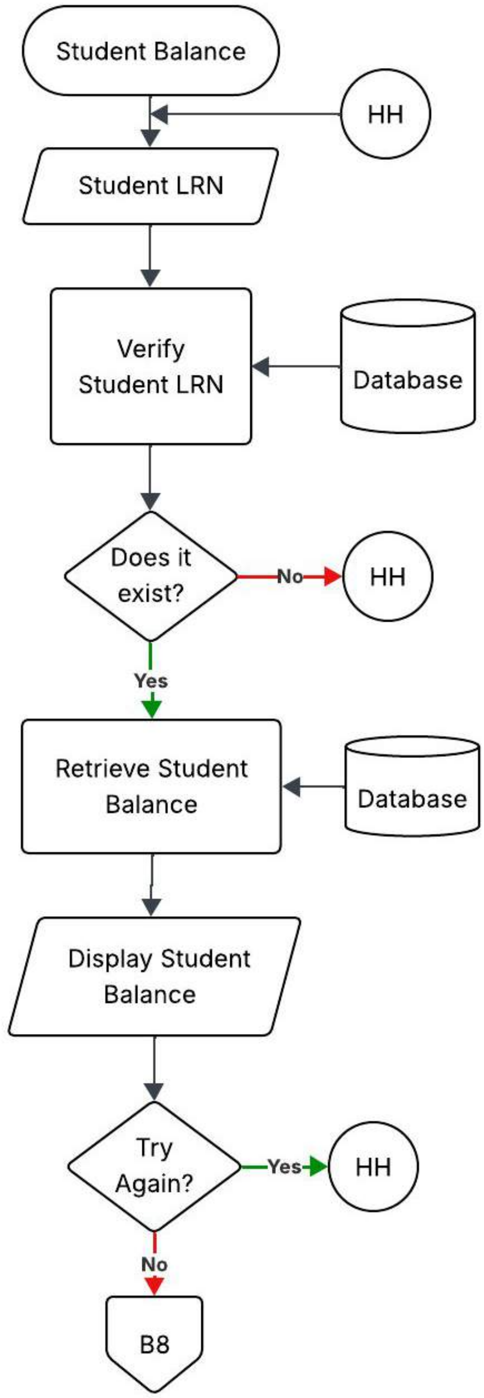


Figure 30 Student Balance Sub-Module Flowchart

This figure shows the retrieval of a student's balance process.

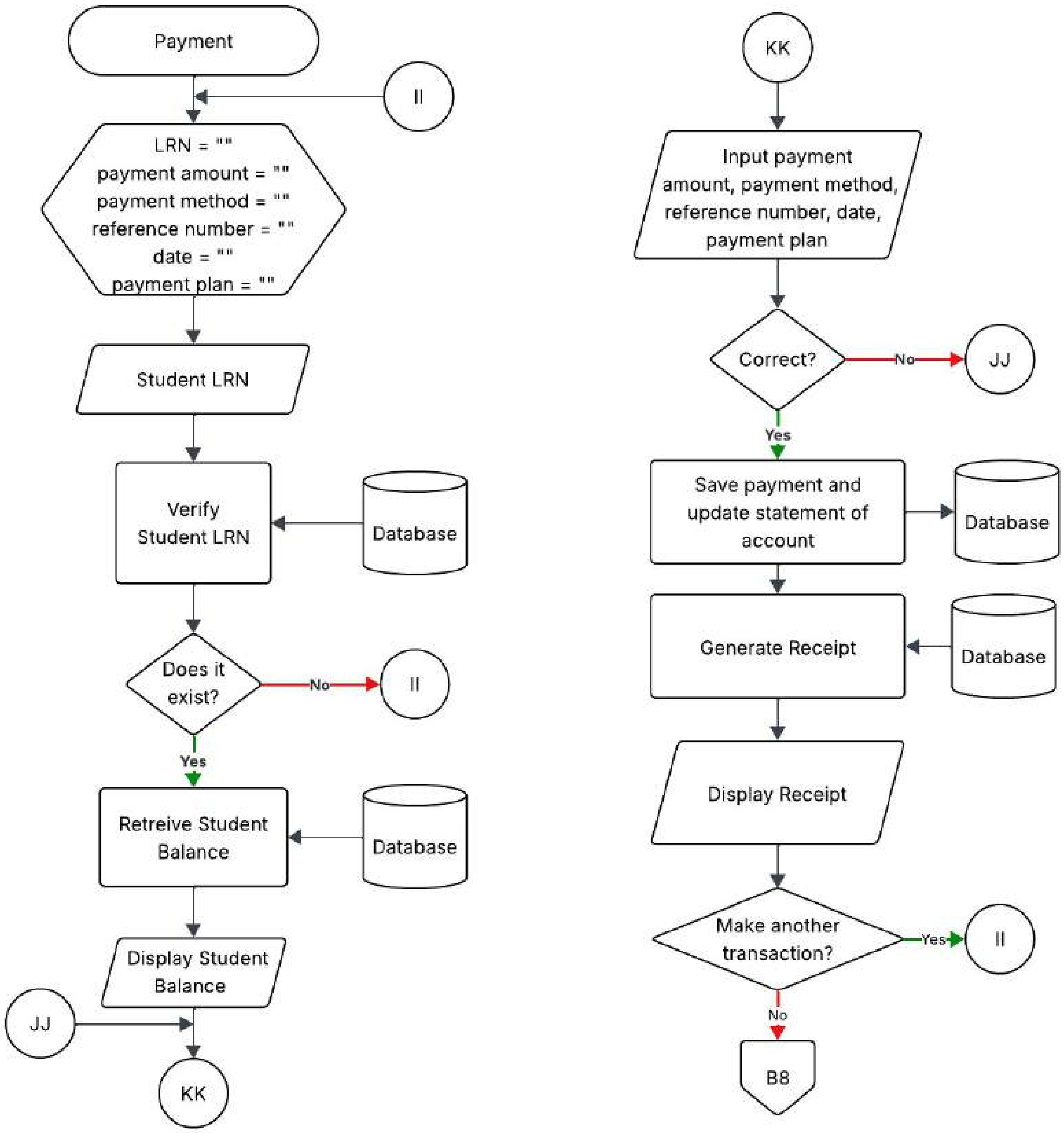


Figure 31 Payment Sub-Module Flowchart

This figure shows the recording of a payment with the generation of receipt process.

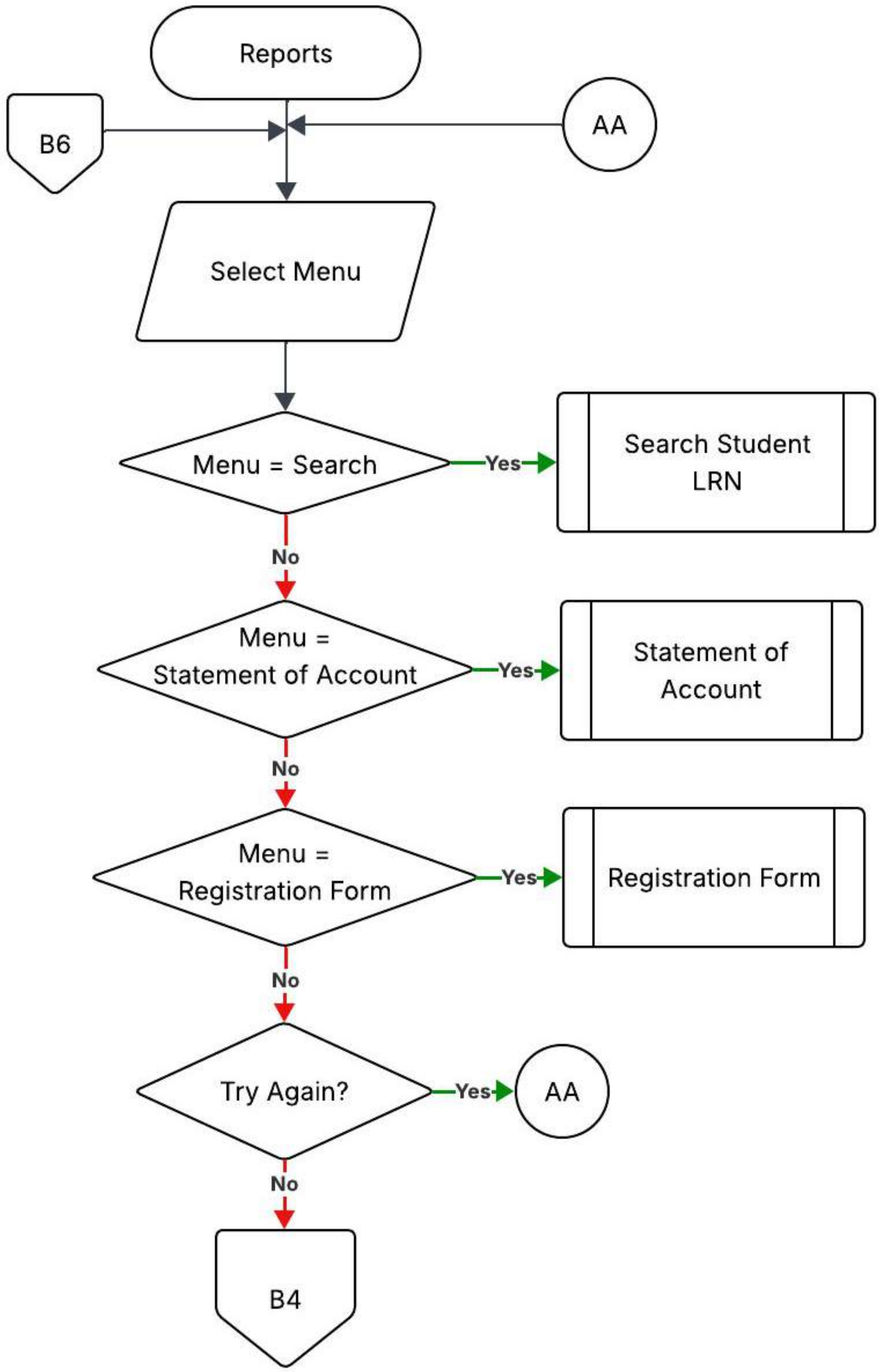


Figure 32 Reports Module Flowchart

This figure shows the reports module of the system where the user can retrieve and generate PDF of specified documents.

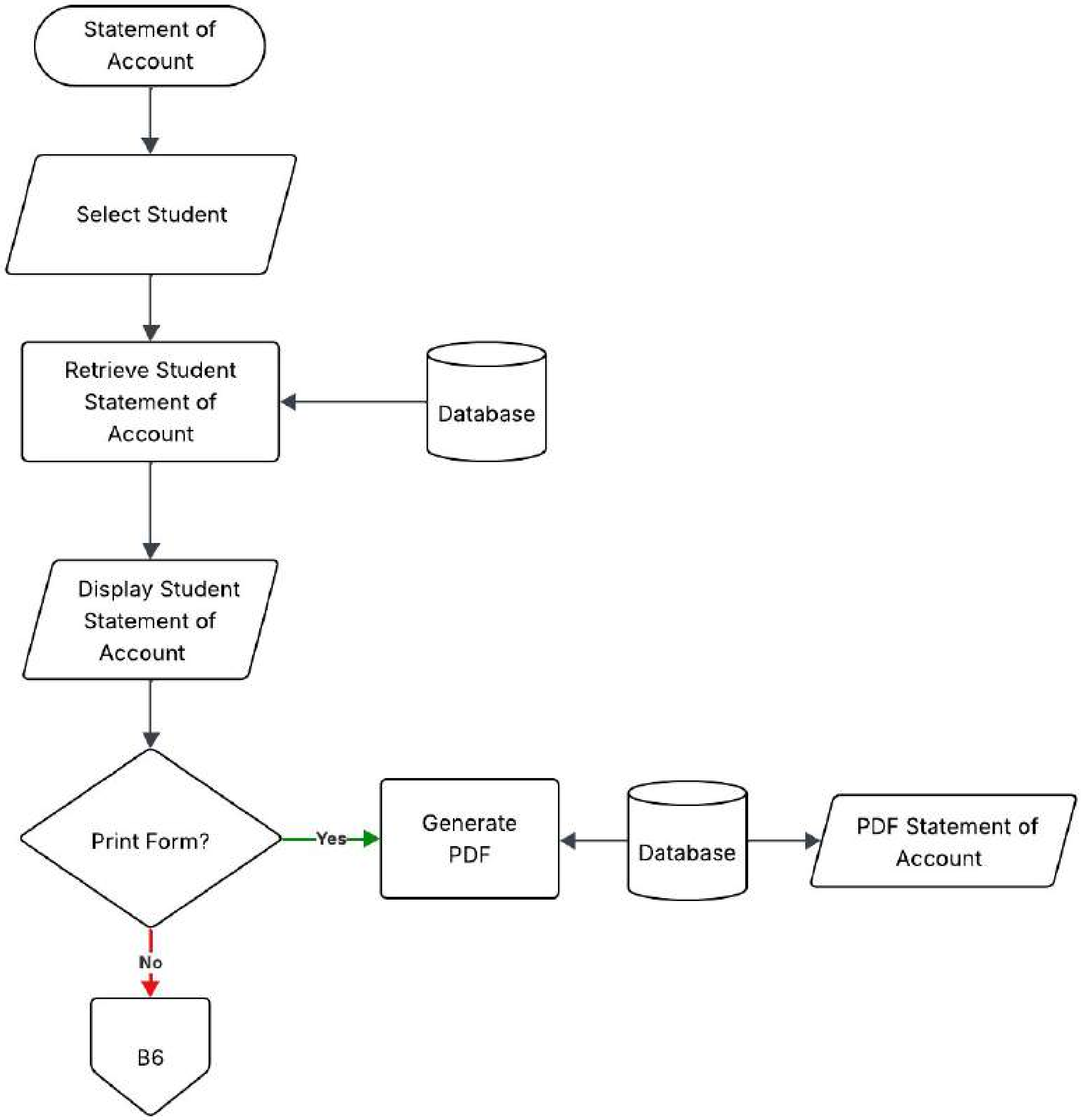


Figure 33 Statement of Account Sub-Module Flowchart

This figure shows the generation of a student's Statement of Account process.

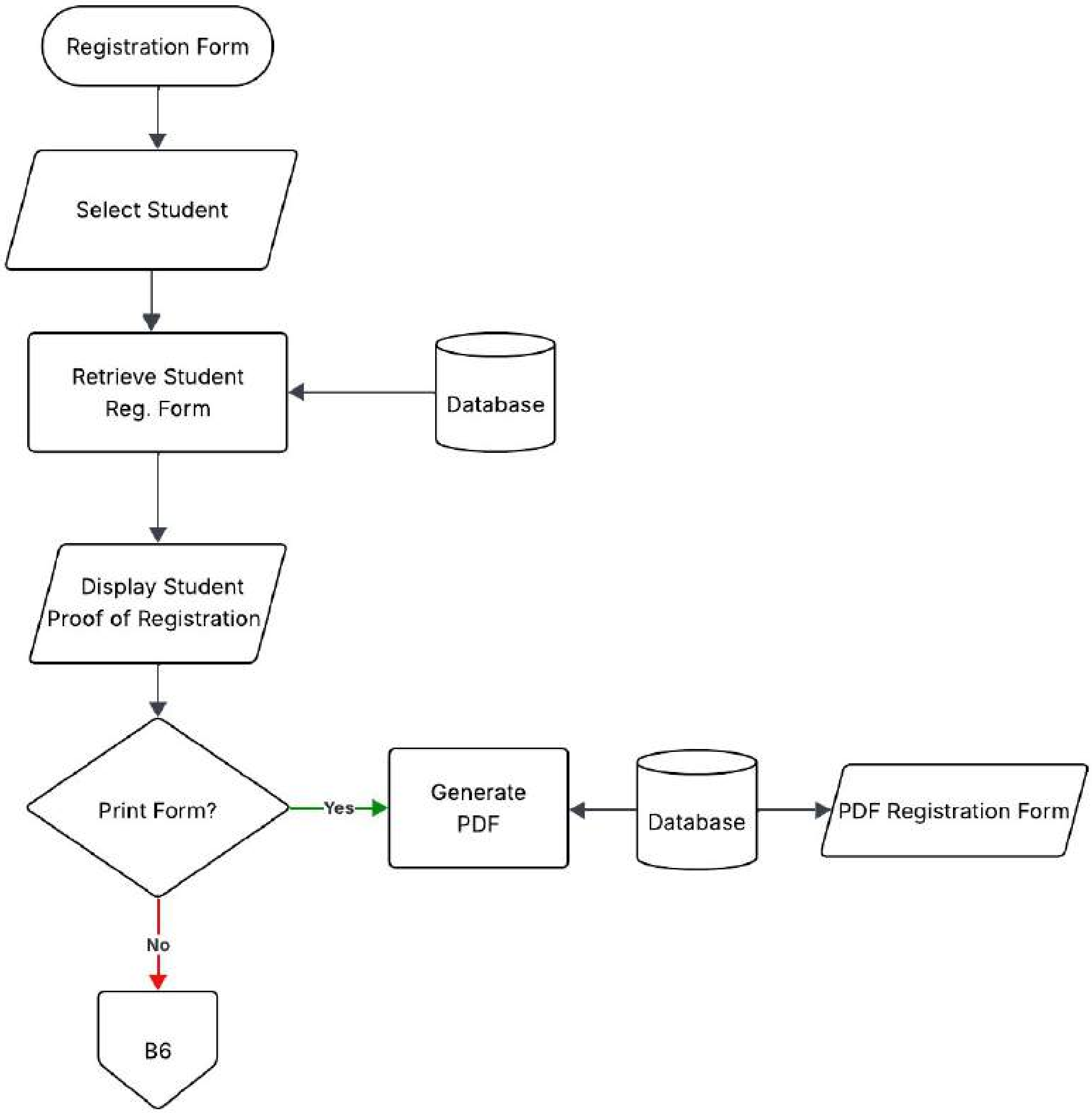


Figure 34 Registration Form Sub-Module Flowchart

This figure shows the generation of a student's Registration Form process.

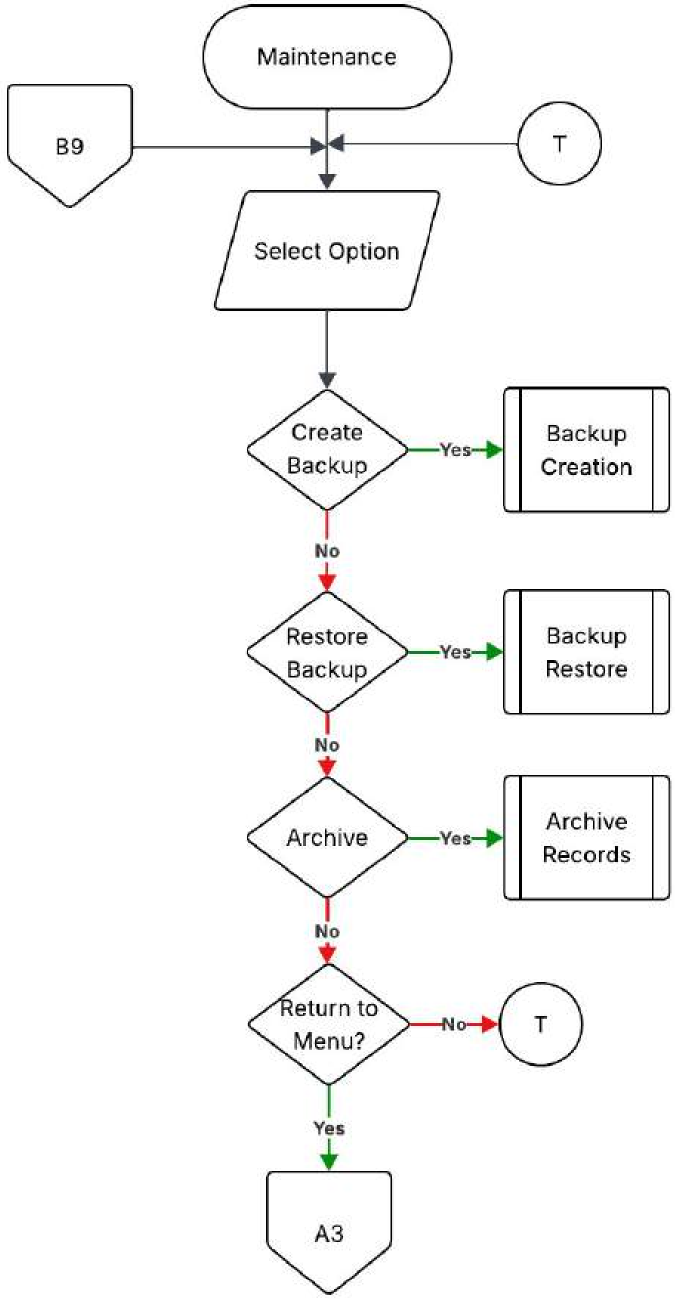


Figure 35 Maintenance Module Flowchart

This figure shows the maintenance module of the system where the user can create or restore backups and archive records of students.

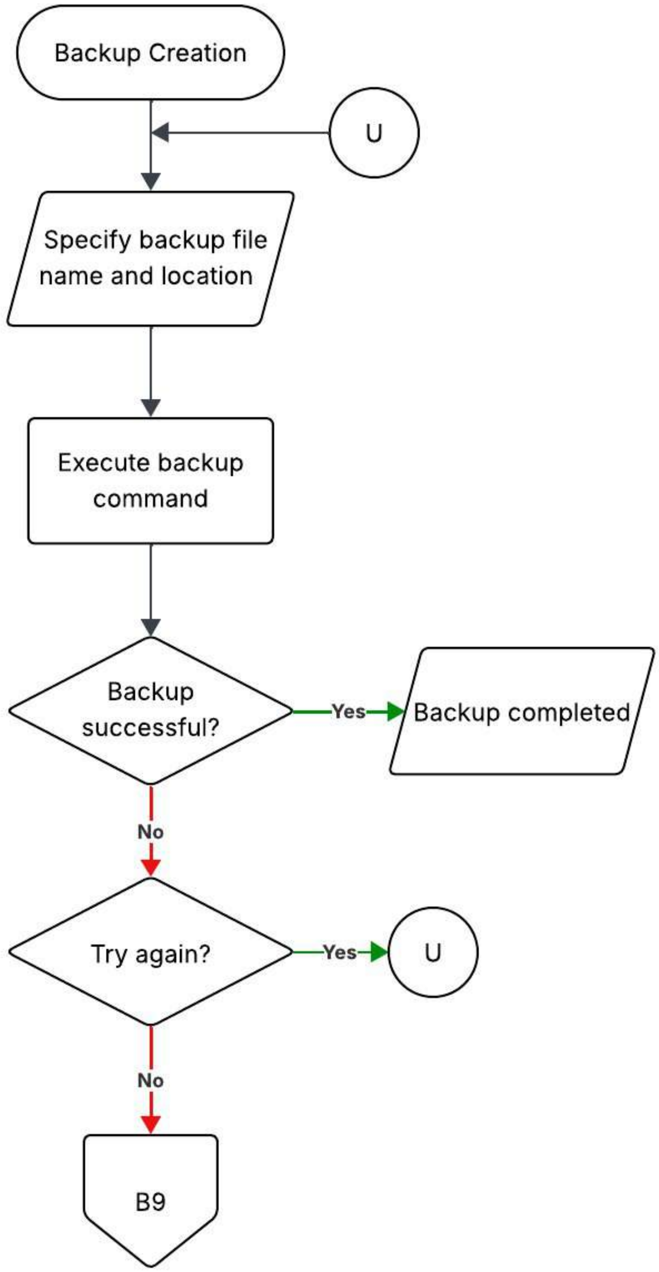


Figure 36 Backup Creation Sub-Module Flowchart

This figure shows the process of creating a backup of the database in the system.

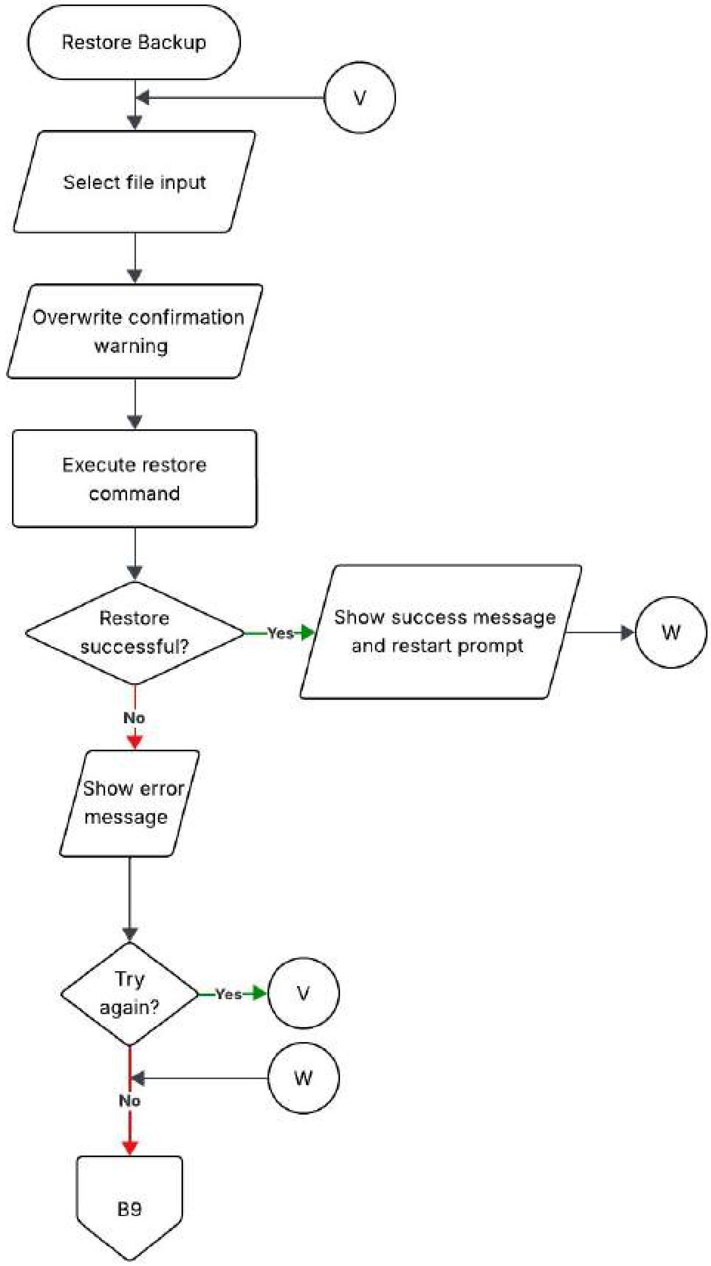


Figure 37 Restore Backup Sub-Module Flowchart

This figure shows the process of restoring a database in the system from a backup.

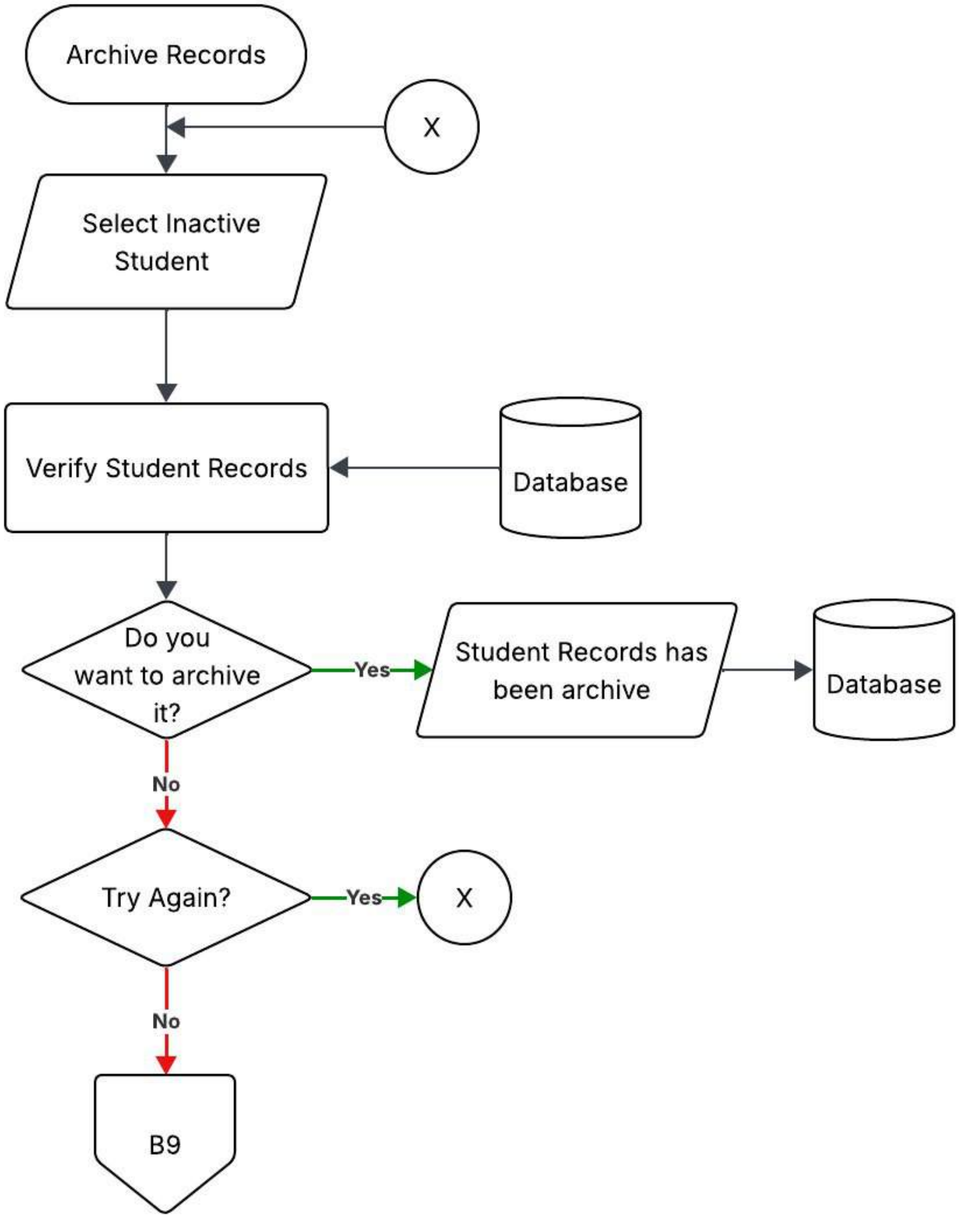


Figure 38 Archive Records Sub-Module Flowchart

This figure shows the process for archiving student records in the system

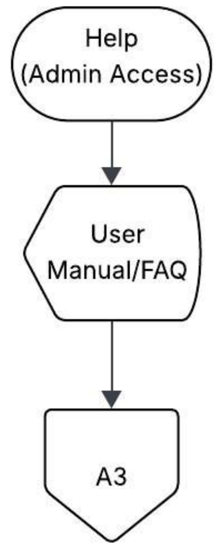


Figure 39 Help (Admin Access) Module Flowchart

This figure shows the help module in the Admin menu, which displays a user manual and a FAQ section.

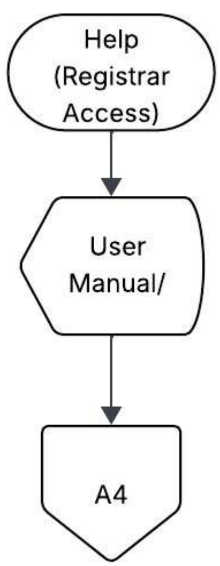


Figure 40 Help (Registrar Access) Module Flowchart

This figure shows the help module in the Registrar menu, which displays a user manual and an FAQ section.

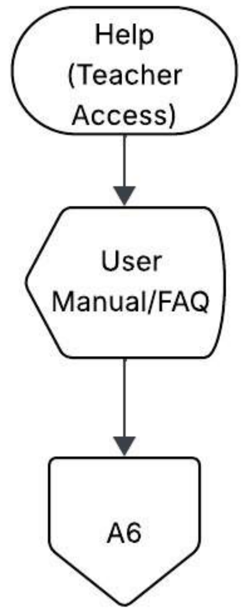


Figure 41 Help (Teacher Access) Module Flowchart

This figure shows the help module in the Teacher menu, which displays a user manual and an FAQ section.

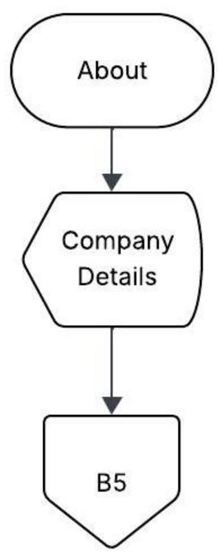


Figure 42 About Module Flowchart

This figure shows the About module of the system that displays the institution's details.

Context Diagram of the Proposed System

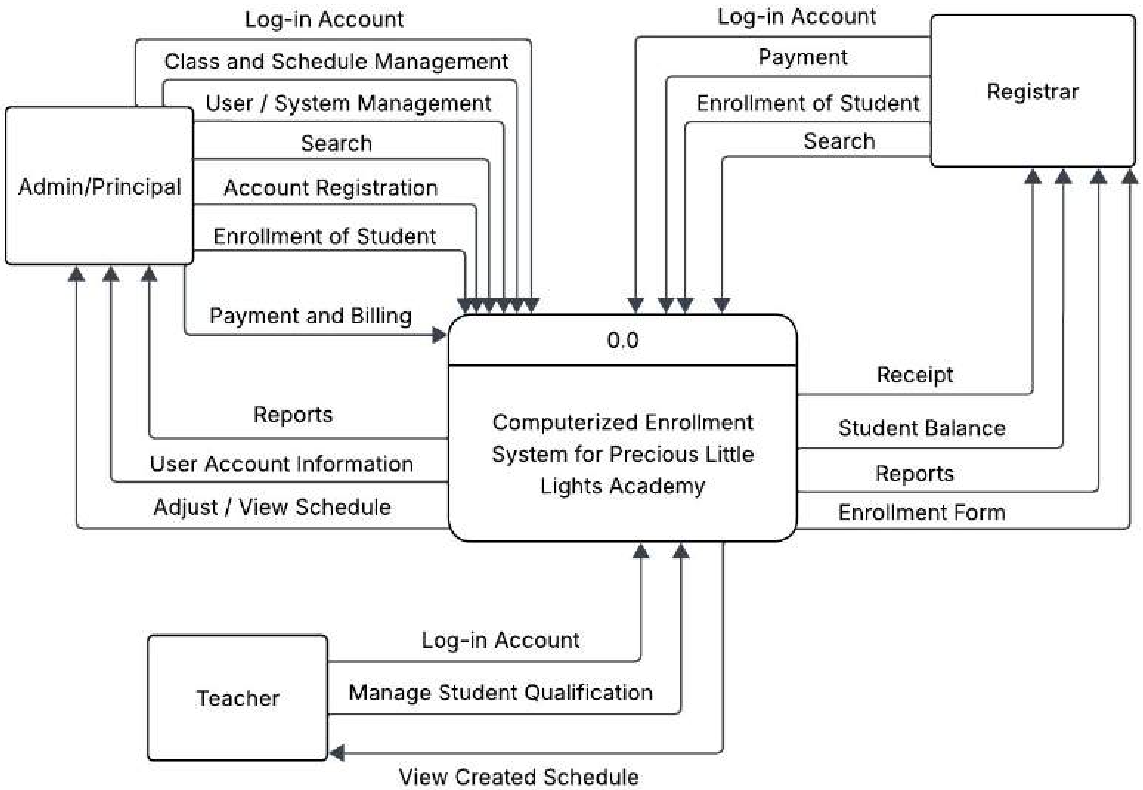


Figure 43 Context Diagram

Data Flow Diagram of the Proposed System

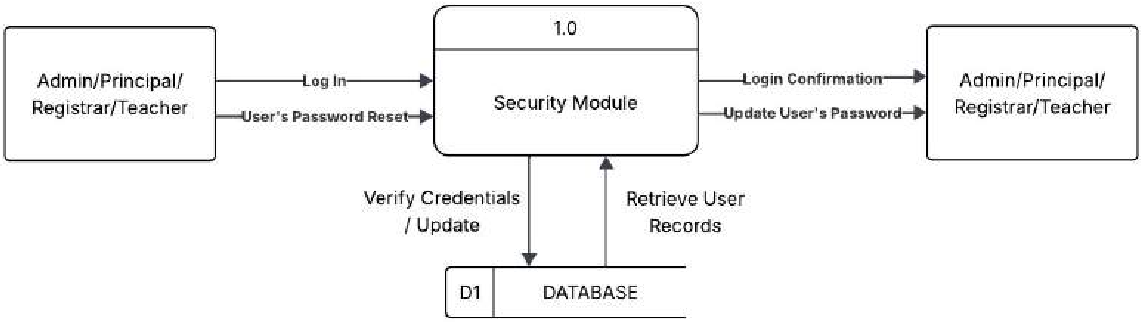


Figure 44 Security Module Dataflow Diagram

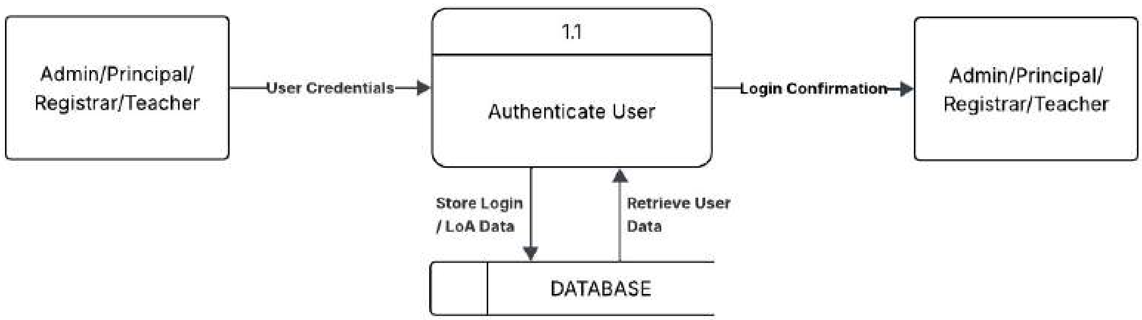


Figure 45 Security Module Sub-process Dataflow Diagram

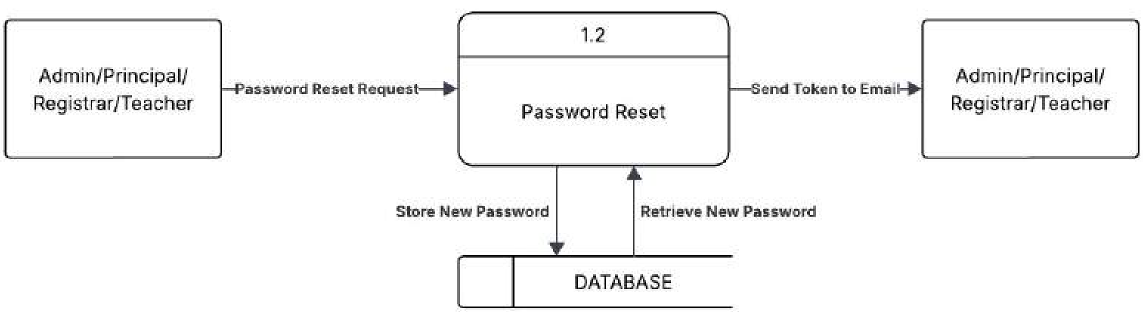


Figure 46 Security Module Sub-process Dataflow Diagram

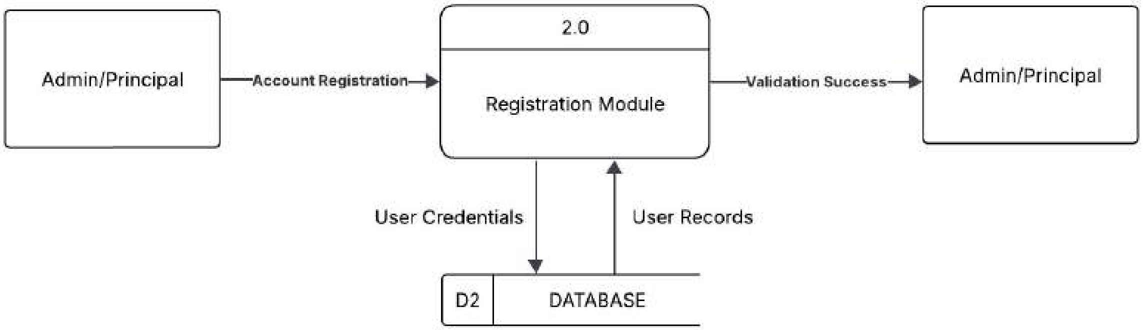


Figure 47 Registration Module Dataflow Diagram

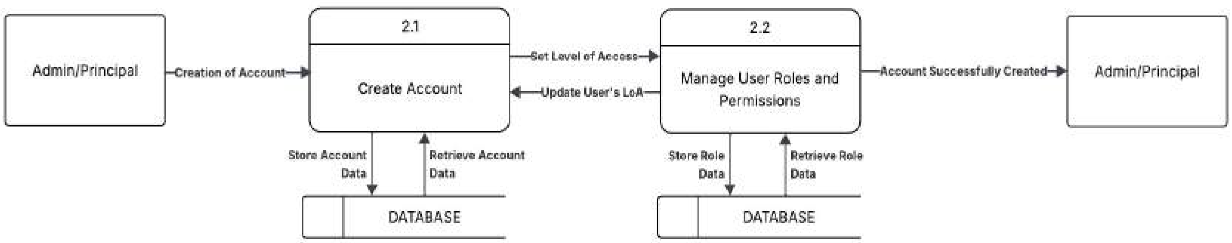


Figure 48 Registration Module Sub-process Dataflow Diagram

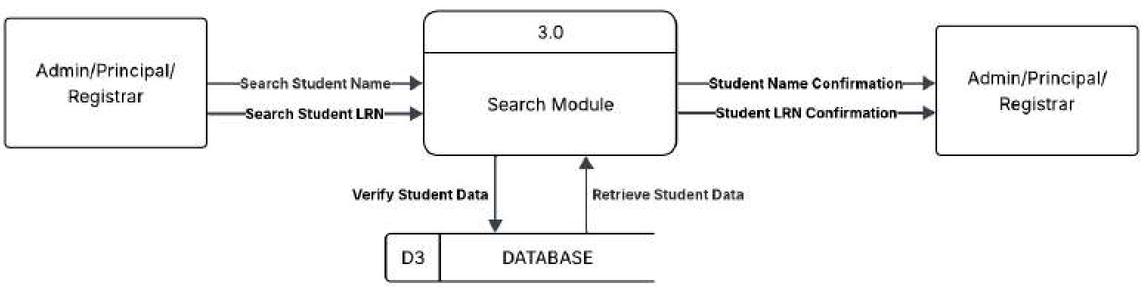


Figure 49 Search Module Dataflow Diagram

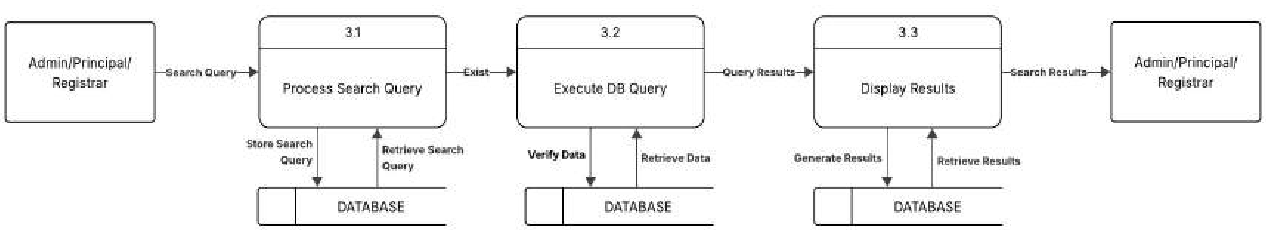


Figure 50 Search Module Sub-process Dataflow Diagram

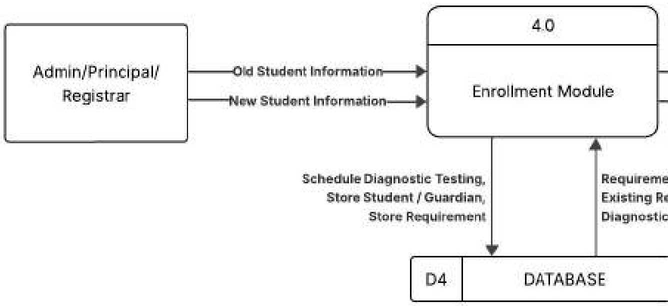


Figure 51 Enrollment Module Dataflow Diagram

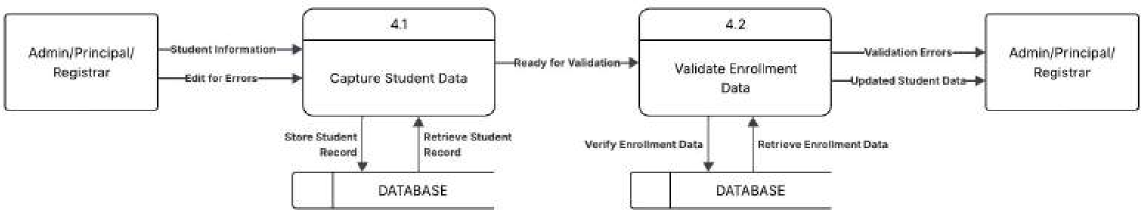


Figure 52 Enrollment Module Sub-process Dataflow Diagram

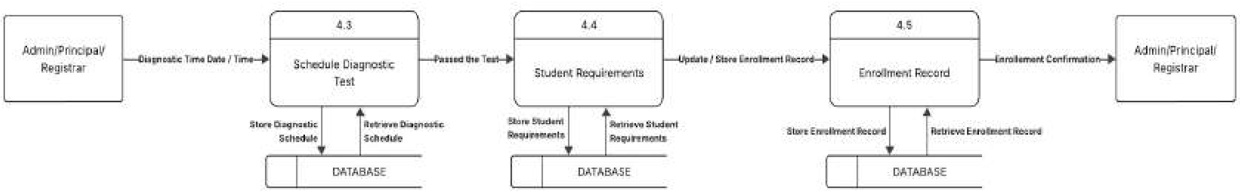


Figure 53 Enrollment Module Sub-process Dataflow Diagram

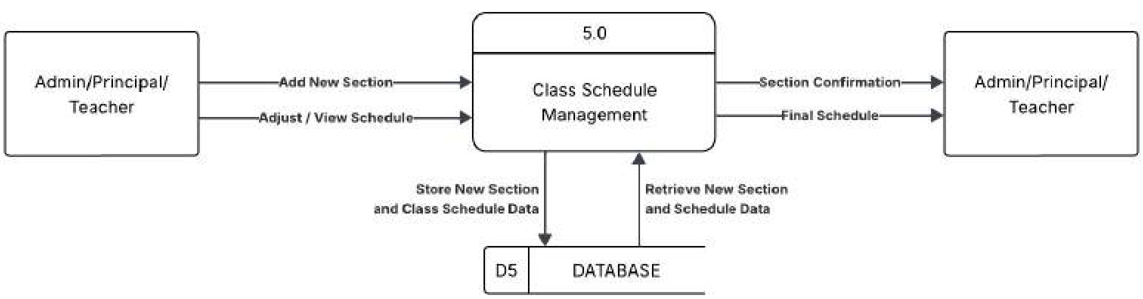


Figure 54 Class Schedule Management Dataflow Diagram

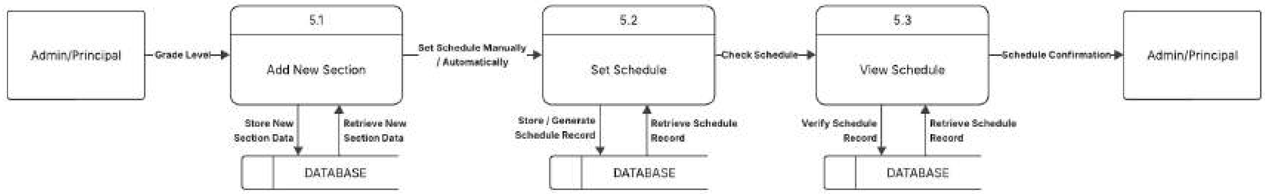


Figure 55 Class Schedule Management Sub-process Dataflow Diagram

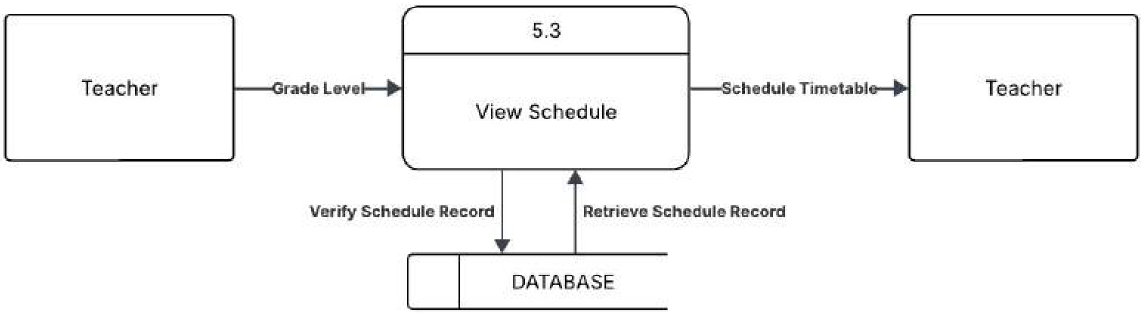


Figure 56 Class Schedule Management Sub-process Dataflow Diagram

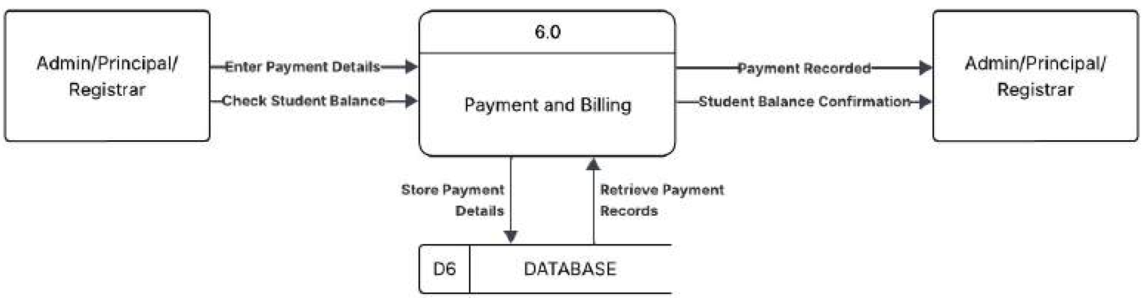


Figure 57 Payment and Billing Dataflow Diagram

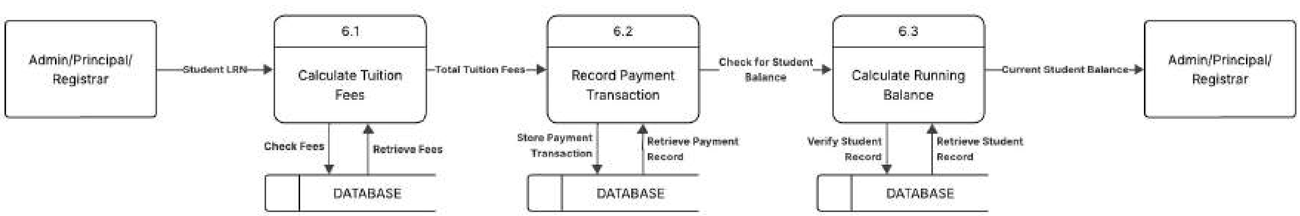


Figure 58 Payment and Billing Sub-process Dataflow Diagram

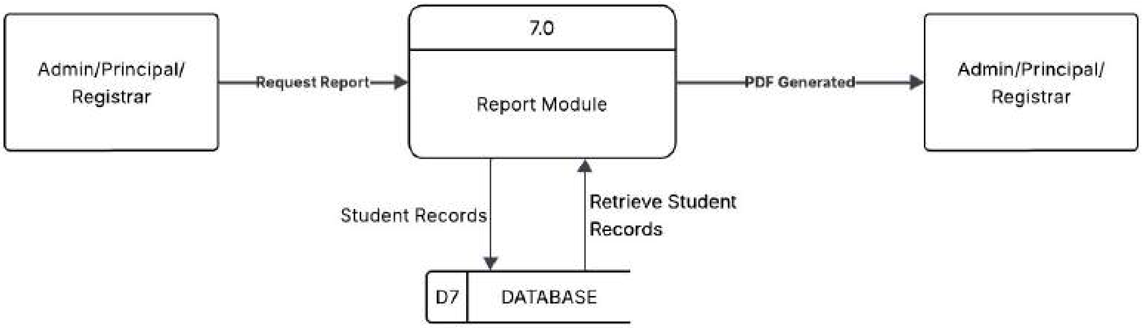


Figure 59 Report Module Dataflow Diagram

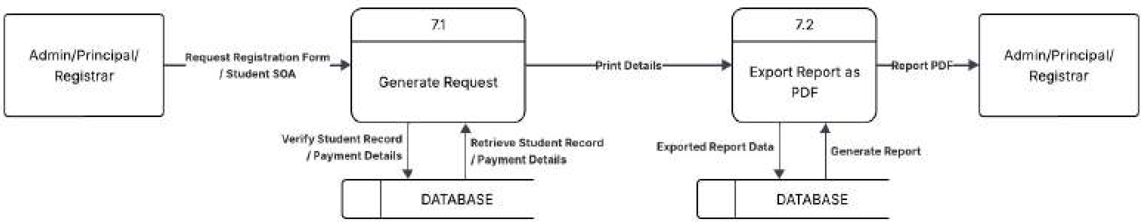


Figure 60 Report Module Sub-process Dataflow Diagram

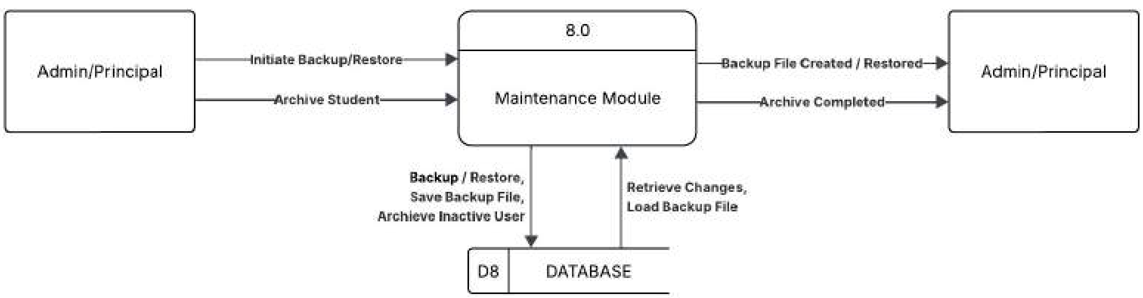


Figure 61 Maintenance Module Dataflow Diagram

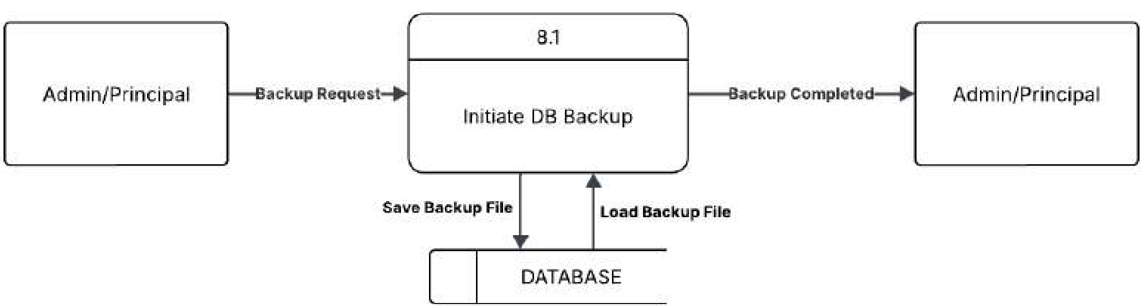


Figure 62 Maintenance Module Sub-process Dataflow Diagram

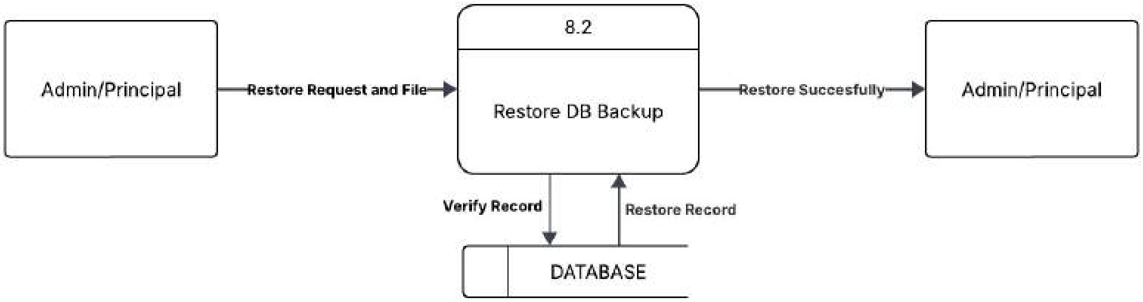


Figure 63 Maintenance Module Sub-process Dataflow Diagram

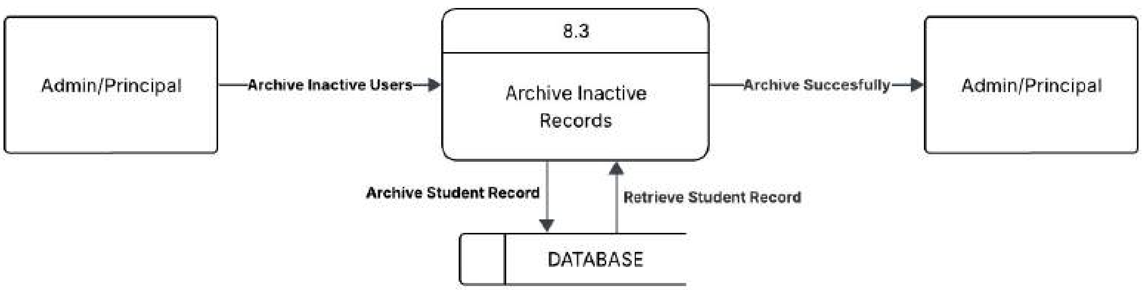


Figure 64 Maintenance Module Sub-process Dataflow Diagram

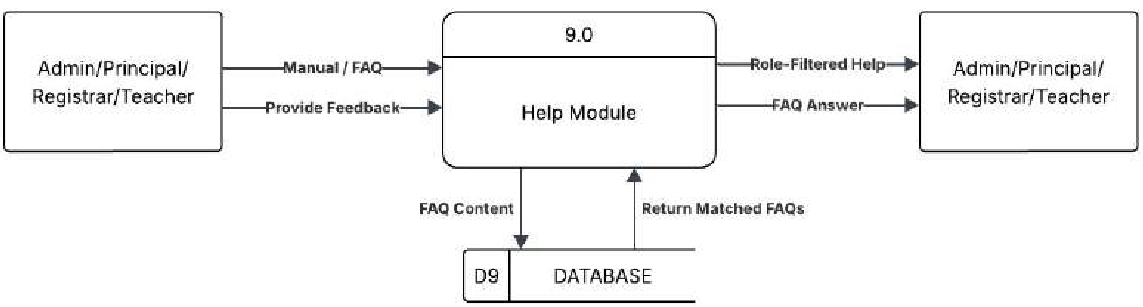


Figure 65 Help Module Dataflow Diagram

Use Case Diagram

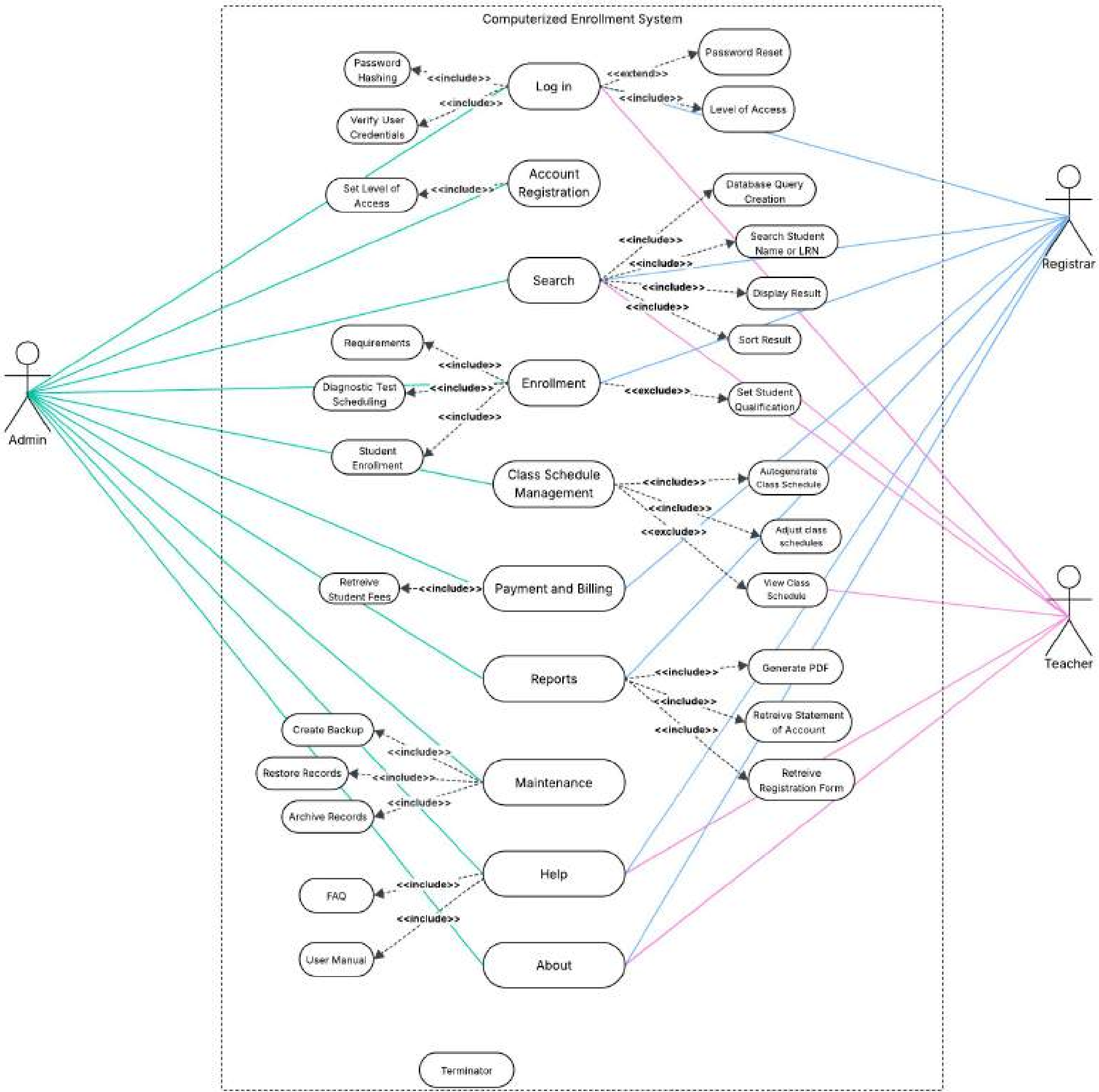


Figure 66 Use Case Diagram

Hierarchical Input Process, and Output

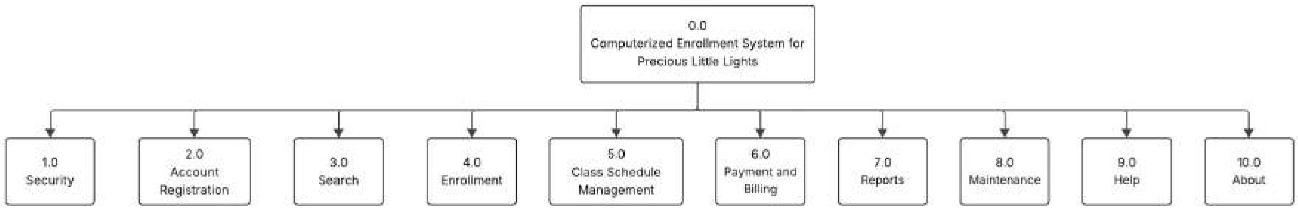


Figure 67 Computerized Enrollment System For Precious Little Lights Module Hierarchy

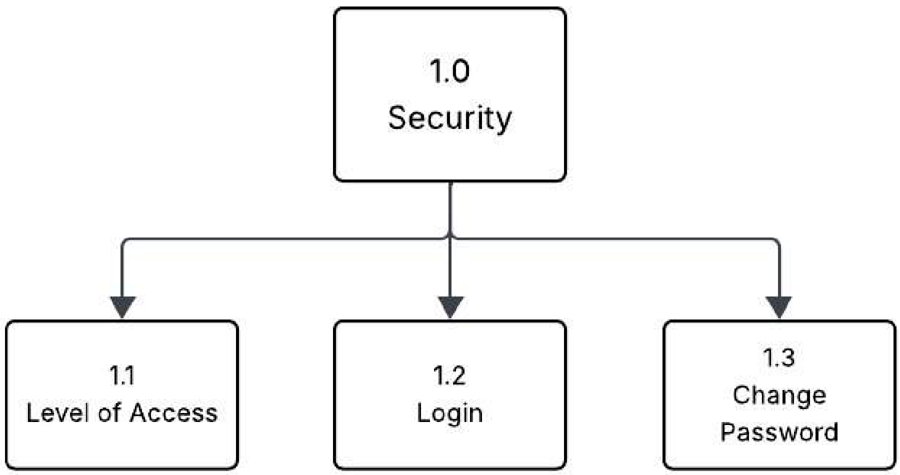


Figure 68 Security

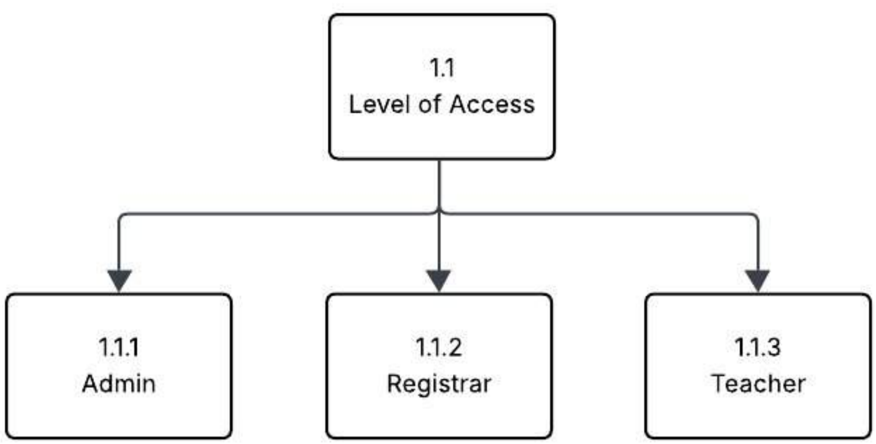


Figure 69 Level of Access

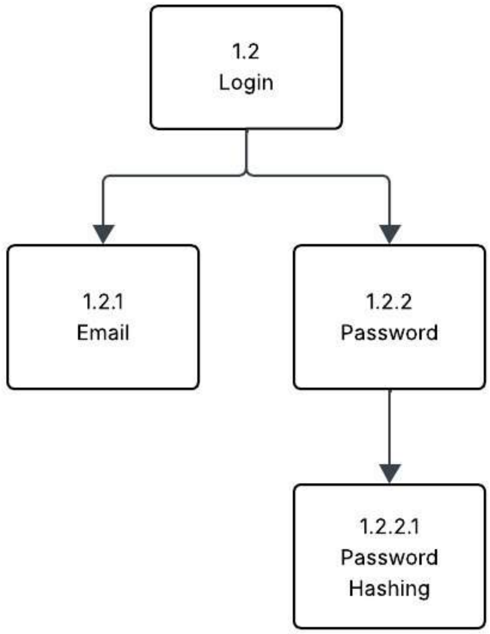


Figure 70 Login

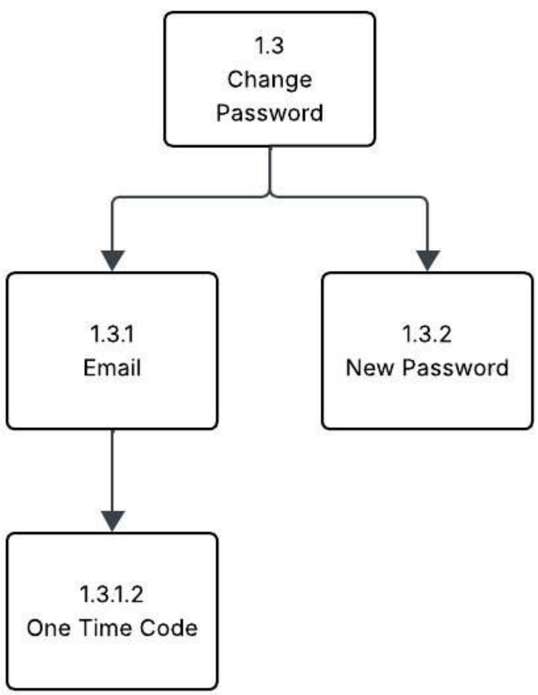


Figure 71 Change Password

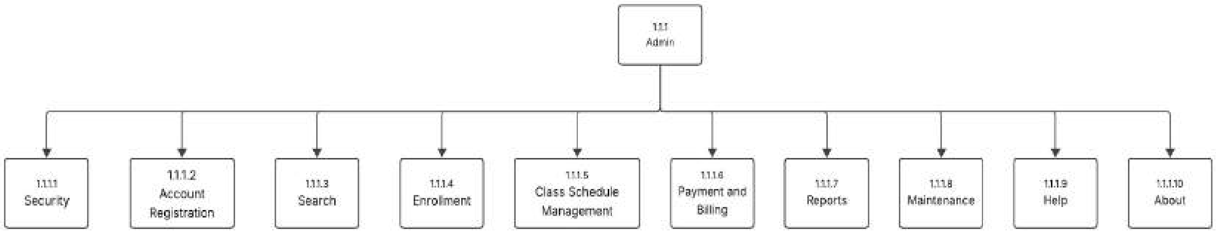


Figure 72 Admin

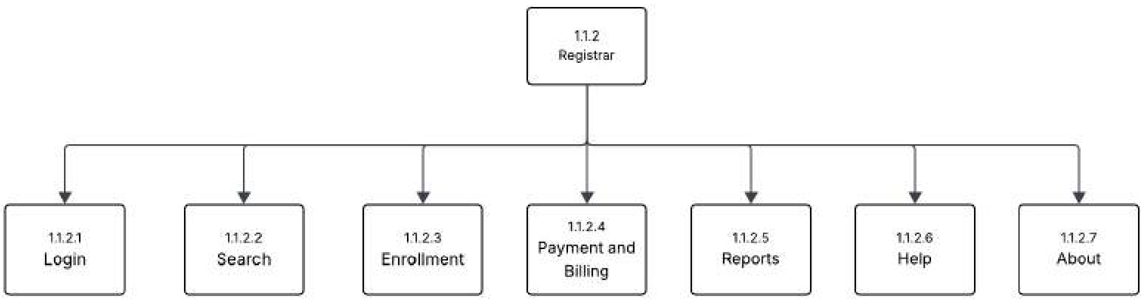


Figure 73 Registrar

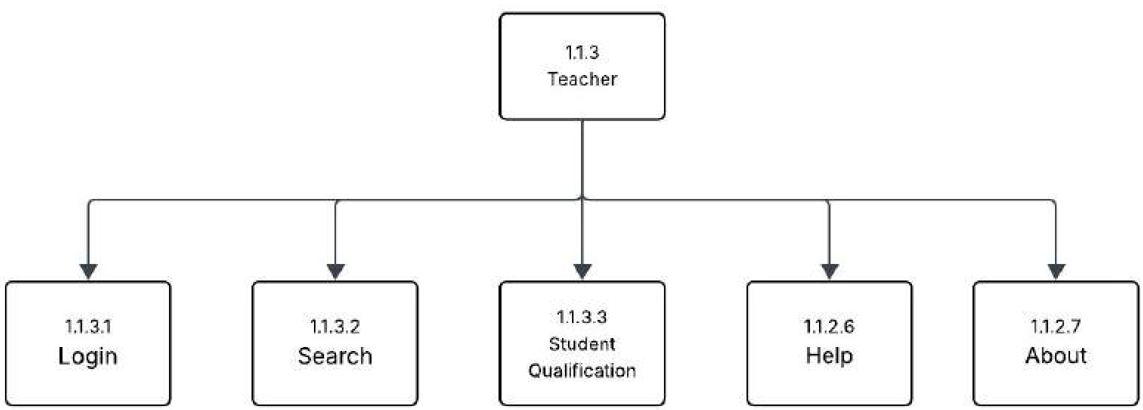


Figure 74 Teacher

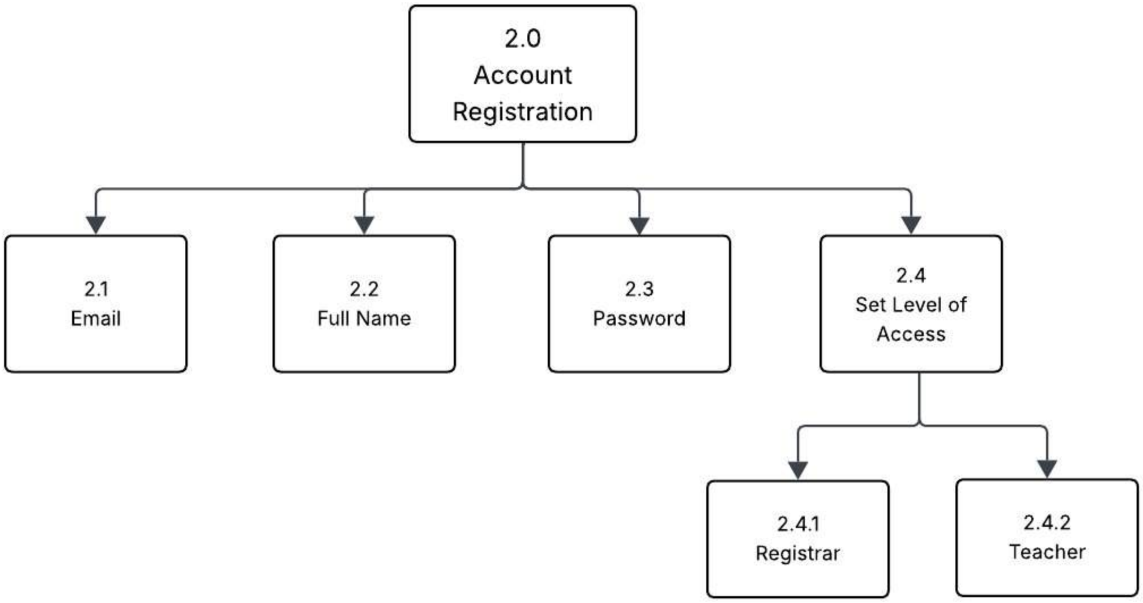


Figure 75 Account Registration

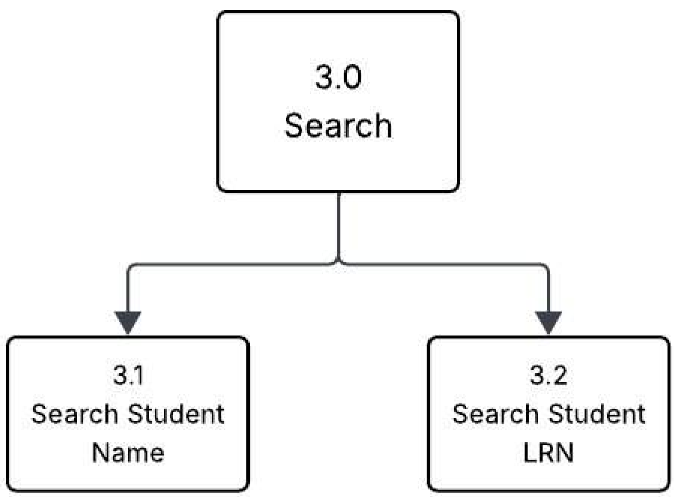


Figure 76 Search

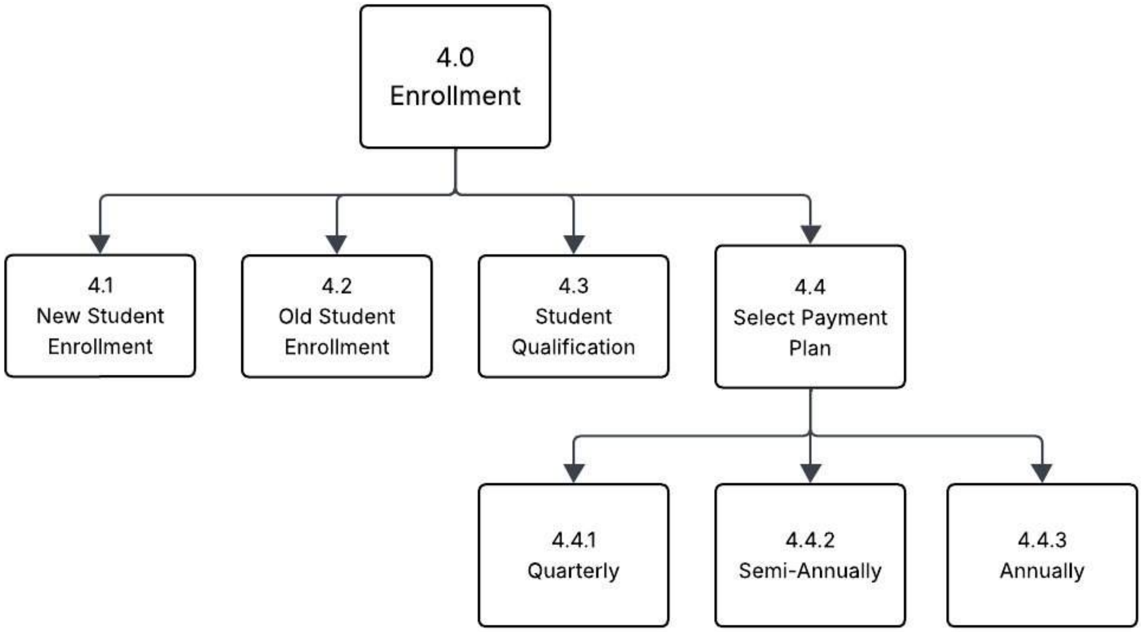


Figure 77 Enrollment

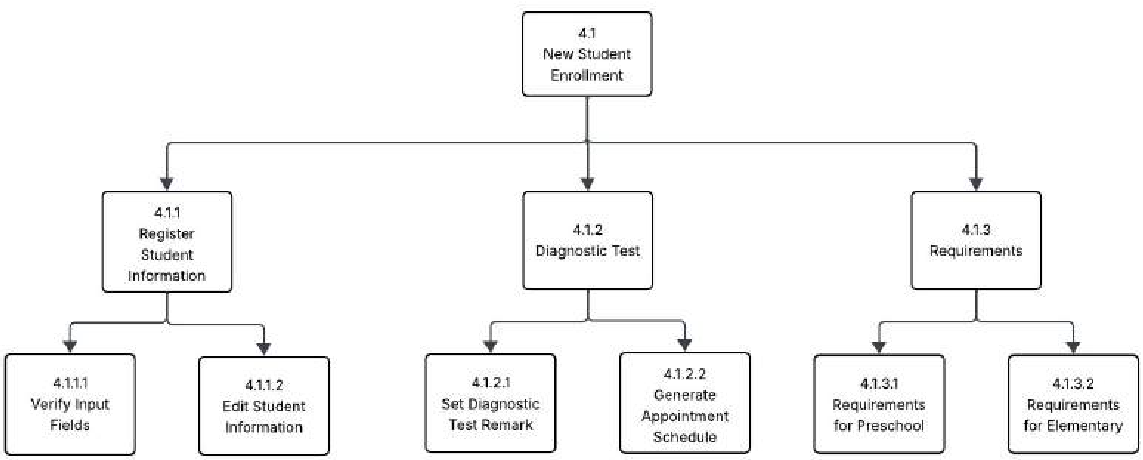


Figure 78 New Student Enrollment

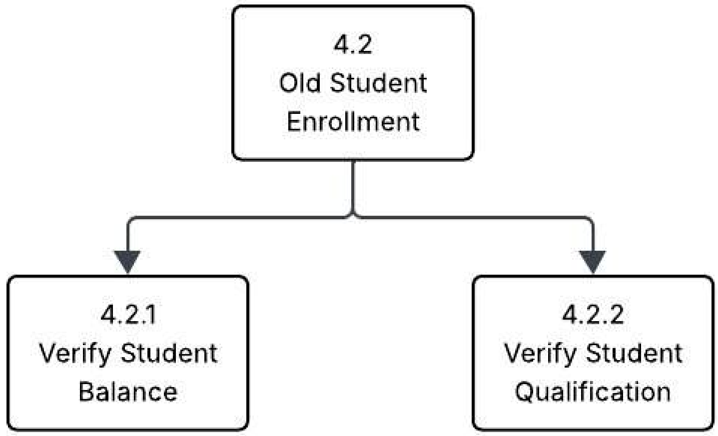


Figure 79 Old Student Enrollment

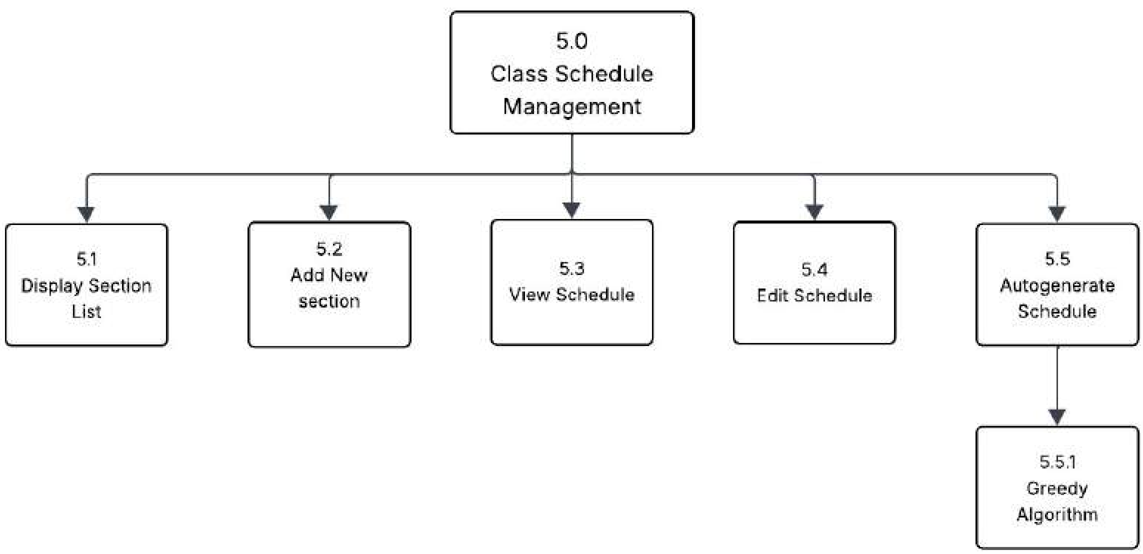


Figure 80 Class Schedule Management

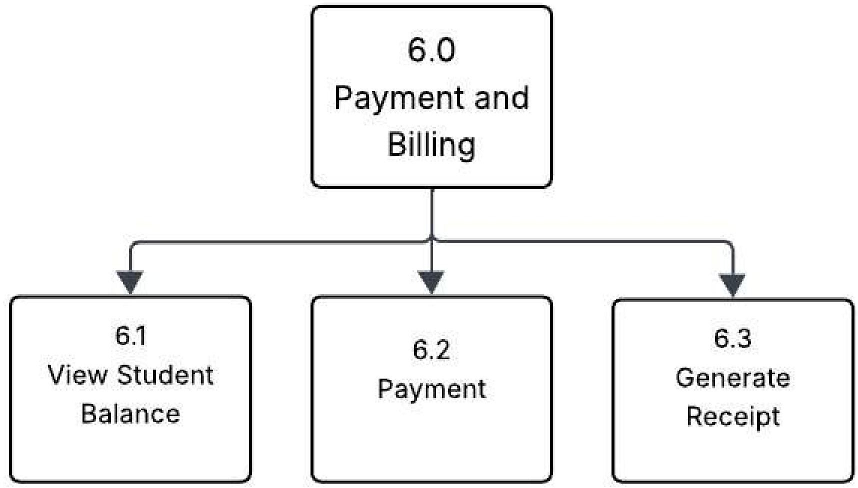


Figure 81 Payment and Billing

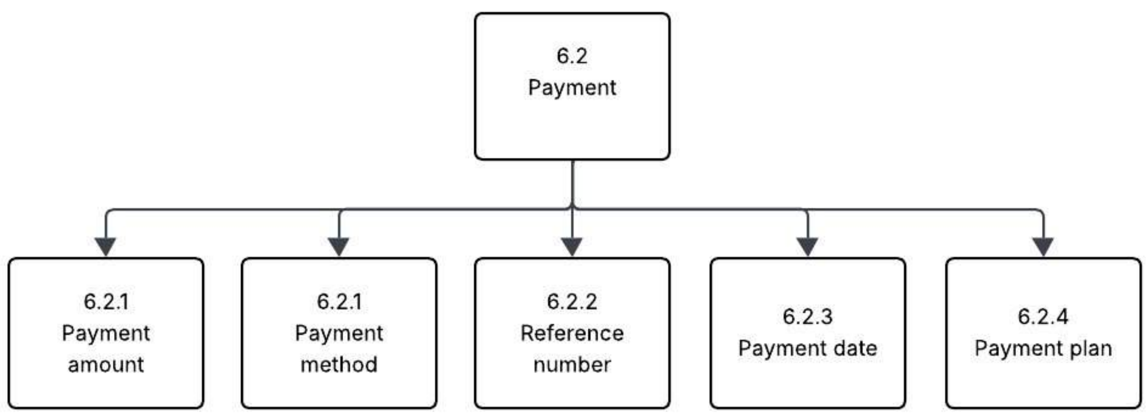


Figure 82 Payment

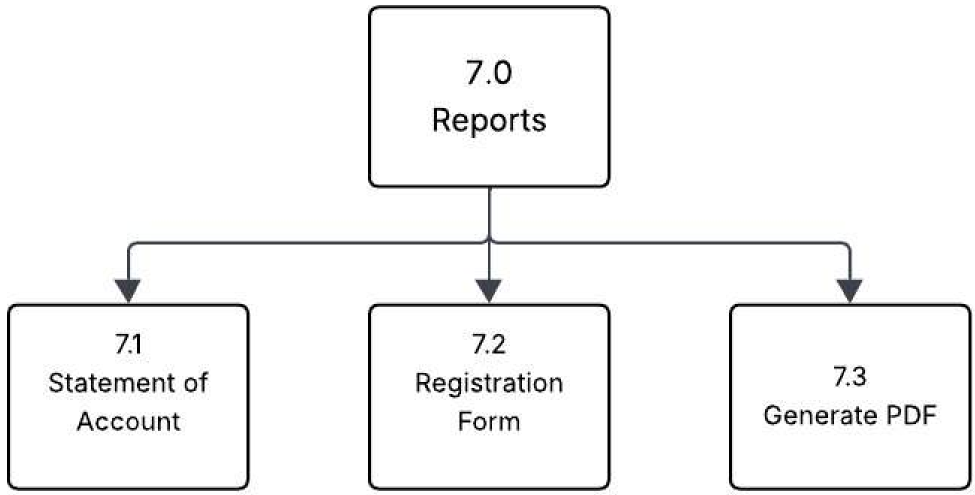


Figure 83 Reports

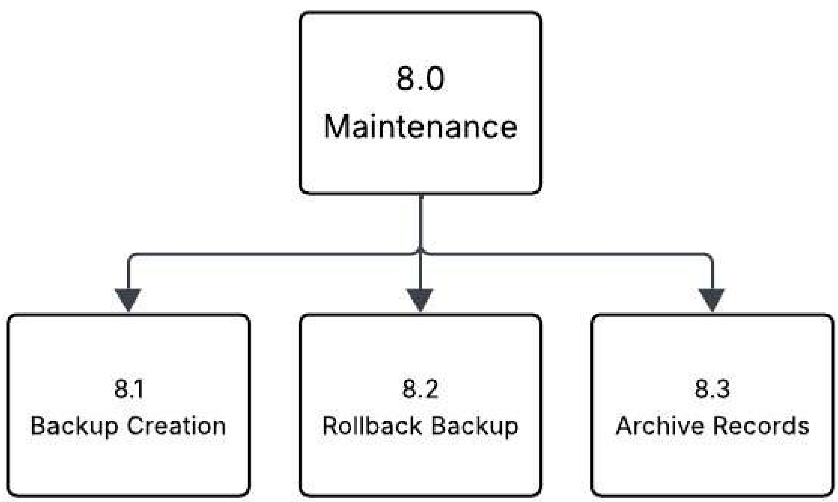


Figure 84 Maintenance

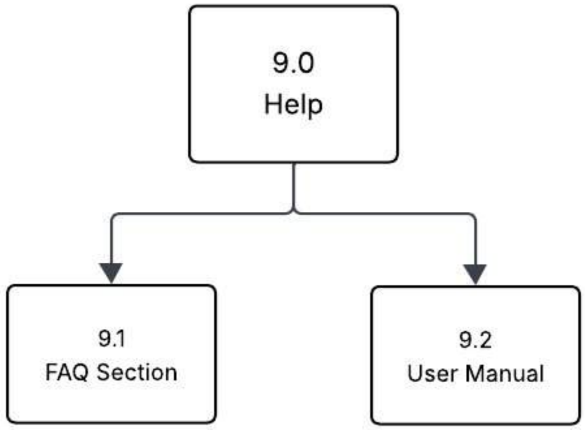


Figure 85 Help

Input, Process, Output

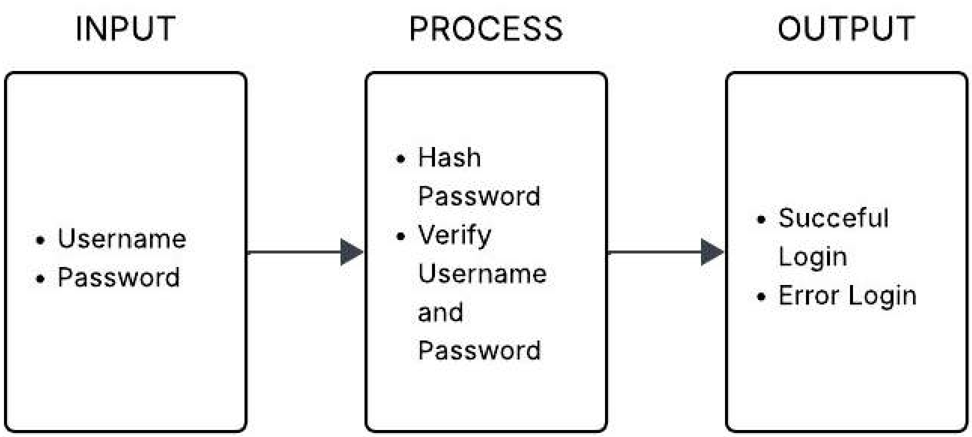


Figure 86 Security IPO

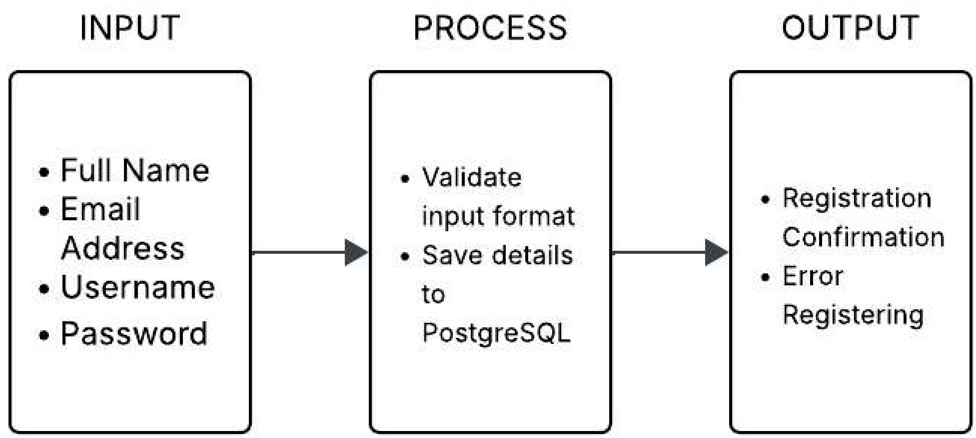


Figure 87 Account Registration IPO

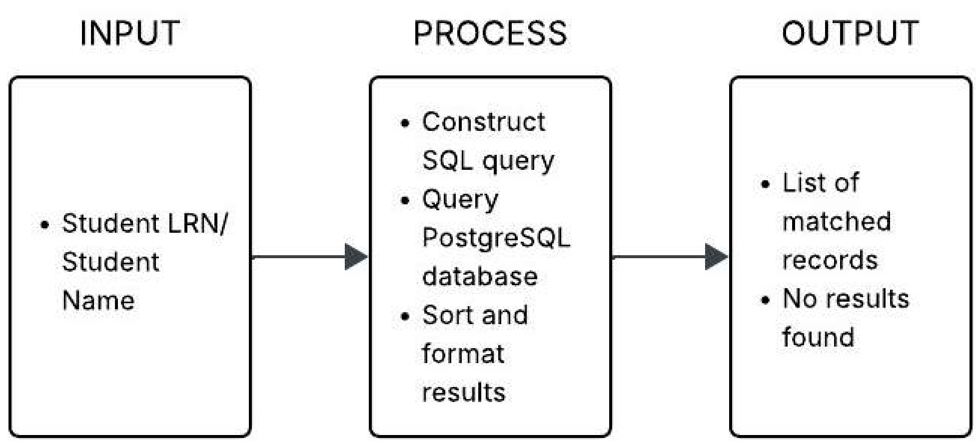


Figure 88 Search IPO

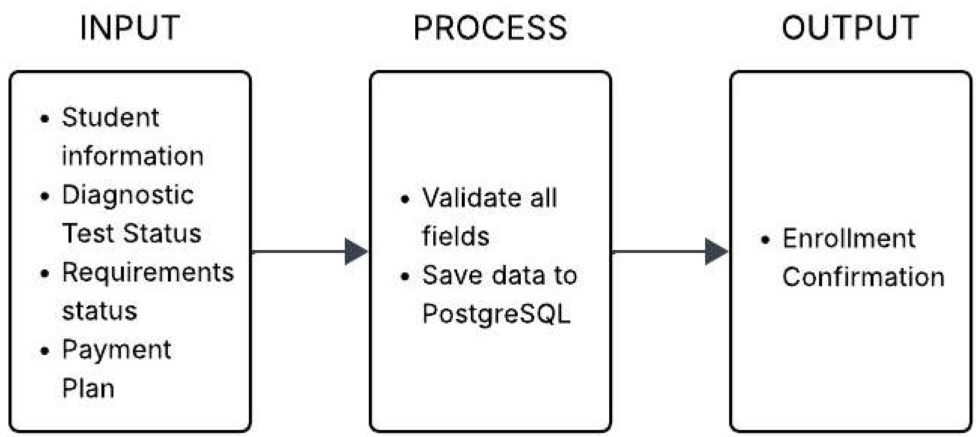


Figure 89 Enrollment IPO

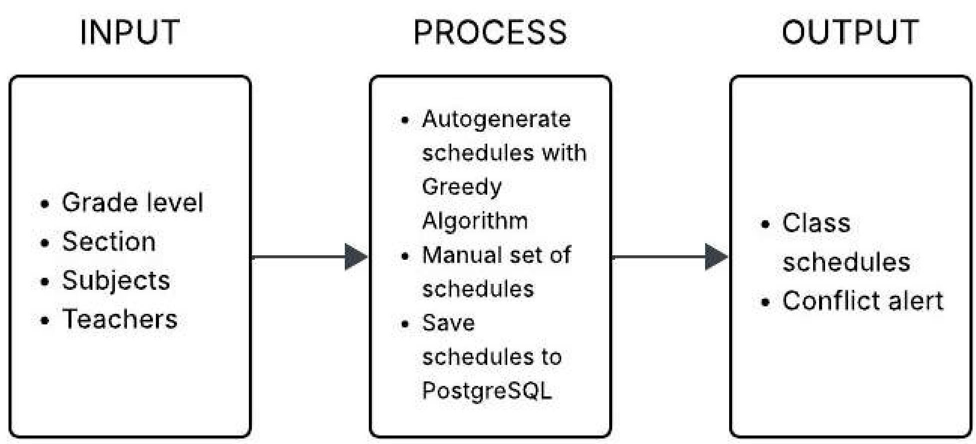


Figure 90 Class and Schedule Management IPO

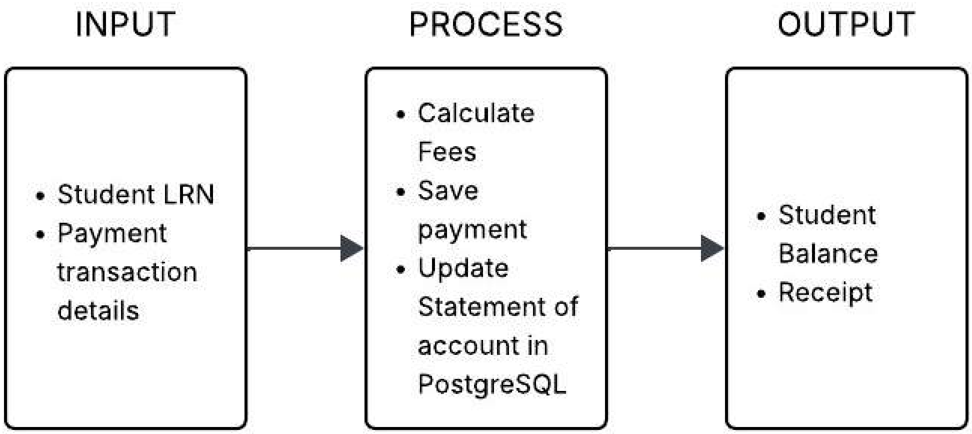


Figure 91 Payment and Billing IPO

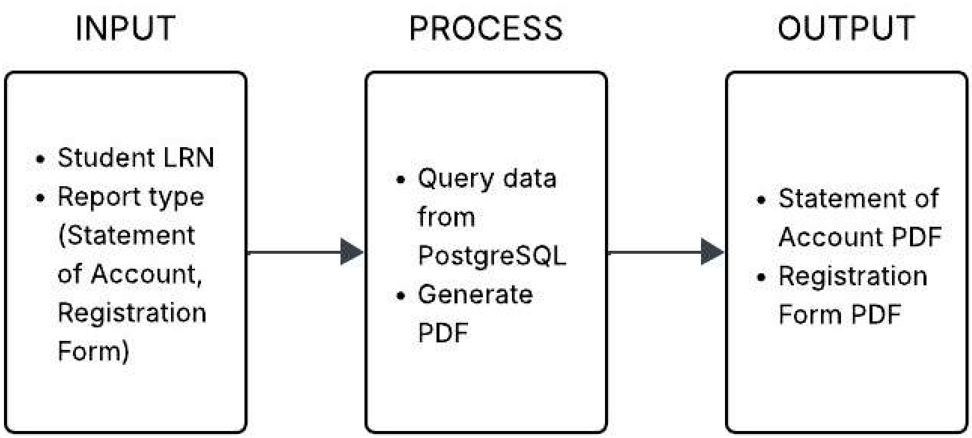


Figure 92 Reports IPO

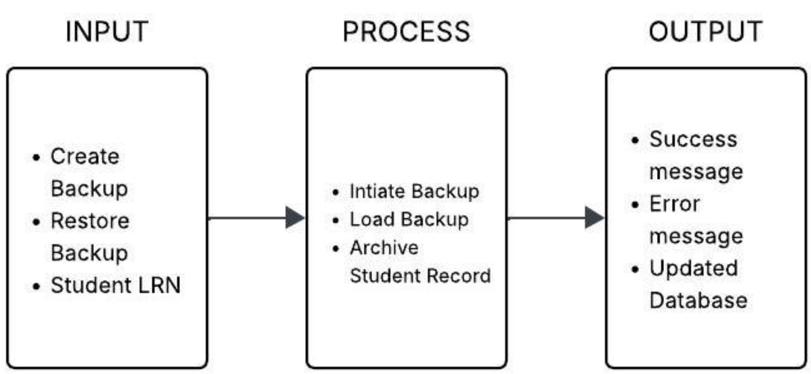


Figure 93 Maintenance IPO

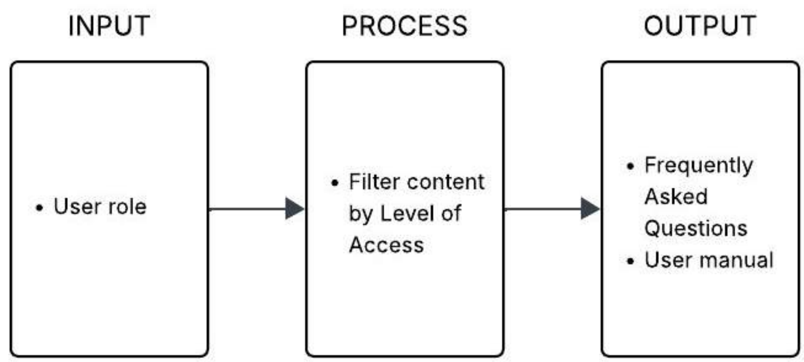


Figure 94 Help IPO

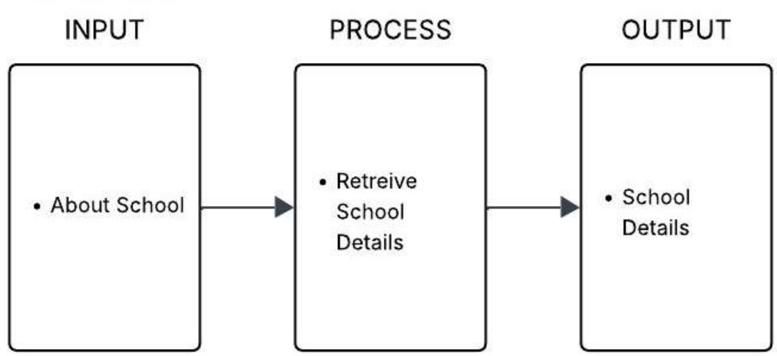


Figure 95 About IPO

Entity Relationship Diagram

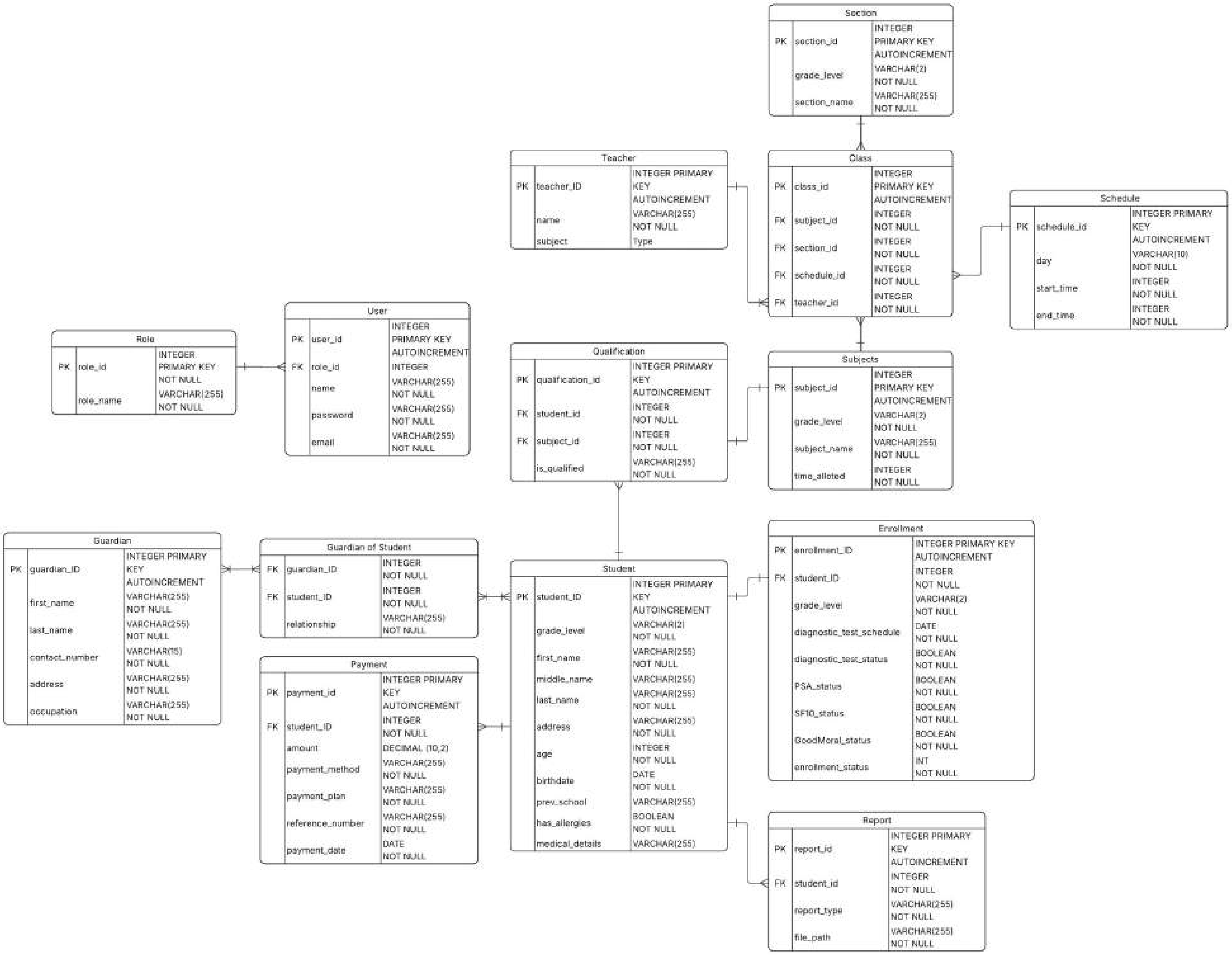


Figure 96 Computerized Enrollment System ERD

Screen Designs

Figure 97 Log-In Screen UI

This screen is the entry point for all users, requiring them to input their credentials to access the system. It provides the initial security layer for the computerized enrollment system.

Figure 98 Forgot Password Screen UI

This interface is for users who have forgotten their password and need to start the recovery process. It asks for an email or username to verify the user's identity.

Figure 99 Forgot Password (Enter OTP) Screen UI

As the next step in password recovery, this screen prompts the user to enter a One-Time Password (OTP) sent to their registered email. This serves as a two-factor authentication method.

Figure 100 Forgot Password (Change Password) Screen UI

After successfully verifying their identity, this screen allows the user to set a new password. It typically requires them to enter the new password twice for confirmation.

Figure 101 Account Registration (Level of Access - Admin) Screen UI

This screen is exclusively for Administrators to create new user accounts for personnel like Registrars or Teachers. It involves inputting new user details and assigning their specific role within the system.

Figure 102 Enrollment Screen UI

This is the main enrollment dashboard for the Administrator, providing a comprehensive view of the enrollment process. From here, the Admin can manage student data, search for students, and access various enrollment-related modules.

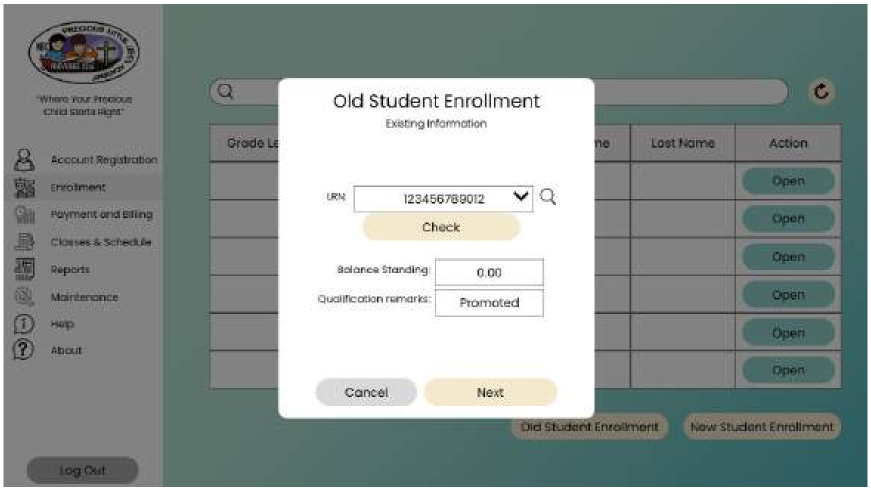


Figure 103 Enrollment (Old Student Enrollment Menu) Screen UI

This popup window appears when an Admin is processing the re-enrollment of an existing student. It displays the student's current information and allows the Admin to update their enrollment status for the new school year.

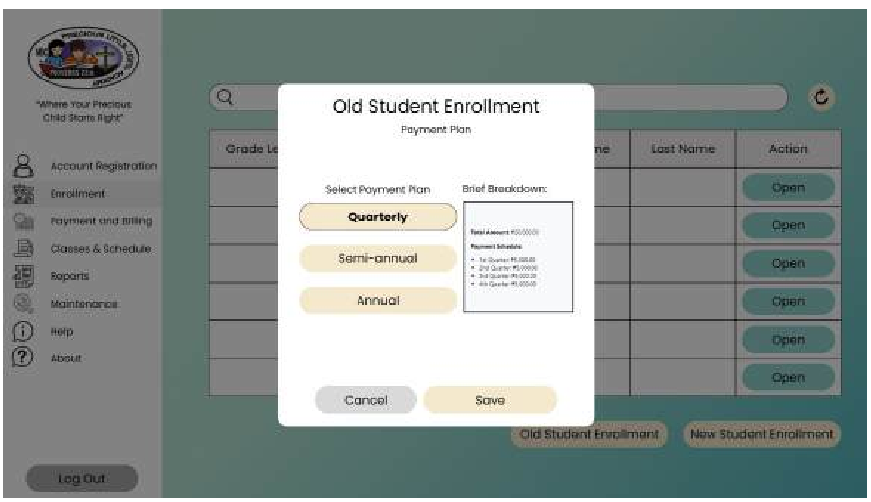


Figure 104 Enrollment (Old Student Enrollment Payment Plan) Screen UI

This screen allows the Administrator to select or update the payment scheme for an old student who is re-enrolling. It is a specific step within the old student enrollment workflow.

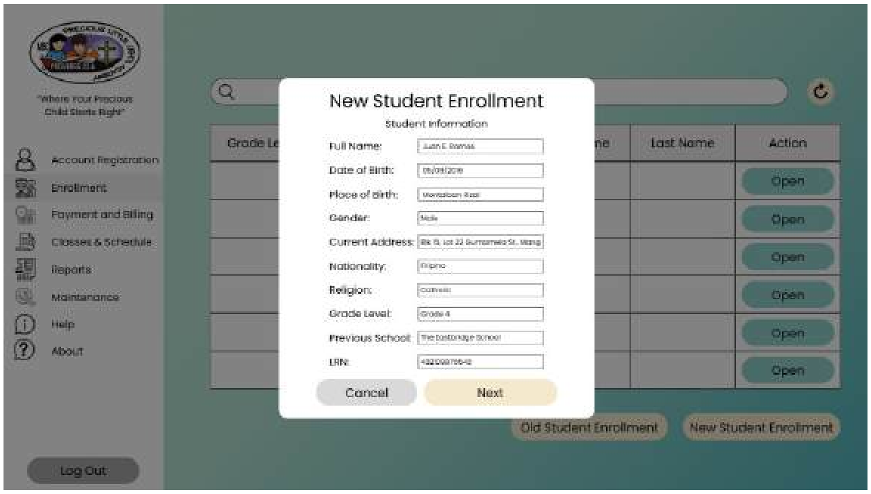


Figure 105 Enrollment (New Student Enrollment Student Information) Screen UI

This form is used by the Admin to input the personal and academic details of a new student. It is the first major step in registering a student who has no prior record in the system.

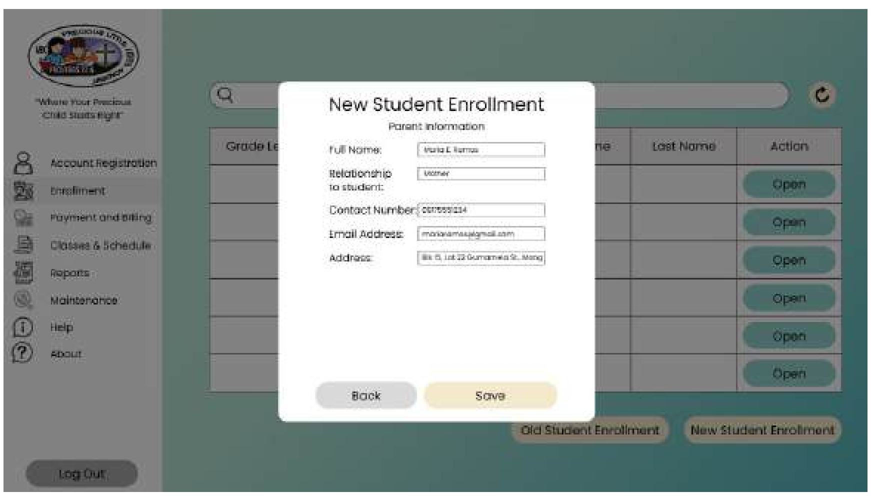


Figure 106 Enrollment (New Student Enrollment Parent Information) Screen UI

This popup is part of the new student registration process, specifically for capturing the guardian's or parent's contact information and details. This information is linked to the new student's record.

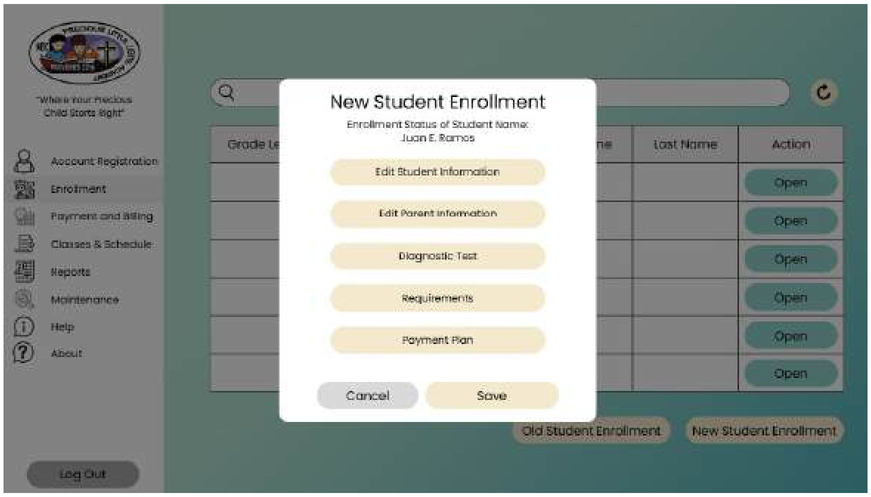


Figure 107 Enrollment (New Student Enrollment Menu) Screen UI

This screen displays a sample or a preview of the completed enrollment form for a new student. It allows the Admin to review all entered information before finalizing the record.

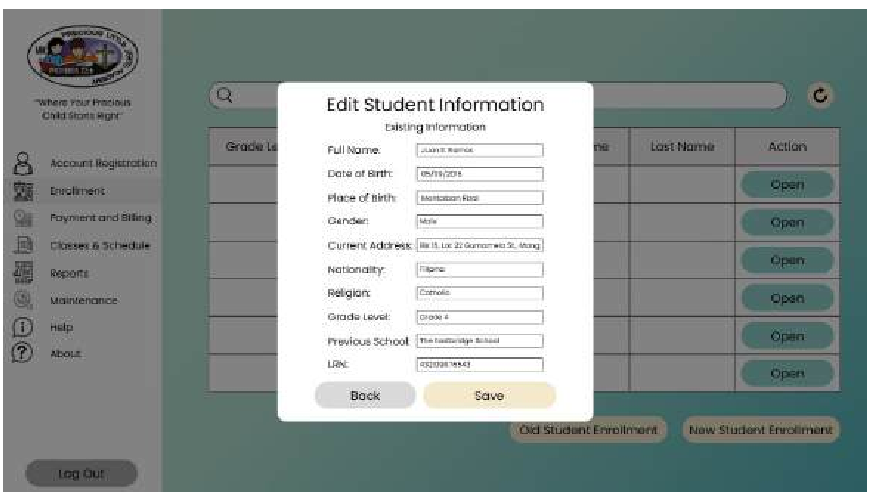


Figure 108 New Student Enrollment Menu (Edit Student Information) Screen UI

This interface allows an Admin to modify or correct a student's existing information. This can be accessed from the main enrollment or student management screen.

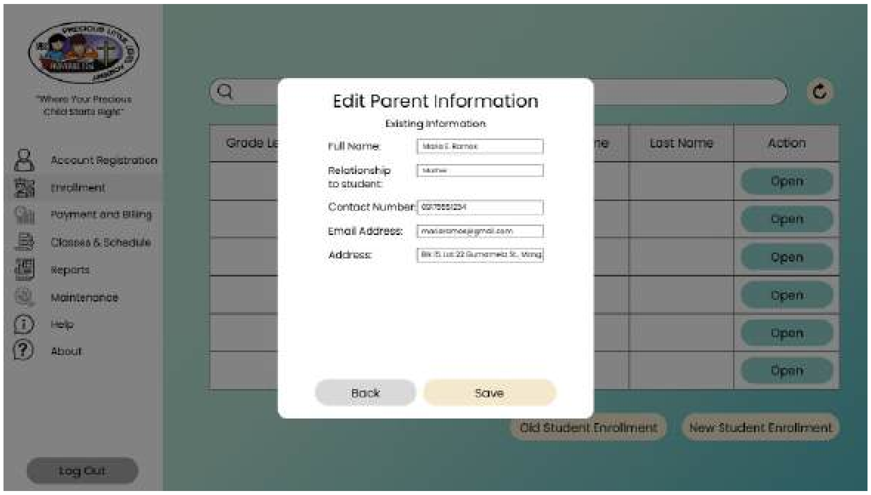


Figure 109 New Student Enrollment Menu (Edit Parent Information) Screen UI

Similar to the student edit screen, this popup is used by the Admin to update or correct a parent's or guardian's details.

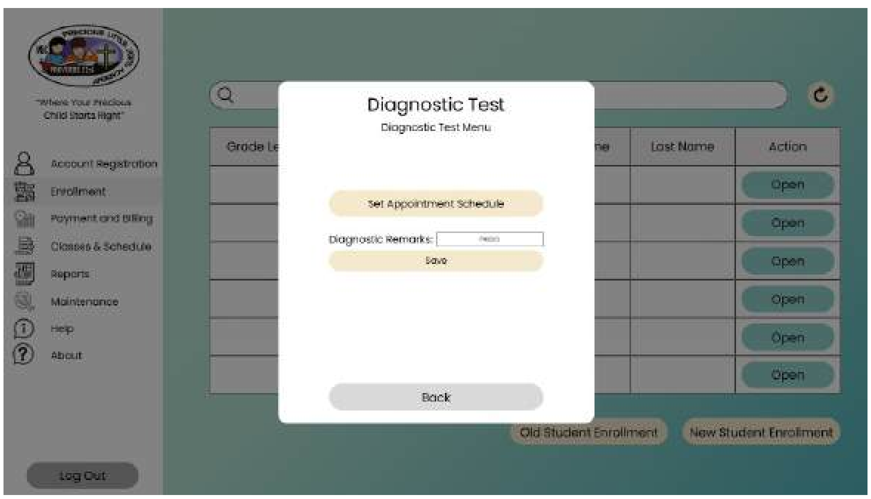


Figure 110 New Student Enrollment Menu (Diagnostic Test) Screen UI

This popup is used by the Admin to manage the diagnostic testing status for new students. It allows setting an appointment or recording the remarks and results of the test.

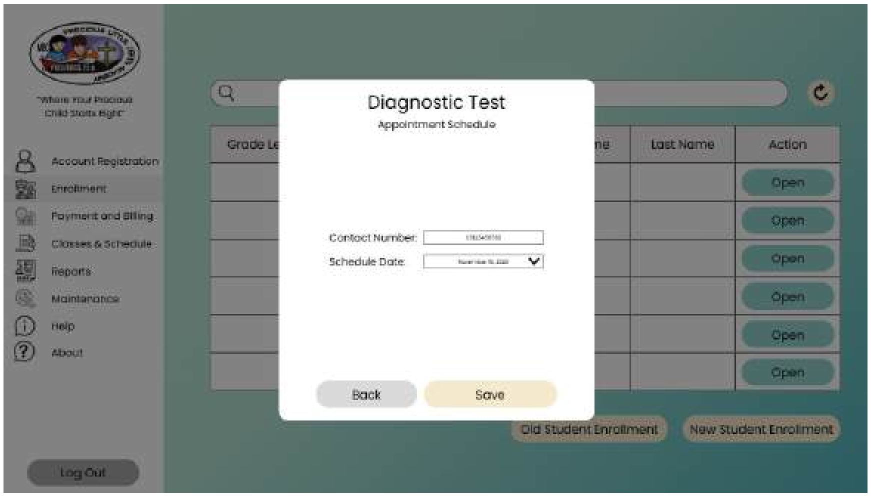


Figure 111 Diagnostic Test (Appointment Schedule) Screen UI

This UI facilitates the creation of a diagnostic test appointment. The user inputs a parent's contact number, which the system then uses to generate and save an appointment schedule to the database.

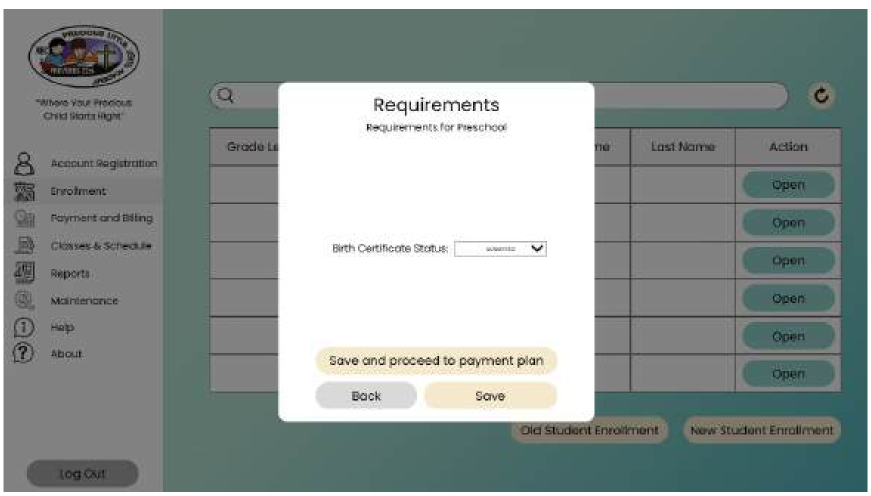


Figure 112 New Student Enrollment Menu (Requirement for Preschool) Screen UI

This interface is for processing Preschool-level requirements. It allows the user to verify the status of the Birth Certificate and subsequently select a payment plan (Quarterly, Semi-annual, or Annual) to save the enrollment record to the database.

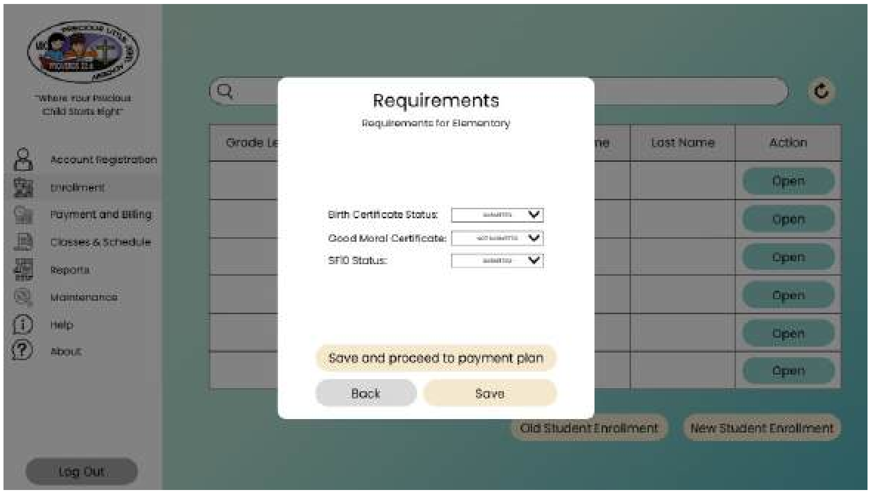


Figure 113 New Student Enrollment Menu (Requirement for Elementary) Screen UI

This UI is for processing Elementary-level enrollment requirements. The user verifies the submission status of the Birth Certificate, Good Moral Certificate, and SF10 before selecting a payment plan to save the enrollment data.

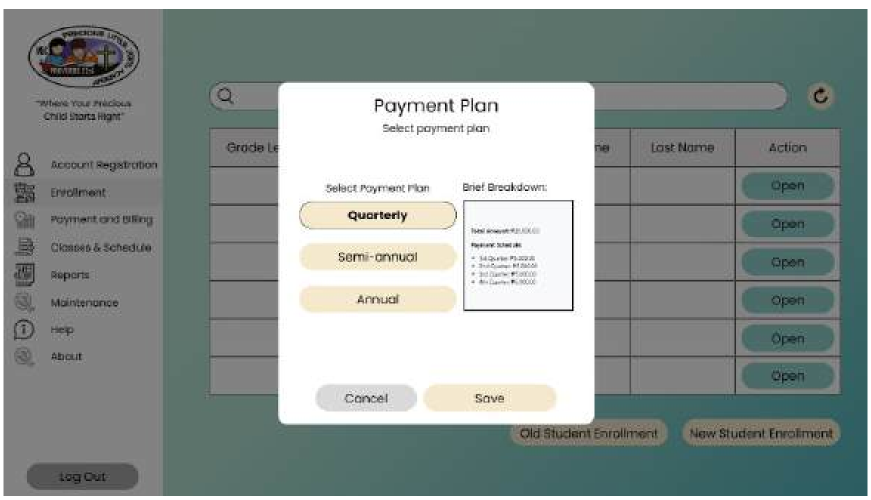


Figure 114 New Student Enrollment Menu (Payment Plan) Screen UI

This UI captures the payment plan selection for a new student, which is a final step in both the Preschool and Elementary enrollment workflows. The user selects Quarterly, Semi-annual, or Annual, and the choice is saved to the database upon completion of the enrollment process.

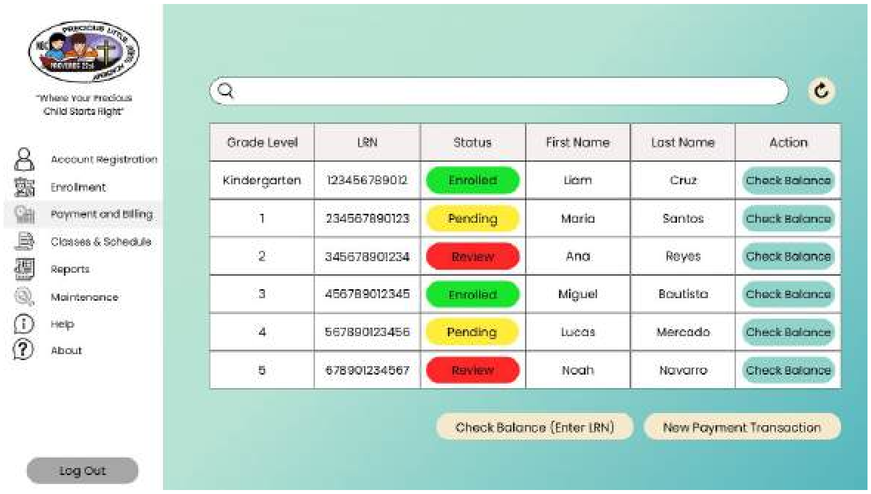


Figure 115 Payment and Billing Screen UI

This is the main dashboard for the Payment and Billing module. It facilitates student record retrieval via an LRN search and provides options to check a student's balance or record a new payment transaction.

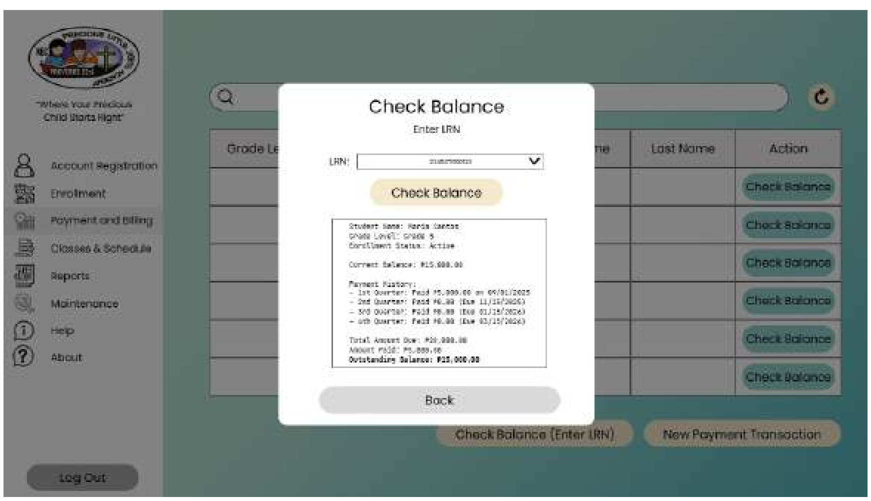


Figure 116 Payment and Billing (Check Balance) Screen UI

This UI displays the results of a student balance check. After verifying a student's LRN, the system retrieves and displays their balance details from the database.

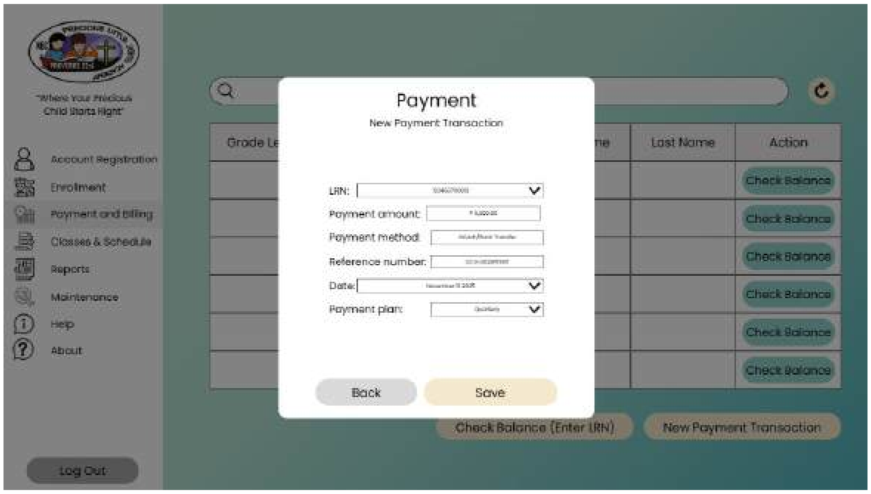


Figure 117 Payment and Billing (Payment) Screen UI

This interface is used to record a student's payment transaction. After LRN verification, the user inputs payment details, which are then saved to the database, updating the student's statement of account.

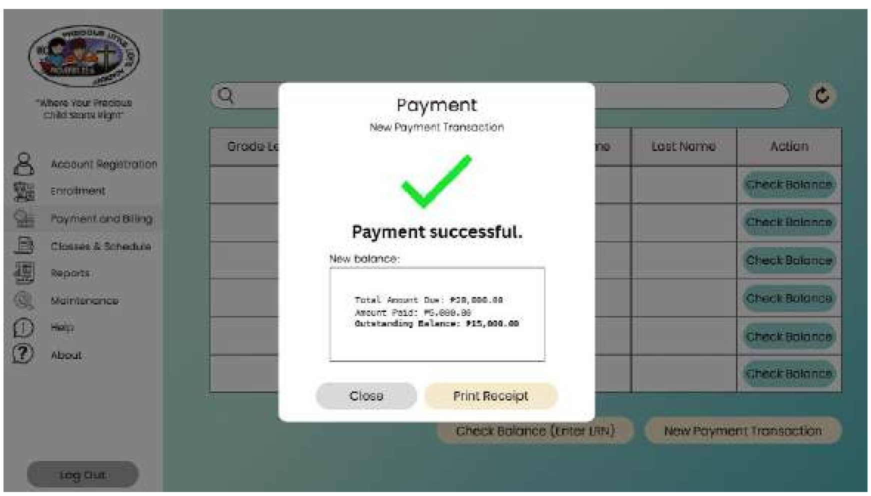


Figure 118 Payment and Billing (Payment Made Sample) Screen UI

This screen displays a generated receipt after a payment is successfully recorded. It shows the transaction details and the student's updated balance, which is retrieved from the database.

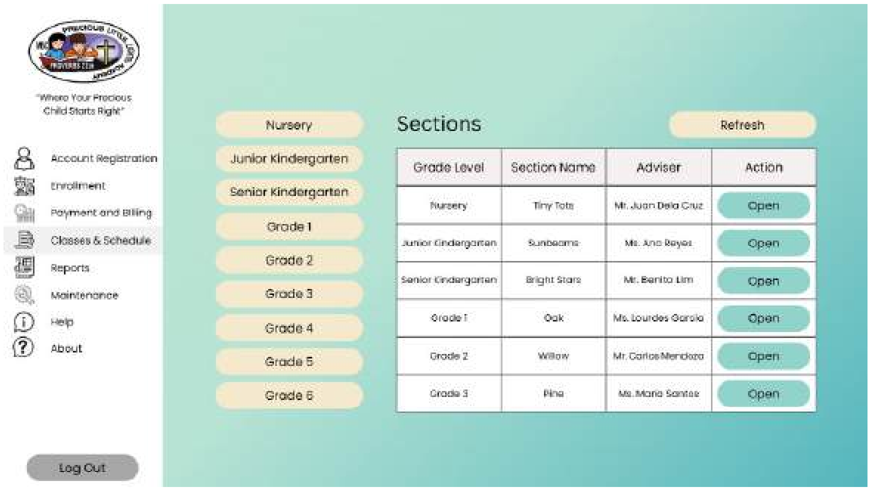


Figure 119 Classes & Schedule (Level of Access - Admin) Screen UI

This screen is the main interface for the Class Schedule Management module. It allows an Admin to select a grade level, retrieve and display its sections from the database, and then select an action to manage schedules.

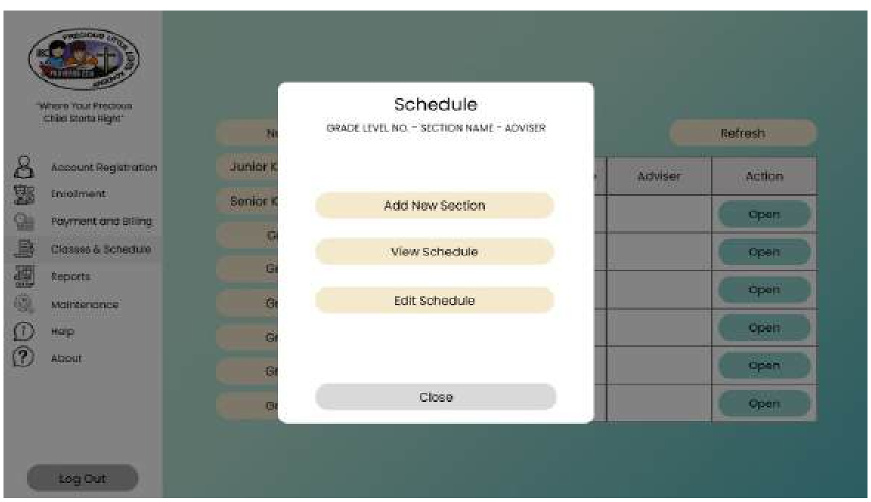


Figure 120 Classes & Schedule (Schedule Menu) Screen UI

This popup menu presents the user with actions for a selected section. The options include viewing the schedule, setting (editing) the schedule, or adding a new section.

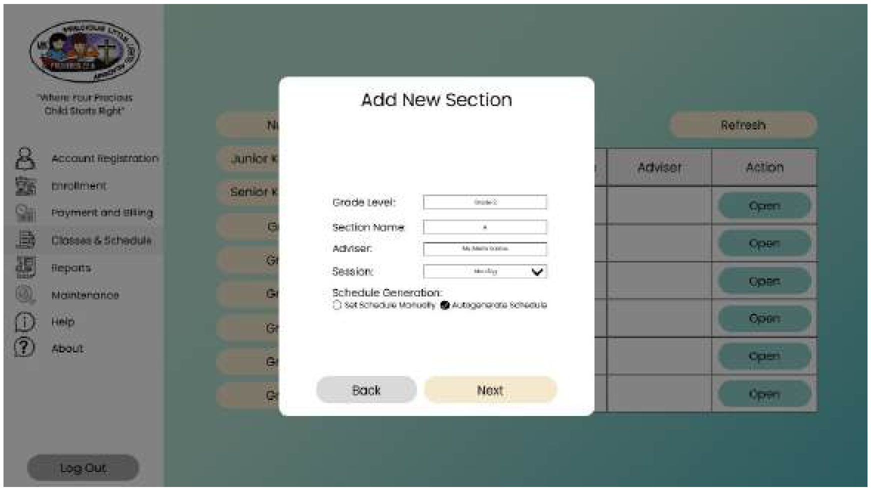


Figure 121 Classes & Schedule (Add New Section) Screen UI

This UI allows an Administrator to create a new section. The user inputs section details and then selects whether to set the schedule manually or use the autogenerate function.

Figure 122 Add New Section (Auto Generate Schedule) Screen UI

This screen displays a schedule timetable generated by the system's Greedy Algorithm. The algorithm retrieves required subjects and verifies the schedule against the database to create a valid, conflict-free timetable.

Figure 123 Classes & Schedule (View Schedule) Screen UI

This UI displays a selected section's schedule timetable. The system retrieves the saved section schedule from the database and presents it to the user in a read-only view.

Figure 124 Reports Screen UI

This screen serves as the main interface for the Reports module. It allows a user to search for a student by LRN and then select to generate either a Statement of Account or a Registration Form as a PDF.

Figure 125 Maintenance Screen UI

This UI is the main dashboard for the Maintenance module, accessible only by the Admin. It provides functions to create a database backup, restore a backup, or archive student records.

Figure 126 Maintenance (Archive Student Record) Screen UI

This UI facilitates the archiving of inactive student records. The Admin selects an inactive student, verifies their records, and confirms the action to archive the data in the database.

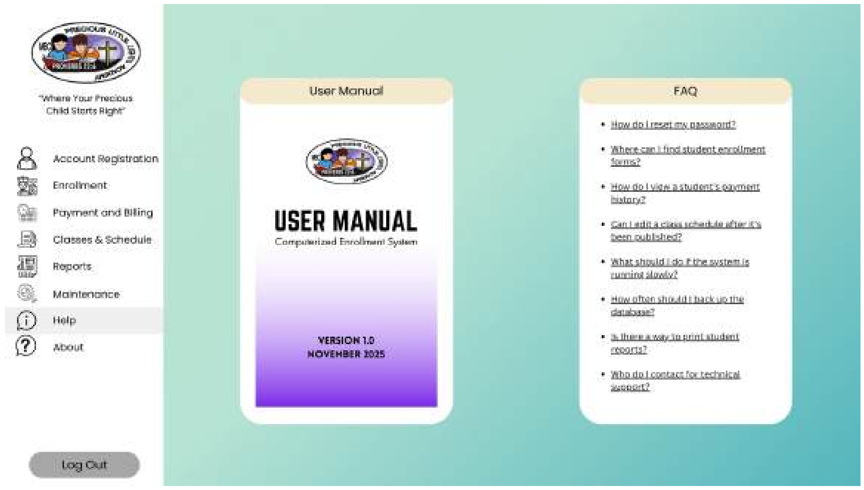


Figure 127 Help Screen UI

This screen provides access to the Help module, which contains a User Manual and an FAQ section. The content presented is filtered based on the user's role (Admin, Registrar, or Teacher).

Figure 128 About Screen UI

This UI displays static institutional details about Precious Little Lights Academy. The system retrieves and displays the school's information, such as its Bible verse, contact details, and address.

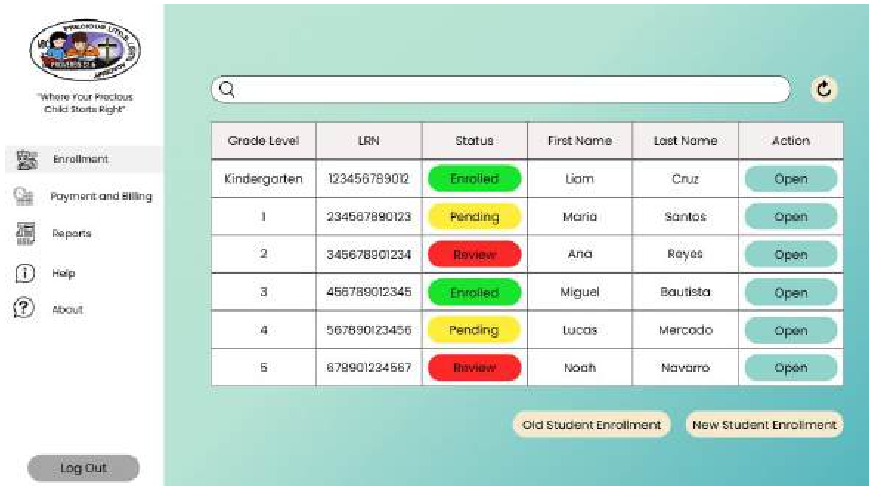


Figure 129 Enrollment (Level of Access - Registrar) Screen UI

This screen shows the Enrollment module dashboard tailored for the Registrar role. The Registrar has access to manage enrollment, billing, and reports, but lacks access to Admin-level modules like Maintenance and Account Registration.

Figure 130 Student Qualification (Level of Access - Teacher) Screen UI

This screen is the main dashboard for the Teacher role, focusing on the Student Qualification module. It allows the Teacher to select a grade level and section to manage the qualifications of their students.

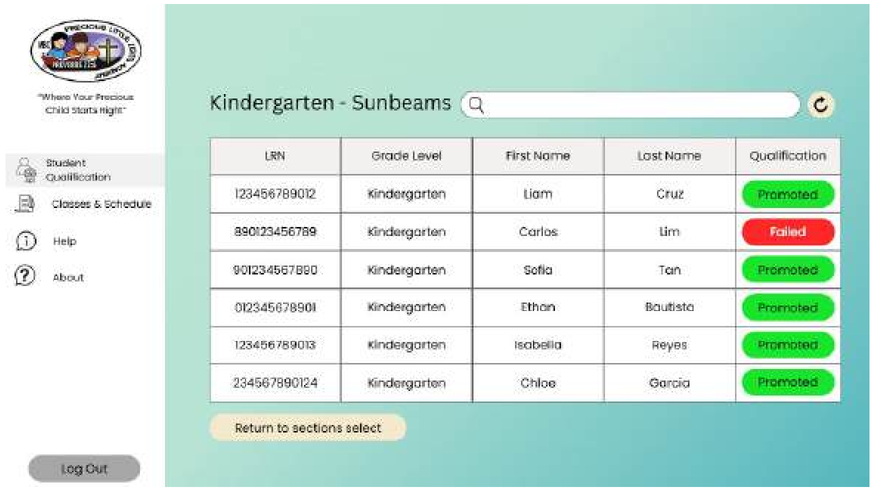


Figure 131 Student Qualification (Selected Grade Level - Section) Screen UI

This UI displays a list of students after a Teacher selects a grade level and section. The Teacher can then update the student's qualification status for promotion, saving the changes to the database.

Figure 132 Classes & Schedule (Level of Access - Teacher) Screen UI

This screen provides Teachers with read-only access to class schedules. The user selects a grade level to retrieve and display the sections from the database.

Figure 133 View Schedule Screen UI

This UI displays the specific schedule timetable for a section selected by the Teacher. The system retrieves the section's schedule from the database and presents it in a read-only format.

CHAPTER 3

Methodology

This chapter presents the Waterfall Model as the selected software development process model, emphasizing the use of a sequential approach for each phase from requirement specification to operation and maintenance.

This chapter also includes the system architecture that explains how the flow of data works from presentation tier to the data tier and the retrieval of data from the database. The feasibility study is also found in this chapter which provides an evaluation of the technological, operational and the economic aspect, a Cost Benefit Analysis is also included where all of the mentioned fields help discern the feasibility of the system. The Testing and Operation Procedure can be found in this chapter, which consist of three testing methods, which are Unit testing, Module testing, and Integration testing. This chapter also contains the Software Evaluation, which is used to evaluate the system by using different standards and sampling tools. A work plan can be found which is represented by a Gantt chart to visualize the timeline of the completion of this study.

All these tools and sections are used in order to evaluate and complete the proposed system in order to provide a functional and reliable software for the beneficiary.

Software Process Model

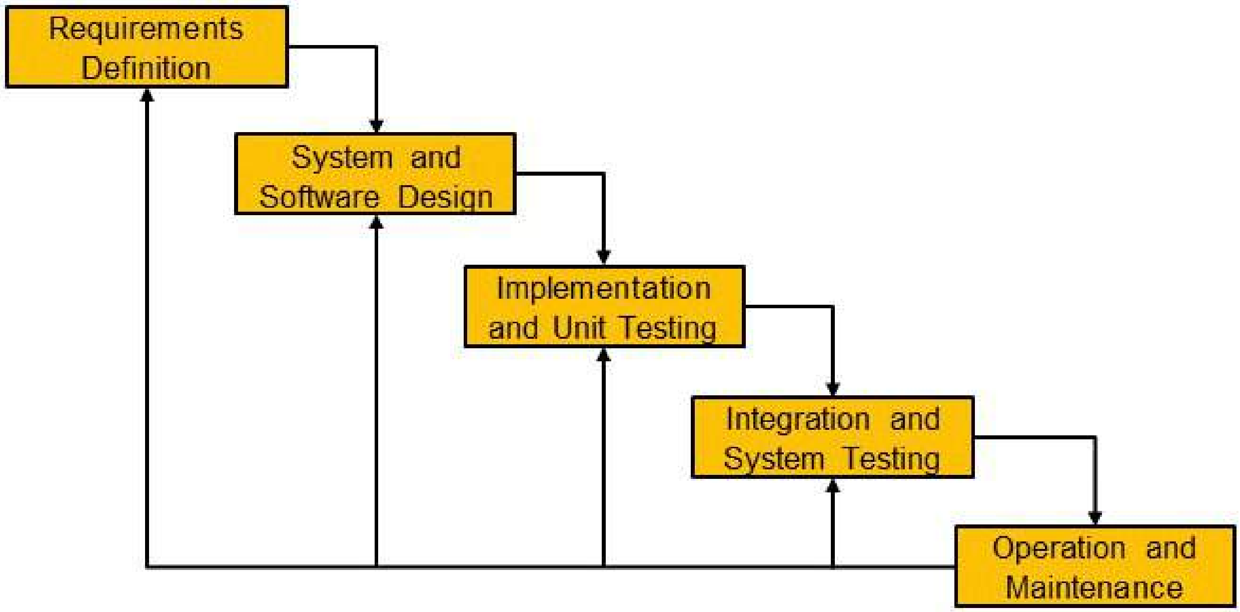


Figure 134 Waterfall Model Diagram

The figure above is the Waterfall Model Diagram, this is a software development model created by Dr. Winston W. Royce in 1970, which prioritizes a sequential approach of all the activities for the project represented as phases to be completed in a linear order (Canevska, 2024). The Waterfall Model consists of five distinct phases, which are Requirement Definition, System and Software Design, Implementation and Unit Testing, Integration and System Testing, and Operation and Maintenance.

The Requirement Definition phase is where all requirements are defined, documented, and finalized before any development begins. In this phase, the proponents gather the functional and non-functional requirements that aligns with the beneficiaries' demand by conducting interviews.

The next phase is the System and Software Design, which includes the designing of the system architecture, software, diagrams, and data based on the requirements defined in the previous phase.

The Implementation and Unit Testing phase includes the production of the software into source code based on the System Design and also includes the testing of each module in the proposed software to ensure functionality. In this phase, the proponents will use Java SE (ver. 25) as the programming language, Netbeans (ver. 27) as the Integrated Development Environment (IDE), and PostgreSQL (ver. 18) for the database.

The next phase is the Integration and System Testing phase, where all the modules of the software are integrated and tested as a whole system. This phase verifies if all the requirements that were defined in the Requirement Definition phase are met.

Lastly, the Operation and Maintenance phase is the final phase of the Waterfall Model which starts after the deployment of the system. This phase is used in order to ensure the completed system is functional, reliable, and up-to-date over time. It includes the maintenance of the system such as Updates, User Support, Bug fixing and more.

System Architecture

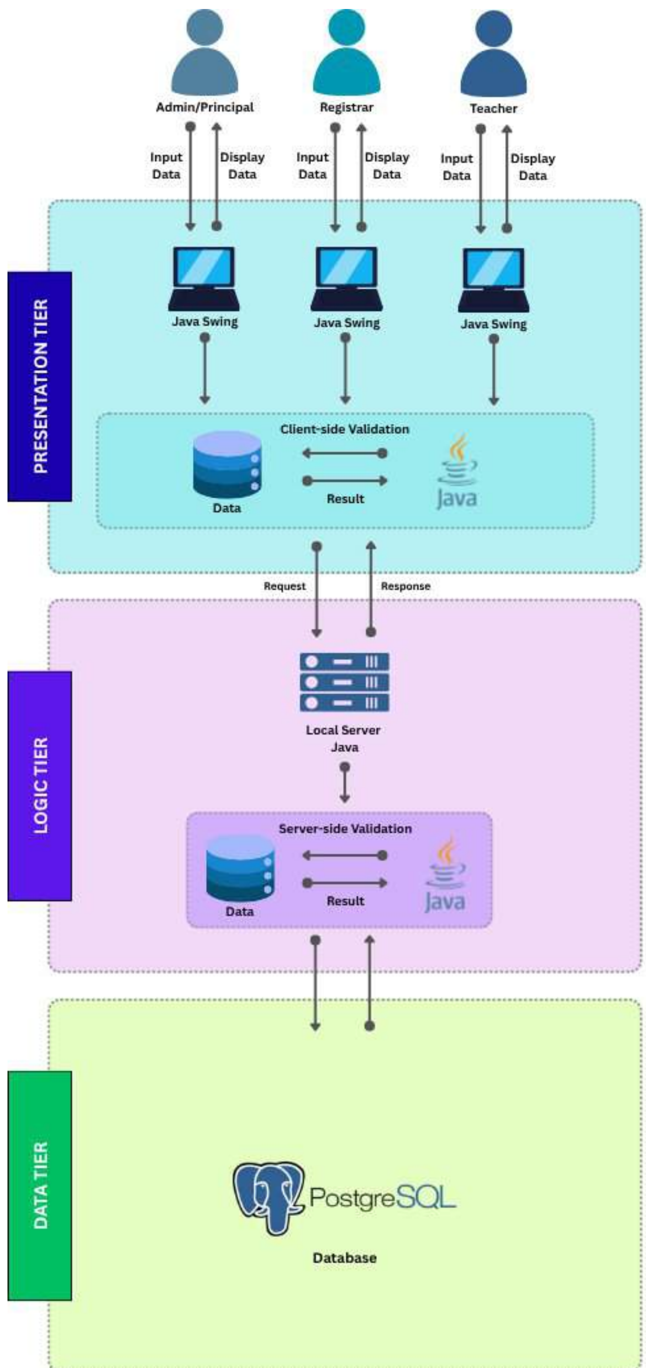


Figure 135 System Architecture

The figure above illustrates the System Architecture for the Computerized Enrollment System for Precious Little Light Academy. This architecture is implemented to streamline the enrollment process, payment and billing, and class schedule management while modernizing and automating this whole procedure to reduce cases of human errors. This system architecture consists of three main tiers which are the Presentation Tier, Logic Tier, and Data Tier.

The Presentation Tier contains the user interface, which is developed using Java Swing for both the graphical user interface and its design. This tier is where the three types of users interact with the system which are the Admin/Principal, Registrar, and Teacher. Each user can input data and view displayed information through their respective interfaces based on their level of access. All data entered by users undergoes initial client-side validation using Java before being sent to the local server, ensuring that only proper formatted data is transmitted to the next tier.

The Logic Tier contains the local server, which connects the database to the user interface. This part uses Java (ver. 25) as its programming language to handle the query requests from the system to the database. This tier also includes server-side validation that all the enrollment details, student information, payment, and schedules data to be updated or retrieved from the database are correct and properly formatted. Additionally, there is a Hashing Algorithm that secures user passwords to the database, ensuring strong security and protecting user accounts from unauthorized access. Moreover, the system implements a Greedy Algorithm for class scheduling by interval, which efficiently generates class schedules by selecting the best time slots without conflicts to maximize the use of available time periods.

The Data Tier is where all enrollment system data is stored and organized. The database uses PostgreSQL (ver. 18) to manage all data for the Computerized Enrollment

System. The database manages different types of data including user credentials, student information, enrollment details, and class schedules while ensuring the security of these data.

Overall, the flow of data begins with an user input through the user interface of the system, where a client-side validation occurs. After this validation, the data is then sent to the local server for a server-side validation to ensure all data saved or retrieved are correct and properly formatted. Then the requested data will be pulled back to the user interface to be displayed to the user. This architecture ensures that the Computerized Enrollment System is easy to use, fast, secure, and functional, providing a smooth enrollment process for both the staff and the enrollees.

Data Gathering Instruments and Procedures

The proponents used three data-gathering instruments in this study: the Interview, the Library, and the Internet. An interview is a formal discussion in which one party asks questions to gather important information from another person. The library, on the other hand, is a place where all books, magazines, and other materials are available for people to use or borrow. The Internet is a worldwide communications system that connects thousands of individual networks. The procedures of the utilized data collection are as follows:

1. Interview - is a formal conversation in which one person asks another for information. After trying to find a company and business properties, the proponents decided to choose Precious Little Light Academy. The proponents prepared an endorsement letter and gave it to the school before proceeding with the interview.

2. Library - is a location where you may find all of the information you need, including books, magazines, and other materials for research or assignments. The proponents used and visited the school library to gather information that was necessary for the study.
3. Internet - It is an online source of information; the proponents used the internet to access specific information for up to date information, in particular for cost-related purposes such as current market prices of hardware and software, as well as estimates for operating expenses, supplies, and utility costs.

Feasibility Study

The project aims to develop a Computerized Enrollment System for Precious Little Lights Academy, which will improve the operational effectiveness of the enrollment process that can be provided by the school. Moreover, this system will employ advanced administrative management by automating processes through efficient management of students' data, enrollment process, data accuracy, and security measures. By substituting manual paper and Excel-driven processes with a unified, easy-to-use enrollment system, this provides a workable, affordable, and manageable solution based on the school's existing infrastructure and requirements.

Technical, Operational, and Economical

The proposed Computerized Enrollment System is technically feasible given the school's existing infrastructure and the proponents' technical capabilities. The system is developed on the basis of Java SE (ver. 25), NetBeans (ver. 27), and PostgreSQL (ver. 18) which are compatible with the standard windows machines. The school already has desktop computers and a LAN system, thus it does not require significant overhaul of hardware. Moreover, the open-source application (PostgreSQL, Java SE) decreases the licensing expenses, whereas the offline, LAN-only architecture of the system is in line with a limited reliance on the internet of the school.

The operational feasibility of this system encompasses the design to fit seamlessly into the current workflow of Precious Little Lights Academy. Staff roles (Admin, Registrar, Teacher) are preserved, and the interface is built using intuitive Java Swing components with role-specific dashboards. A Help Module with offline User Manual and FAQs ensures usability even for staff with limited technical experience. Training sessions which are included in operational cost will further support smooth adoption. Because the system replaces error-prone paper and Excel-based processes, it is expected to be welcomed by staff seeking efficiency and accuracy.

In assessing the economical feasibility of this system, the total development and deployment cost is reasonable for a small private school, and the long-term savings in time, paper, labor, and error correction justify the investment. The system eliminates recurring costs associated with printing forms, manual record-keeping, and schedule coordination. With a projected payback period of under 15 months and an ROI of over 80%, the system is economically viable and sustainable.

Cost Benefit Analysis (CBA)

1. **Hardware Cost**

Recommended Requirements	Specifications	Quantity	Unit Price	Costs
Printer	Epson EcoTank L1210 A4 Ink Tank Printer (Pre-Owned)	1	7,945.00	0.00
System Unit	GreatWall Chuga All in One PC (Pre-Owned)	6	10,500.00	0.00
Mouse and Keyboard	A4Tech KK-3330 Keyboard and Mouse Combo (Pre-Owned)	6	470.00	0.00

Table 1 Hardware Cost

Total: ₱0.00

Source:

https://www.epson.com.ph/

https://easypc.com.ph/

2. **Software Development Cost**

Personnel	No. of Personnel	Salary
Programmer	3	26,847.00

Table 2 Hardware Cost

Total: ₱89,301.00

Personnel Salary for 30 Days:

Programmer: 26,847.00 / 30 = ₱ 894.90

System Analyst: 35,607.00 / 30 = ₱ 1,186.90

Source:

https://ph.indeed.com/

III. Operational Cost

A. System Cost

Items	Specifications	Costs
Operating System	Windows 11 64-bit	9,500.00
Database Engine	PostgreSQL Version 18	Free
Programming Language	Java SE Version 25	Free

Table 3 System Cost

Total: ₱9,500.00

Source:

https://pcx.com.ph/

https://www.postgresql.org/

https://www.oracle.com/

B. Stationaries and Supplies

Item	Quantity	Price	Cost
Short Bond Paper	2	554.00	554.00
A4 Bond Paper	2	576.00	576.00
Ink Refill Set	3	1,392.00	4,176.00
Ballpen Set	2	61.00	122.00

Table 4 Stationaries and Supplies

Total: ₱5,428.00

Source:

https://www.nationalbookstore.com/

C. Utility Expenses

Particulars	Cost
Electricity	723.99

Table 5 Utility Expenses

Total: <u>₱723.99</u>

Electricity Usage 2.47/day for 9 hours in a month (22 working days)

Source:

https://appliancecalculator.meralco.com.ph/

D. Training Cost

Personnel	Amount Per Day	Days	Hours	Total
Administrator	485.00	5	6	2,425.00
Registrar	485.00	3	6	1,455.00
Teacher	485.00	3	6	1,455.00

Table 6 Training Cost

Total: <u>₱5,335.00</u>

(80.83 / hour * 6 Hours) * 5 Days = **<u>₱2,425.00</u>**

(80.83 / hour * 6 Hours) * 3 Days = **<u>₱1,455.00</u>**

Summary Cost

Description	Cost
I. Hardware Cost	0.00
II. Software Development Cost	89,301.00
III. A. System Cost	9,500.00
III. B. Stationeries and Supplies	5,428.00
III. C. Utility Expenses	723.99

III. D. Training Cost	5,335.00

Table 7 Summary Cost

Total Costs: ₱110,287.99

Estimated Benefits

Accuracy and efficiency of the software at approximately 80%

Total estimated Benefits = 110,287.99 * 80%

Total = 88,230.392

Payback Period

Payback Period = (Total Cost / Total Estimated Benefits) * 12

= (110,287.99/88,230.392) * 12

Total = 15 months or 1 year and 3 months

Return of Investments (ROI)

Return of Investment = (Total Estimated Benefits / Total Cost) * 100

= (110,287.99 / 88,230.392) * 100

Total = 80%

The total project cost including hardware, software development, system, stationery, supplies, utilities, and training is ₱110,287.99. The investment is expected to be returned within 15 months or 1 year and 3 months, by which time approximately 80% of the cost, which is ₱88,230.392, will have been recouped through estimated benefits.

Materials

The system will utilize various software and hardware. For hardware, a computer will be used to access the system and process the student enrollment. Moreover, a printer will be used for printing registration and statement of account forms. For software, the system will be programmed using Java SE (ver. 25) programming language for the reason that it is easy to use and the proponents have extensive experience with this programming language. Additionally, it offers a graphic user interface (GUI) capabilities that can help lessen the difficulty of designing the system and Java Database Connectivity (JDBC), which ensures a smooth connection to the database; PostgreSQL (ver. 18) for the database.

Testing and Operating Procedure

Unit Testing

In this project, the unit testing will be to check the functionality of each program unit separately and to make sure that each unit is working as it should. The basic procedures involved in these units are email and phone number validation, hashing passwords using bcrypt, PDF generation with iText 7, and database code used through JDBC and PostgreSQL. Every one of these functions is individually tested when developing in NetBeans through manual checking. As an example, the system should not admit an invalid email but only a valid email from trusted emails such as Gmail. On the same note, the password hashing is authenticated by making sure that different hashes are irreversible. This step is aimed at making sure that there is sound reasoning at the base level before any other party can be integrated.

Module Testing

After the unit tests have been passed, modules are grouped into related units, and modules level testing is performed. The modules, such as Enrollment, Payment and Billing, Class Schedule Management, Security, and Reports, are then individually tested as a unit to ensure internal data processing, user interface behavior, and business policy compliance. An example is the Enrollment Module, where new student and guardian information are properly entered into the Java Swing interface and verified on both the client and server sides, in addition to the correct storage of such information in the PostgreSQL database with the correct constraints, including NOT NULL and UNIQUE. The Security Module is tested to check that only authorized access (Admin, Registrar, Teacher) can be successfully logged in, and the change password module works as intended. This stage would ensure that each module is functioning properly on its own before it is integrated with others.

Integration Testing

Integration testing checks the interface of fully tested modules in the LAN environment of the school. This stage will make sure that the data movements throughout the system are smooth and that inter-module dependencies work accordingly. Some of the most important test scenarios would be to ensure that a new student in enrolment is present in the Search Module, he is billable in the Payment and Billing Module, and he is reflected in official reports like the Statement of Account. The other essential test is used to ensure that the schedules that the Admin had prepared in the Class Schedule Management Module properly appear in the read-only form of the Teachers. Also, a successful login should load role-specific dashboards properly without revealing unauthorized capabilities. All the integration tests are done on Windows machines that have PostgreSQL running locally, which resembles the real deployment environment. Communication errors or inconsistencies in data are recorded, fixed, and again tested until the system is running as an integrated, stable solution that meets the specifications in Chapter 1.

Software Evaluation

The following are the software criteria obtained from ISO 25010 standards:

1. Functionality - The level of functionality of a system in terms of fulfilling the requirements of users under given conditions. The system will be evaluated based on its capability to manage correct enrollment, billing, scheduling, and generation of reports.
2. Security - Security helps prevent unauthorized access or alteration of data. The system implemented bcrypt password hashing and role-based access to ensure that sensitive student records are secured.
3. Reliability - Reliability implies the ability to perform reliably. This is guaranteed by constant maintenance of the software before and after deployment.
4. Operability - Operability implies to which a software or system is easy to operate and control.
5. Learnability - Learnability implies to which the functions of a software or system can be learnt to be used by specified users within a specified amount of time.

Scale	**Options**	**Score Range**	**Level**
6	Strongly Agree	5.17 – 6.00	Very High
5	Agree	4.33 – 5.16	High
4	Partially Agree	3.49 – 4.32	Average
3	Partially Disagree	2.67 – 3.50	Average
2	Disagree	1.83 – 2.66	Low
1	Strongly Disagree	1.00 – 1.82	Very Low

Table 8 6 Point-Likert Scale

Sampling Tools

The study employs cluster sampling in which participants are chosen if they are an IT professional or a Non-IT professional to select twenty respondents who will evaluate the system's functionality, usability, and overall effectiveness. The sample is divided into two groups to gather balanced perspectives: ten (10) IT professionals and ten (10) Non-IT professionals.

The group of IT professionals are individuals who graduated in any computer-related programs such as Computer Science (CS), Information Technology (IT), Computer Engineering (CpE), and Information Systems (IS) with at least one (1) year experience in the technology industry.

While the group of Non-IT professionals consist of educators, school administrators, and staff from institutions similar to Precious Little Lights Academy like private elementary schools in Rizal. This group represents the actual end-users who would operate the system in a real-world setting.

This sampling method ensures that the respondents are able to provide valuable insights and feedback from their experiences, where the proponents can use these responses in order to evaluate the functionality of the system.

Work Plan

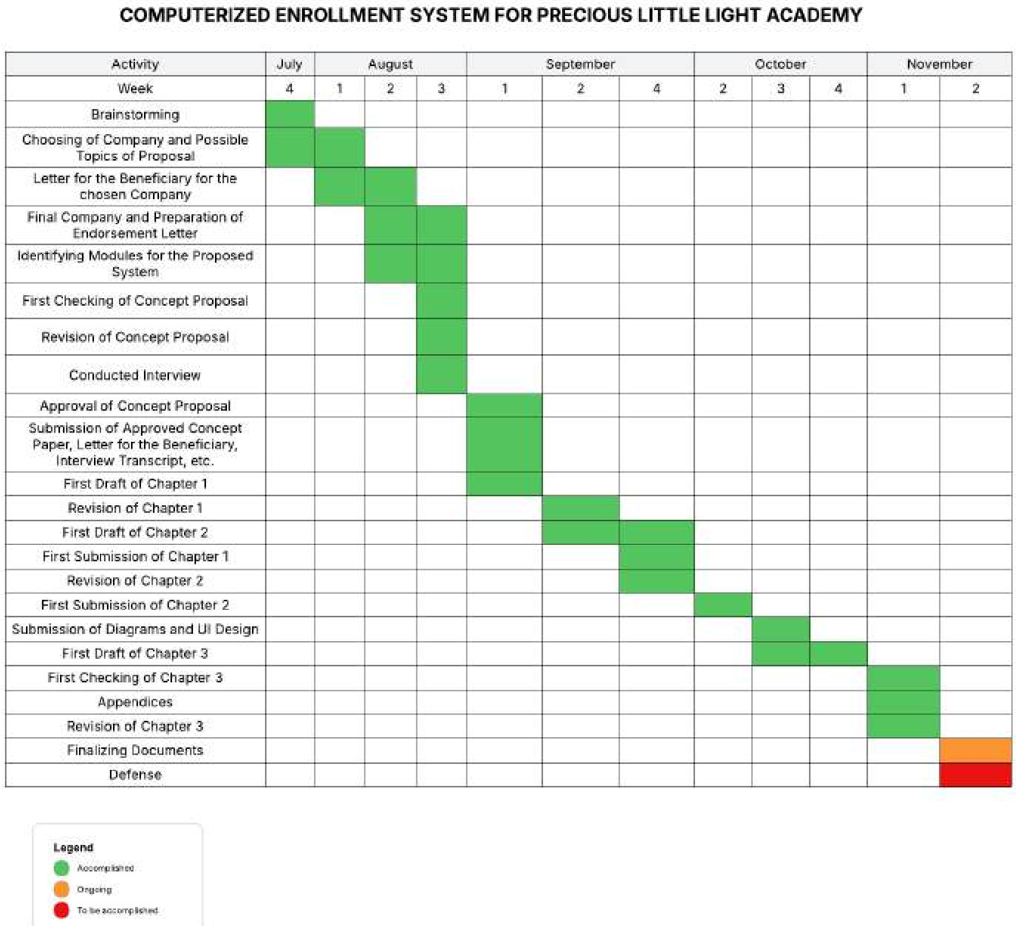


Figure 136 Gantt Chart

Algorithm Implementation

The Computerized Enrollment System will implement two algorithms, which are Hashing Algorithm and Greedy Algorithm. In the system, the registered password is not stored as plain text to the database because this is not secure in scenarios of system attacks. By implementing Hashing Algorithm, the registered password will first undergo transformation by using the bcrypt hashing function, which will generate a 72 character hash to be stored in the database. This stored hash will be used to compare to the inputted password if the user wishes to log in to the system, the same hashing function will also be applied to the inputted password which will generate a hashed password. If the stored hashed password matches the inputted hashed password, the user will be granted access on the system based on their level of access.

Additionally, the system will also use the Greedy algorithm for the scheduling of classes per section. The implementation of this algorithm will provide auto-generation of class schedule timetables, which would help lessen the task of creating schedules that the Principal or Admin is tasked with.

System Architecture of Algorithm

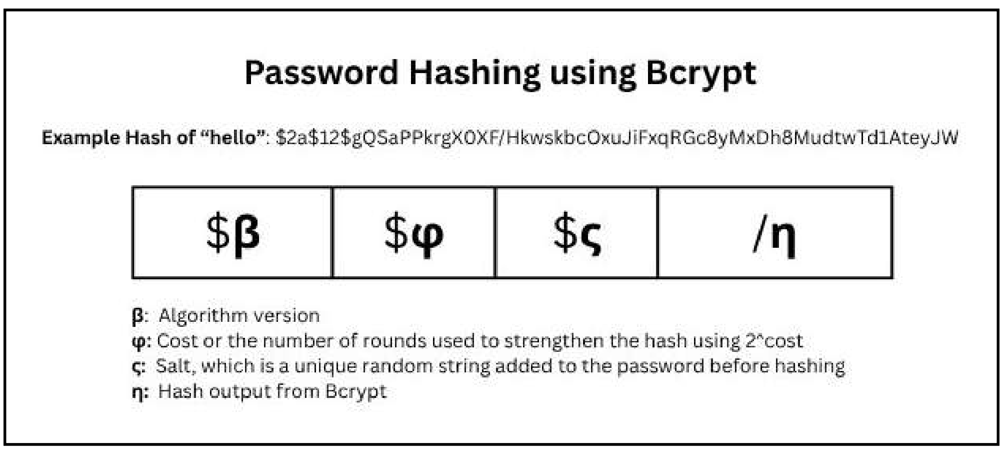


Figure 137 Password Hashing Algorithm using Bcrypt Architecture

The diagram shown above shows the bcrypt hashing function that will be implemented in the log-in and account registration module of the system for securing user passwords. Bcrypt is a cryptographic hash function that transforms a plain text password into a fixed 72 byte-length, irreversible hash value. It incorporates a salt and a configurable cost to strengthen the hash against malicious attacks. The algorithm performs multiple iterations of key expansion and encryption based on the Blowfish cipher, ensuring that even small changes in the input result in completely different hash outputs. The final hash output consists of four parts: the version, the cost value, the salt, and the hashed password. This process ensures that stored passwords are protected and cannot be easily reversed or matched.

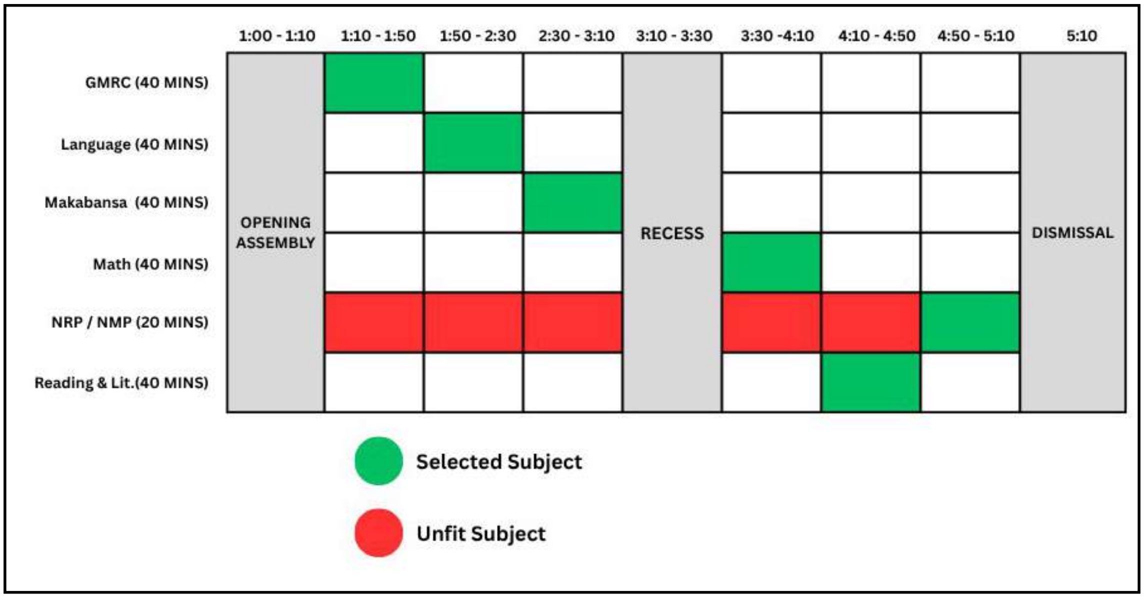


Figure 138 Greedy Algorithm Architecture

The diagram shown above represents the Greedy Algorithm Architecture that will be implemented for the Class Schedule Management System. This algorithm is used to efficiently assign subjects and time slots while minimizing schedule conflicts. The greedy algorithm works by selecting the best first available time slot for a time slot without revisiting past choices.

Appendices

Appendix A: Concept Paper

Technological Institute of the Philippines
Quezon City

College of Computer Studies
Computer Science Department

CS 301 – Software Engineering 1
First Semester S.Y. 2025-2026

CONCEPT PROPOSAL

(1) Proponents	
Name	**Roles** *(Programmer, System Analyst, Designer, etc., please specify)*
1. Hofileña, Paul April A.	Programmer, Designer
2. Mejia, Neil Sean D.	Programmer, Debugger
3. Raoet, Christian Louie C.	Programmer, System Analyst

(2) Title

Computerized Enrollment System for Precious Little Lights Academy

(3) Background of the Study *(Brief description of the target beneficiary. The proponents will present the problems encountered by the end users in the procedure of the existing system.)*

Precious Little Lights Academy is a Christian educational institution providing quality education from preschool to elementary. As of 2025, the school is solely based in Burgos, Rodriguez, Rizal and currently accommodates 119 students with an average of 11 students per section.

The institution currently uses both paper and Excel record-keeping systems to store records of the students personal information, statement of accounts, and forms that are used for grades, an example is SF9, which contains the grades of the student which is to be shown to the parents at a certain time period.

Traditional record-keeping systems deal with a lot of challenges since they are manually encoded. The problems that they encounter are from the student's personal information forms, where some parents have yet to fill up the required fields of information for reasons such as they do not have free time.

Their enrollment procedure requires a diagnostic test since they are not certified to educate students with psychological disorders like ADHD or etc. The problem they encounter here is the scheduling of the execution of this test since some prefer to do it at a later date.

The scheduling of the teacher's assigned subject and room and the viewing of the SF9 of the student for the parents are manually done. And the beneficiary also has stated that the school is in need of a system that can automatically generate the forms that are ready for printing due to the huge number of documents that are stored within a file cabinet and the Excel sheets.

(4) Project Objectives *(What do you want to accomplish in your investigation? The proponents will present the solutions that will answer all the problems in the existing system.)*

This project aims to develop a digital enrollment system for Precious Little Light Academy, addressing the issue of manually enrolling students. The primary objectives are:

- To develop a user-friendly and computerized enrollment system that can record, update, and retrieve student information.
- To reduce the time and effort for the enrollment process of students.
- To ensure data accuracy and minimize errors in student records.
- To implement security measures that protect sensitive student information from unauthorized access.

(5) Expected Output *(Indicate the specific products, processes, or services which the project is expected to produce. Final project must be reliable, efficient, user-friendly, and accessible to all users, especially for PWD personnel.)*

The proposed system will utilize the Greedy algorithm and Hashing algorithm as its core for scheduling and security. The Greedy Search algorithm will aid in the scheduling of class schedules of the enrollees for optimal schedules. And the Hashing algorithm for the security and encryption of the user credentials. The following are the modules to be delivered with the system:

The User Management module, this module handles account creation and the restriction of access of certain features according to their role/position. This also includes the registration and login page.

The Payment and Billing module, which manages the tuition and other school-related fees. It is responsible for fee calculation and breakdown which is to be added to the student's statement of account and tracking. It also ensures the proper recording of payments from the parents.

The Enrollment and Registration Module, which is the main module for processing both new and returning student enrollments. It provides functions for filling out new student information forms, re-enrolling returning students, in-person diagnostic tests scheduling, and monitoring the enrollment status (pending, approved, rejected). It also generates the official registration and enrollment forms.

The Class and Schedule Management module handles the organization of classes, subjects, and timetables. It allows for the creation of classes and sections, assignment of subjects and teachers, and scheduling of class timetables. This ensures proper coordination between students, teachers, and subjects.

The Reports Module, and this module generates the soft-copy reports and forms of the student's records and documents. It generates a PDF copy of the document for easy printing and portability.

(6) Significance of the Study *(Contributions of the project/study. This section justifies the study. It should include a statement of the contribution of the study in pushing back the existing frontiers of knowledge. It should also include the potential beneficiaries of the research and the possible uses of the results.)*

The study is significant to educational institutions, particularly smaller schools, which face the common problems mentioned in this paper, to provide a practical solution. By developing a Computerized Enrollment System, it will offer a practical, modern solution that directly addresses these issues.

The system is designed to streamline and simplify the enrollment process, making it faster, more accurate, and easier for administrators, registrars, teachers, and parents. It will also reduce the manual work, minimize errors, and enhance the security of student information. Moreover, it will provide openness and convenience for parents in enrolling their children. The system will help school administrators and staff better coordinate student records, payments, and class scheduling. For instructors, it reduces administrative obligations, providing them more time to concentrate on teaching.

Overall, this study not only brings technological improvements but also enables smaller schools to improve their operations and create a smoother, more reliable enrollment experience that benefits both schools and parents.

(7) Target Beneficiaries *(Who the clientele are and what are the expected outcomes/effects of the use of the project output?)*

The development of the digital enrollment system for Precious Little Light Academy will benefit the following:

- **For School Administration**, the development of the system will streamline enrollment-related tasks, significantly reducing the time and manual effort required to process applications.
- **For the Registrar and Staff**, the system will be an effective tool for centralizing and organizing information, enrollment applications, and payment processes.
- **For Parents and Students**, the system will provide an enrollment process that is more convenient for them and ensure transparency in the enrollment process and maintain the security of personal information.
- **For Proponents**, this project gives an opportunity to apply their knowledge while also increasing their skills in system development, problem solving, and teamwork. Moreover, it also gives the students with real-world experience that will help them prepare for future jobs, as well as the opportunity to design a meaningful solution for the school.
- **For Future Researchers**, this project can be used as a reference or foundation for implementing similar enrollment systems in other educational organizations. Its structure, modular layout, and security features can help future computer science students develop an enrollment system for a specific company.

(8) Remarks *(To Be Filled out by the Faculty In-Charge)*

Approved by:

9/4/2025

Dr. Karren V. de Lara
Faculty In-Charge

Appendix B: Endorsement Letter

September 3, 2025

Mrs. Annalie T. Amadore, M.A.Ed
School Principal
Precious Little Lights Academy
Block 5 Lot 8 St. Peter Ext., King David Subd., Burgos, Rodriguez, Rizal

Dear Ma'am,

We, the undersigned students of the Technological Institute of the Philippines (TIP), currently enrolled in the Bachelor of Science in Computer Science (BSCS) program, would like to respectfully request your permission to conduct an interview with your company.

This activity is part of a research study for our course Software Engineering 1 (CS 301), which involves analyzing an existing manual enrollment system. The objective of this study is to gather accurate and relevant information through interviews and observation, which will help us in designing a digital enrollment system tailored to the needs of your institution.

The system we intend to propose aims to enhance the security, reliability, and efficiency of processing information within your institution. It also seeks to improve the generation of timely reports to support better decision-making.

We assure you that all information gathered will be used strictly for academic purposes and will be treated with the utmost confidentiality.

We would be deeply grateful for your approval of this request and for the opportunity to learn from your organization.

Thank you very much for your kind consideration.

Respectfully yours,

Hofileña, Paul April A.

Mejia, Neil Sean D.

Raoet, Christian Louie C.

Noted by:

Dr. Karren V. de Lara
Faculty In-Charge

Received by:

ANNALIE T. AMADORE, M.A.Ed

Appendix C: Interview Transcription

Date: August 29, 2025

Interviewers: Paul April Hofileñ, Neil Sean Mejia

Notetaker: Christian Louie C. Raoet

Interviewee: Principal Amadore

Interviewer (Paul): Can we know the enrollment process at your school?

Interviewee (Principal): The enrollment process starts with parents booking an appointment and registering/enrolling in the school in person or they can inquire through our website, facebook or our email in the .

Interviewer (Paul): What are the requirements needed to submit during enrollment?

Interviewee (Principal): We require submission of requirements like SF 10 (Form 137) for transferees, which is the same format as public schools. We also require the submission of a photocopy of the PSA, and there is another requirement for the kindergarten; we require enrollees must be 5 years old or above, and an entrance exam is required.

Interviewer (Neil): In addition, how does the school process the student's ID?

Interviewee (Principal): We use the student's LRN, which is generated by DEPED after we process the student through them.

Interviewer (Neil): Do you conduct interviews during enrollment and an entrance exam for the enrollees?

Interviewee (Principal): Yes, we conduct an interview before enrollment, often on the same day as the test or beforehand. Moreover, we also have diagnostic testing that is either done on the same day or before enrollment, to assess their health and intelligence, or if they have some sort of mental health conditions like ADHD. We do not have any certification to teach those kids.

Interviewer (Paul): What happened to the children who didn't pass the diagnostic or like to prove that the child has special needs?

Interviewee (Principal): Sometimes, kids with special needs slip through, which can be a problem, but we do refund if a student is pulled out.

Interviewer (Paul): Do you think a digital enrollment system can improve the enrollment process? Which part is digital and paper-based?

Interviewee (Principal): Yes, definitely. We currently use both manual and Excel. Our printed form that we store in a file cabinet is both the student their parent's information and their statement of account. And we process the student's SF 9 or their grades on Excel, which will be printed out for the parents. A digital system would be very helpful for our daily work in the academy.

Interviewer (Paul): May we ask how many students are currently enrolled in Precious Little Lights Academy?

Interviewee (Principal): We have 119 students enrolled right now.

Interviewer (Paul): How many sections are there per grade level?

Interviewee (Principal): We combine preschool to Grade 1 in sections. Specifically, Kinder has 2 sections, and the others have 1 section each: Grade 1, Grade 2, Grade 4, Grade 5, and Grade 6.

Interviewer (Paul): How many students per section on average this day, and do you provide a limit per section?

Interviewee (Principal): On average, about 11 students per section. Our limit is 15 per section. Current numbers: Kinder - 15 total (across 2 sections), Grade 1 - 12, Grade 2 - 12, Grade 4 - 10, Grade 5 - 15, Grade 6 - 13.

Interviewer (Paul): What is the basis of enrolling a student in a section? Is it first-come-first serve or based on their grades?

Interviewee (Principal): We combine the children who were prior classmates since they are already friends or are acquainted with each other, but if the student is new, it's primarily first-come-first serve.

Interviewer (Paul): How many school days are there in the school's academic year?

Interviewee (Principal): DepEd provides the school days each year via a department order. For example, for 2025 to 2026, we follow the required 109 days. This order also includes the activities that they recommend the schools to conduct, but we make some changes to those activities if they do not apply to our school. You can see the school calendar in DepEd Order No. 12 per school year.

Interviewer (Paul): How long does each subject go for in a student's schedule?

Interviewee (Principal): Classes are scheduled at 40 minutes per subject, and I do the initial manual scheduling before dedicating it to teachers.

Interviewer (Paul): What are your payment methods? Do you give discounts and accept government discounts?

Interviewee (Principal): For payment methods, we accept GCash, bank-to-bank transfers, and cash. We issue two types of receipts: acknowledgement and official. For plans and installments, we handle quarterly changes for convenience. We also provide a 50% discount for church students and scholars. And we do not take any government subsidies.

Interviewer (Paul): How many facilities does your school have? Do you have extracurricular activities and fees?

Interviewee (Principal): We have 6 classrooms, a clinic, a playground for preschool, and labs for computer and science. We adjust activities based on DepEd's calendar if needed, since some of the extracurricular activities are not appropriate for our school.

Interviewer (Paul): If the school were to commission a digital system project, what would be the scope?

Interviewee (Principal): The scope would focus on an enrollment system and accounting, replacing our current manual and Excel listing. We'd want hard copies for staff, soft copies per grade level ready for print, and online viewing of grades for limited periods.

Appendix D: Photo with Company

Appendix E: References

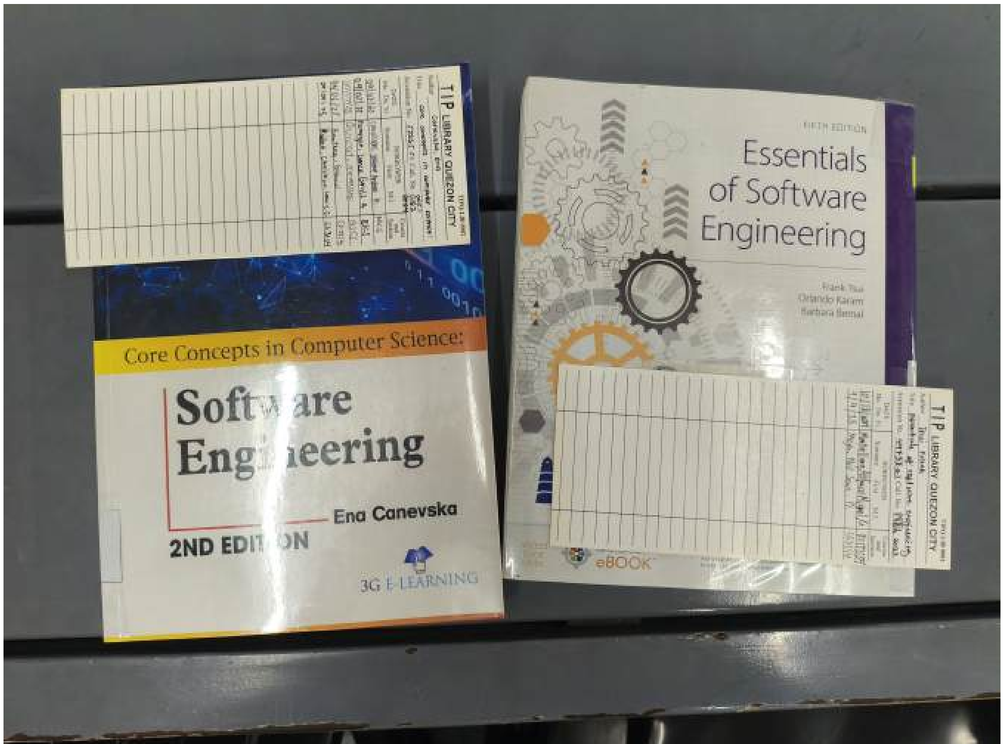
Ceveska, E. (2024). *Core Concepts in Computer Science: Software Engineering (2nd ed.)*. 3G E-Learning

Tsui, F., Karam, O., & Bernal, B. (2023). *Essentials of software engineering (5th ed.)*. Jones & Bartlett Learning

Wolny, J. (2024). *Career Guide: Software Designer (3rd ed.)*. 3G E-Learning

ISO 25010. (n.d.). https://iso25000.com/index.php/en/iso-25000-standards/iso-25010

GeeksforGeeks. (2024, May 10). *NPM bcrypt*. GeeksforGeeks. https://www.geeksforgeeks.org/node-js/npm-bcrypt/


TIP LIBRARY QUEZON CITY
Core Concepts in Computer Science:
Ena Canevska
3G E-LEARNING
FIFTH EDITION
Essentials of Software Engineering
Frank Tsui
Orlando Karam
Barbara Bernal
TIP LIBRARY QUEZON CITY
eBOOK

Appendix F: Permit

TIPQ-FIN-3004 Revision Status/Date: 3/2016 Feb. 10

TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES
938 Aurora Boulevard, Cubao, Quezon City
EXAMINATION PERMIT

FIRST SEMESTER 2025-2026

STUDENT NO. 2311588
STUDENT NAME MEJIA, NEIL SEAN DE OCAMPO
PROGRAM / STRAND / YEAR LEVEL BSCS / 2

FACULTY SIGNATURE AND DATE

COURSE CODE	FINALS	Date Taken
CS 002		11/18
CS 301		
CS 004		11/18
SOCSC 005		
GEE 003		11/19
ITE 013		
MATH 009CS		11-18
CS 403		

NOV 17 2025

TIPQ-FIN-3004 Revision Status/Date: 3/2016 Feb. 10

TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES
938 Aurora Boulevard, Cubao, Quezon City
EXAMINATION PERMIT

FIRST SEMESTER 2025-2026

STUDENT NO. 2311154
STUDENT NAME HOFILEÑA, PAUL APRIL AMADORE
PROGRAM / STRAND / YEAR LEVEL BSCS / 3

FACULTY SIGNATURE AND DATE

COURSE CODE	FINALS	Date Taken
CS 403		
CS 300		11/17/25
CS 301		
MATH 009CS		
GEC 007		
CS 004		11/18
SOCSC 005		
ITE 013		

NOV 18 2025

TIPQ-FIN-3004 Revision Status/Date: 3/2016 Feb. 10

TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES

938 Aurora Boulevard, Cubao, Quezon City

EXAMINATION PERMIT

FIRST SEMESTER 2025-2026

STUDENT NO. 2311621
STUDENT NAME RAOET, CHRISTIAN LOUIE CAPISTRANO
PROGRAM / STRAND / YEAR LEVEL BSCS / 3

FACULTY SIGNATURE AND DATE

COURSE CODE	FINALS	Date Taken
CS 300		11/19/25
GEC 007		
SOCSC 005		[illegible]
CS 004		
ITE 013		
CS 301		
MATH 009CS		11-18
CS 403		

NOV 12 2025